

Appendix A

Fundamentals and Complements of Linear Algebra

This appendix recalls some fundamental concepts of Linear Algebra, and discusses in greater details some topics that are normally not included in the math classes that are typically taught before Systems Theory. The here recalled concepts are indeed particularly useful for a better understanding of the concepts underlying Systems Theory. Readers that are interested in even further details may also refer to the texts listed in the bibliography of this appendix.

A.1 Basics on vector spaces

The central concept in the axiomatic development of Linear Algebra is that of *vector space over a field*. Historically, vector spaces have been studied by first assuming the scalar field to be $\mathbb{R}$ (the real field) or $\mathbb{C}$ (the complex field). Soon it became clear that most of the results remain valid for any field, and today the term “Linear Algebra” denotes the study of vector spaces over arbitrary fields.

Definition A.1.1 (VECTOR SPACE) *Let K be a field. A vector space over K is a set $\mathbf{V}$ on which a binary operation is defined, called sum and denoted by the symbol $+$, so that $\mathbf{V}$ is a commutative group with respect to the sum, namely*

- *the operation is associative: $(\mathbf{v}_1 + \mathbf{v}_2) + \mathbf{v}_3 = \mathbf{v}_1 + (\mathbf{v}_2 + \mathbf{v}_3)$, $\forall \mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3 \in \mathbf{V}$;*
- *the operation is commutative: $\mathbf{v}_1 + \mathbf{v}_2 = \mathbf{v}_2 + \mathbf{v}_1$, $\forall \mathbf{v}_1, \mathbf{v}_2 \in \mathbf{V}$;*
- *there exists the null element $\mathbf{0}$ with respect to (w.r.t.) the operation: $\mathbf{0} + \mathbf{v} = \mathbf{v} + \mathbf{0} = \mathbf{v}$, $\forall \mathbf{v} \in \mathbf{V}$;*
- *any element has an inverse: $\forall \mathbf{v} \in \mathbf{V}, \exists (-\mathbf{v})$ such that $\mathbf{v} + (-\mathbf{v}) = \mathbf{0}$.*

Moreover, a function $K \times \mathbf{V} \rightarrow \mathbf{V} : (\alpha, \mathbf{v}) \mapsto \alpha\mathbf{v}$ is defined, called scalar product, that associates to a scalar α and a vector $\mathbf{v}$ the vector $\alpha\mathbf{v}$, such that, for any choice of α and β in K , and of $\mathbf{v}$ and $\tilde{\mathbf{v}}$ in $\mathbf{V}$, the following properties hold:

$$\begin{aligned}\alpha(\mathbf{v} + \tilde{\mathbf{v}}) &= \alpha\mathbf{v} + \alpha\tilde{\mathbf{v}}; \\ (\alpha + \beta)\mathbf{v} &= \alpha\mathbf{v} + \beta\mathbf{v};\end{aligned}$$

$$(\alpha\beta)\mathbf{v} = \alpha(\beta\mathbf{v});$$

$$1\mathbf{v} = \mathbf{v}, \text{ where } 1 \text{ the unit in } K.$$

Note that the symbol $+$ indicates both the sum over the vector space $\mathbf{V}$ and the sum over the field K .

Example A.1.1 (i) The set of n -tuples of elements belonging to the field K

$$K^n = \left\{ \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_n \end{bmatrix}, \alpha_i \in K \right\}$$

is a vector space over K with the usual operations.

(ii) The set of $p \times m$ matrices with entries in the field K

$$K^{p \times m} = \left\{ \begin{bmatrix} \alpha_{11} & \alpha_{12} & \dots & \alpha_{1m} \\ \alpha_{21} & \alpha_{22} & \dots & \alpha_{2m} \\ \dots & \dots & \dots & \dots \\ \alpha_{p1} & \alpha_{p2} & \dots & \alpha_{pm} \end{bmatrix}, \alpha_{ij} \in K \right\}$$

is a vector space over K . How are the *sum* and *scalar product* defined?

(iii) The set of polynomials in the variable z , with coefficients belonging to the field K

$$K[z] = \{\alpha_0 + \alpha_1 z + \dots + \alpha_\nu z^\nu, \alpha_i \in K, \nu \in \mathbb{N}\}$$

takes the structure of vector space over K , when sum and scalar product by an element of the field are the usual ones defined in the algebra of polynomials.

(iv) The set of real functions of a real variable, continuous on the interval $[0, 1]$

$$\mathcal{C}_{[0,1]} = \{f : [0, 1] \rightarrow \mathbb{R}, f \text{ continuous}\}$$

is a vector space once the sum of two functions f and g is defined as

$$f + g : [0, 1] \rightarrow \mathbb{R} : x \mapsto f(x) + g(x)$$

and the product of the function f by the scalar α is defined as

$$\alpha f : [0, 1] \rightarrow \mathbb{R} : x \mapsto \alpha f(x).$$

(v) $\mathbb{C}$ can be regarded as a vector space over $\mathbb{R}$: the sum of two vectors $\alpha + i\beta$ and $\gamma + i\delta$, $\alpha, \beta, \gamma, \delta \in \mathbb{R}$ is performed as the sum over $\mathbb{C}$ and the scalar product of $\rho \in \mathbb{R}$ by the vector $\alpha + i\beta$ is computed as the product in $\mathbb{C}$ of $\rho + i0$ by $\alpha + i\beta$.

Definition A.1.2 (VECTOR SUBSPACE) Let $\mathbf{V}$ be a vector space over the field K . A subset $\mathbf{W}$ of $\mathbf{V}$ is a (vector) subspace of $\mathbf{V}$ if, for any pair of vectors $\mathbf{w}_1, \mathbf{w}_2$ in $\mathbf{W}$ and for any pair of scalar numbers α, β in K , the vector $\alpha\mathbf{w}_1 + \beta\mathbf{w}_2$ is in $\mathbf{W}$.

Example A.1.2 (i) The subset $\mathbf{W}$ of K^n defined as

$$\mathbf{W} := \left\{ \begin{bmatrix} 0 \\ \alpha_2 \\ \vdots \\ \alpha_n \end{bmatrix}, \alpha_i \in K \right\}$$

is a vector subspace of K^n .

(ii) The set $\mathbf{D}$ of diagonal matrices of dimension $m \times m$ with entries in the field K

$$\mathbf{D} = \left\{ \begin{bmatrix} \alpha_{11} & 0 & \dots & 0 \\ 0 & \alpha_{22} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \alpha_{mm} \end{bmatrix}, \alpha_{ii} \in K \right\}$$

is a vector subspace of $K^{m \times m}$.

(iii) The set $K[z^2]$ of *even polynomials* in the variable z

$$K[z^2] = \{\alpha_0 + \alpha_2 z^2 + \dots + \alpha_{2\nu} z^{2\nu}, \alpha_i \in K, \nu \in \mathbb{N}\}$$

is a subspace of $K[z]$.

(iv) The set of real functions of real variable, differentiable on the interval $[0, 1]$, is a subspace of $\mathcal{C}_{[0,1]}$.

(v) The set of purely imaginary complex numbers is a subspace of $\mathbb{C}$, when $\mathbb{C}$ is considered as a vector space over the field $\mathbb{R}$.

Definition A.1.3 (LINEAR DEPENDENCE) m vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m$ of the vector space $\mathbf{V}$ over K can be

- linearly dependent, if there exist $\alpha_1, \alpha_2, \dots, \alpha_m$ in K , not all equal to zero, such that $\sum_{i=1}^m \alpha_i \mathbf{v}_i = \mathbf{0}$, or
- linearly independent, if $\sum_{i=1}^m \alpha_i \mathbf{v}_i = \mathbf{0}$ implies that the scalars $\alpha_1, \alpha_2, \dots, \alpha_m$ are all equal to zero.

An arbitrary subset $\mathbf{S}$ of $\mathbf{V}$ is linearly independent if any finite subset of vectors taken from $\mathbf{S}$ is linearly independent.

If $\mathbf{S} = \{\mathbf{s}_\gamma, \gamma \in \Gamma\}$ is a subset of $\mathbf{V}$ indexed on set Γ , the set

$$\text{span}(\mathbf{S}) = \langle \mathbf{S} \rangle := \left\{ \sum_{i \in F} \alpha_i \mathbf{s}_i, F \subseteq \Gamma, F \text{ finite}, \alpha_i \in K, \mathbf{s}_i \in \mathbf{S} \right\}$$

of all finite linear combinations of elements of $\mathbf{S}$ is a subspace of $\mathbf{V}$, called *(sub)space generated by $\mathbf{S}$* . If $\text{span}(\mathbf{S}) = \mathbf{V}$, the set $\mathbf{S}$ is a *set of generators* for $\mathbf{V}$ and if $\mathbf{S}$ is a finite set, $\mathbf{V}$ is said to be *finitely generated*.

Example A.1.3 (i) The set of all vectors in $\mathbb{R}^n$ whose components take exclusively the values 0 or 1 is a set of generators for $\mathbb{R}^n$.

(ii) The set of all polynomials in the variable z with real coefficients is a set of generators for $\mathbb{R}[z]$.

Definition A.1.4 [BASIS OF A VECTOR SPACE] A subset $\mathcal{B}$ of a vector space $\mathbf{V}$ over K is a basis of $\mathbf{V}$ if the following two conditions simultaneously hold

- i) $\mathcal{B}$ is a set of generators for $\mathbf{V}$;
- ii) $\mathcal{B}$ is a linearly independent set.

The following facts can be proved¹:

1. Any vector space has a basis.
2. All bases of the same vector space have the same cardinality, which is called *space dimension*.
3. If $\mathbf{S}$ is a linearly independent subset of a vector space $\mathbf{V}$ and $\mathcal{B}$ is a basis of $\mathbf{V}$, there exists a subset $\mathbf{C} \subset \mathcal{B}$ such that $\mathbf{C} \cup \mathbf{S}$ is a basis of $\mathbf{V}$. All completions of $\mathbf{S}$ to a basis of $\mathbf{V}$ have the same cardinality.

¹For a formal proof, we refer the interested reader to reference 4. of section A.15.

4. If $\mathbf{S}$ is a subset of generators of the space $\mathbf{V}$, from $\mathbf{S}$ a basis can be chosen, namely there exists a subset $\mathbf{P} \subseteq \mathbf{S}$ that is a basis of $\mathbf{V}$.
5. If $\mathcal{B}$ is a basis of $\mathbf{V}$, any vector of $\mathbf{V}$ can be expressed in one and only one way as a linear combination of a finite number of vectors of $\mathcal{B}$.

Example A.1.4 (i) The subset of $\mathbb{R}^n$ consisting of the vectors

$$\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{bmatrix}, \dots, \mathbf{e}_n = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix}$$

is a basis (called *canonical basis* or *standard basis*) for $\mathbb{R}^n$.

(ii) Let us consider, for $i = 1, 2, \dots, p$ and $j = 1, 2, \dots, m$, matrices $E^{(ij)} \in K^{p \times m}$ defined as

$$E_{hk}^{(i,j)} = \begin{cases} 1 & \text{if } h = i \text{ and } k = j, \\ 0 & \text{for any other value of } h \text{ and } k. \end{cases}$$

Since the set of these matrices is linearly independent and generates $K^{p \times m}$, it is a basis for the space $K^{p \times m}$, which is hence of dimension pm .

(iii) The monomials $1, z, z^2, z^3, \dots$ constitute a (countable) basis for the space $K[z]$ over the field K .

(iv) The complex numbers $1 = 1 + i0$ and $i = 0 + i1$ are a basis of $\mathbb{C}$, considered as a vector space over the field $\mathbb{R}$.

If the space $\mathbf{V}$ over K has finite dimension n and if $(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n)$ is a *ordered* n -tuple of vectors that constitute a basis of $\mathbf{V}$, any vector $\mathbf{v} \in \mathbf{V}$ can be represented in a unique way as $\mathbf{v} = \sum_{i=1}^n \xi_i \mathbf{v}_i$ or as

$$\mathbf{v} = (\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n) \begin{bmatrix} \xi_1 \\ \xi_2 \\ \vdots \\ \xi_n \end{bmatrix}, \quad (\text{A.1})$$

where it is meant that the basis vectors need to be combined in order with the elements of the n -tuple of K^n . The numbers ξ_i are the *components* of vector $\mathbf{v}$ w.r.t. the chosen basis. In this way, every n -tuple of K^n bijectively corresponds to a vector of $\mathbf{V}$ and it is immediate to verify that this correspondence is an isomorphism between $\mathbf{V}$ and K^n , in the sense that the sum of two vectors corresponds to the sum of the associated n -tuples of components and the product of a scalar $\alpha \in K$ by a vector $\mathbf{v}$ in $\mathbf{V}$ corresponds to the product of α by the n -tuple of the components of $\mathbf{v}$. Note that by replacing the sum and product operations in the “abstract” space $\mathbf{V}$ with operations among n -tuples in K^n we implicitly refer to a specific choice of the basis of $\mathbf{V}$.

Choosing a different basis $(\tilde{\mathbf{v}}_1, \tilde{\mathbf{v}}_2, \dots, \tilde{\mathbf{v}}_n)$ yields a different correspondence between the vectors of $\mathbf{V}$ and the n -tuples of K^n : in other words, the components of the same vector $\mathbf{v}$ w.r.t. different bases are in general different.

In many situations it is useful to change the reference basis in a vector space and it is therefore necessary to understand how the components of an assigned vector $\mathbf{v}$ will change as the reference basis varies.

Let $(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n)$ and $(\tilde{\mathbf{v}}_1, \tilde{\mathbf{v}}_2, \dots, \tilde{\mathbf{v}}_n)$ be two ordered bases of $\mathbf{V}$, called, to ease the

discussion, “old” basis and “new” basis, respectively. Any vector of the new basis can be expressed as a linear combination of the vectors of the old basis, that is

$$\tilde{\mathbf{v}}_j = \sum_{i=1}^n \mathbf{v}_i t_{ij}, \quad j = 1, 2, \dots, n.$$

Upon denoting by T the matrix whose j -th column is given by the n -tuple of the components of the vector $\tilde{\mathbf{v}}_j$, belonging to the new basis, w.r.t. the old basis, $j = 1, 2, \dots, n$, we can express in a compact way the new basis with reference to the old one:

$$(\tilde{\mathbf{v}}_1, \tilde{\mathbf{v}}_2, \dots, \tilde{\mathbf{v}}_n) = (\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n) \begin{bmatrix} t_{11} & t_{12} & \dots & t_{1n} \\ t_{21} & t_{22} & \dots & t_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ t_{n1} & t_{n2} & \dots & t_{nn} \end{bmatrix} = (\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n) T. \quad (\text{A.2})$$

Similarly, a matrix S of size $n \times n$ can be obtained to express the old basis with respect to the new one

$$(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n) = (\tilde{\mathbf{v}}_1, \tilde{\mathbf{v}}_2, \dots, \tilde{\mathbf{v}}_n) S \quad (\text{A.3})$$

and it is straightforward to obtain, by replacing (A.2) in (A.3),

$$(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n) = (\tilde{\mathbf{v}}_1, \tilde{\mathbf{v}}_2, \dots, \tilde{\mathbf{v}}_n) S = (\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n) T S. \quad (\text{A.4})$$

Since the components of any vector w.r.t. an assigned basis are only determined, in (A.4) it must hold that $TS = I_n$. It follows that T and S are invertible matrices and $S = T^{-1}$.

From the n -tuple $[\xi_1 \ \xi_2 \ \dots \ \xi_n]^T$ of the components of $\mathbf{v}$ w.r.t. the old basis, the n -tuple $[\tilde{\xi}_1 \ \tilde{\xi}_2 \ \dots \ \tilde{\xi}_n]^T$ of the components of $\mathbf{v}$ w.r.t. the new basis can be computed by resorting to (A.1) and (A.3). Indeed, it holds

$$\mathbf{v} = (\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n) \begin{bmatrix} \xi_1 \\ \xi_2 \\ \vdots \\ \xi_n \end{bmatrix} = (\tilde{\mathbf{v}}_1, \tilde{\mathbf{v}}_2, \dots, \tilde{\mathbf{v}}_n) T^{-1} \begin{bmatrix} \xi_1 \\ \xi_2 \\ \vdots \\ \xi_n \end{bmatrix}$$

and thus

$$\begin{bmatrix} \tilde{\xi}_1 \\ \tilde{\xi}_2 \\ \vdots \\ \tilde{\xi}_n \end{bmatrix} = T^{-1} \begin{bmatrix} \xi_1 \\ \xi_2 \\ \vdots \\ \xi_n \end{bmatrix}.$$

- EXERCISE A.1.1 [n -TUPLES SPACES] (i) If $\mathbf{x} = \begin{bmatrix} \xi_1 \\ \vdots \\ \xi_n \end{bmatrix}$ is an n -tuple of K^n , the coefficients ξ_i can be interpreted as the components of $\mathbf{x}$ w.r.t. the canonical basis in K^n . Let us consider a set of linearly independent n -tuples

$$\mathbf{x}^{(1)} = \begin{bmatrix} \xi_1^{(1)} \\ \vdots \\ \xi_n^{(1)} \end{bmatrix}, \mathbf{x}^{(2)} = \begin{bmatrix} \xi_1^{(2)} \\ \vdots \\ \xi_n^{(2)} \end{bmatrix}, \dots, \mathbf{x}^{(n)} = \begin{bmatrix} \xi_1^{(n)} \\ \vdots \\ \xi_n^{(n)} \end{bmatrix}. \quad (\text{A.5})$$

These can be seen as a “new” basis in K^n . If we introduce the matrix $X = [\mathbf{x}^{(1)} \ \dots \ \mathbf{x}^{(n)}]$, the components of the n -tuple $\mathbf{x}$ w.r.t. the “new” basis (A.5) of K^n are given by $X^{-1}\mathbf{x}$ (indeed, by combining the n -tuples of (A.5) with the components of $X^{-1}\mathbf{x}$ we get $\mathbf{x}$).

(ii) If the space $\mathbf{V}$ over the field K is referred to the basis $(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n)$ and if $\mathbf{x} \in K^n$ gives the components of $\mathbf{v} \in \mathbf{V}$ w.r.t. such basis, then $X^{-1}\mathbf{x}$ gives the components of the same vector w.r.t. the basis:

$$(\tilde{\mathbf{v}}_1, \tilde{\mathbf{v}}_2, \dots, \tilde{\mathbf{v}}_n) := (\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n)X.$$

In this sense, selecting a new basis (A.5) in K^n corresponds to choosing a new basis in $\mathbf{V}$ and the components of the n -tuple $\mathbf{x}$ w.r.t. the new basis in K^n coincide with the components w.r.t. the new basis in $\mathbf{V}$ of the vector that was represented by the n -tuple $\mathbf{x}$ w.r.t. the old basis of $\mathbf{V}$.

Definition A.1.5 [SUM OF VECTOR SPACES] *The sum*

$$\mathbf{W}^{(1)} + \mathbf{W}^{(2)} + \dots + \mathbf{W}^{(h)} = \sum_{i=1}^h \mathbf{W}^{(i)}$$

of the subspaces $\mathbf{W}^{(1)}, \mathbf{W}^{(2)}, \dots, \mathbf{W}^{(h)}$ of the vector space $\mathbf{V}$ is the space generated by $\mathbf{W}^{(1)} \cup \mathbf{W}^{(2)} \cup \dots \cup \mathbf{W}^{(h)}$.

It is easy to verify that

1. $\sum_{i=1}^h \mathbf{W}^{(i)} = \{\mathbf{w}^{(1)} + \mathbf{w}^{(2)} + \dots + \mathbf{w}^{(h)}, \mathbf{w}^{(i)} \in \mathbf{W}^{(i)}, i = 1, 2, \dots, h\};$
2. if $\mathcal{B}^{(i)}$ is a basis of $\mathbf{W}^{(i)}$ for $i = 1, 2, \dots, h$, then $\cup_{i=1}^h \mathcal{B}^{(i)}$ is a set of generators for $\sum_{i=1}^h \mathbf{W}^{(i)}$.

Proposition A.1.6 [DIRECT SUM OF VECTOR SPACES] *Let $\mathbf{W}^{(i)}, i = 1, 2, \dots, h$, be subspaces of $\mathbf{V}$. The following facts are equivalent*

- i) *given any h -tuple of non-zero vectors $(\mathbf{v}^{(1)}, \mathbf{v}^{(2)}, \dots, \mathbf{v}^{(h)})$ such that $\mathbf{v}^{(i)} \in \mathbf{W}^{(i)}$, the vectors of the h -tuple are linearly independent;*
- ii) *the bases $\mathcal{B}^{(i)}, i = 1, 2, \dots, h$, for $\mathbf{W}^{(i)}$ are pairwise disjoint and $\cup_{i=1}^h \mathcal{B}^{(i)}$ is a linearly independent set;*
- iii) *the intersection between every subspace and the sum of the others is the null vector:*

$$\mathbf{W}^{(\nu)} \cap \left(\sum_{i \neq \nu} \mathbf{W}^{(i)} \right) = \{\mathbf{0}\}, \quad \nu = 1, 2, \dots, h. \quad (\text{A.6})$$

PROOF (i) $\Rightarrow$ (ii) The basis are disjoint: if two of them, e.g. $\mathcal{B}^{(1)}$ and $\mathcal{B}^{(2)}$, had a (necessarily non-null) vector $\mathbf{v}^{(12)}$ in common, any h -tuple $(\mathbf{v}^{(1)}, \mathbf{v}^{(2)}, \dots, \mathbf{v}^{(h)})$ with $\mathbf{v}^{(1)} = \mathbf{v}^{(2)} = \mathbf{v}^{(12)}$ would have non-zero vectors that are linearly dependent.

Let us arbitrarily choose n vectors in $\cup_{i=1}^h \mathcal{B}^{(i)}$. Among these, n_1 will be in $\mathcal{B}^{(1)}$, say $\mathbf{v}_1^{(1)}, \dots, \mathbf{v}_{n_1}^{(1)}$, n_2 in $\mathcal{B}^{(2)}$, say $\mathbf{v}_1^{(2)}, \dots, \mathbf{v}_{n_2}^{(2)}$, and so on, up to a set of n_h vectors in $\mathcal{B}^{(h)}$, say $\mathbf{v}_1^{(h)}, \dots, \mathbf{v}_{n_h}^{(h)}$. If it results

$$\mathbf{0} = \sum_{i=1}^{n_1} \alpha_i^{(1)} \mathbf{v}_i^{(1)} + \dots + \sum_{i=1}^{n_h} \alpha_i^{(h)} \mathbf{v}_i^{(h)}, \quad (\text{A.7})$$

for assumption (i) to hold it must be

$$\sum_{i=1}^{n_1} \alpha_i^{(1)} \mathbf{v}_i^{(1)} = \dots = \sum_{i=1}^{n_h} \alpha_i^{(h)} \mathbf{v}_i^{(h)} = \mathbf{0}.$$

Then, it follows that $0 = \alpha_i^{(1)}, i = 1, 2, \dots, n_1$, because the vectors $\mathbf{v}_i^{(1)}$ are linearly independent because they are elements of the basis $\mathcal{B}_1$. Similarly, $0 = \alpha_i^{(j)}, i = 1, 2, \dots, n_j$ per $j = 2, 3, \dots, h$. Therefore, in (A.7) all coefficients are zero and $\cup_{i=1}^h \mathcal{B}^{(i)}$ is a linearly independent set.

(ii) $\Rightarrow$ (iii) If a non-zero vector $\mathbf{v}$ would, for instance, belong to a $\mathbf{W}^{(1)}$ and to $\sum_{i=2}^h \mathbf{W}^{(i)}$, it would be possible to represent it both as a combination of vectors of $\mathcal{B}^{(1)}$, and as a combination of vectors of $\cup_{i=2}^h \mathcal{B}^{(i)}$. By subtracting the second combination from the first, the null vector $\mathbf{0}$ would be expressed as a combination (with non-zero coefficients) of vectors in $\cup_{i=1}^h \mathcal{B}^{(i)}$.

(iii) $\Rightarrow$ (i) Let $\mathbf{0} = \sum_{i=1}^h \alpha_i \mathbf{v}^{(i)}$, with $\mathbf{v}^{(i)} \in \mathbf{W}^{(i)}$. If at least one $\mathbf{v}^{(i)}$ was not null, say $\mathbf{v}^{(1)} \neq \mathbf{0}$, then $\mathbf{v}^{(1)} \in \mathbf{W}^{(1)} \cap \left(\sum_{i \neq 1} \mathbf{W}^{(i)} \right)$ would contradict (A.6). ■

When one (and hence all) of the aforementioned properties holds, the subspaces $\mathbf{W}^{(i)}$ are said to be *independent* and the subspace sum is called *direct sum*. The following notation is adopted to highlight this fact $\oplus_{i=1}^h \mathbf{W}^{(i)}$ or $\mathbf{W}^{(1)} \oplus \mathbf{W}^{(2)} \oplus \dots \oplus \mathbf{W}^{(h)}$.

Based on the definition of independent subspaces, also the following definition holds.

Definition A.1.7 [VECTOR INDEPENDENCE W.R.T. A SUBSPACE] *Let $\mathbf{U}$ be a subspace of $\mathbf{V}$ and let $\mathbf{G}$ be a set of generators of $\mathbf{U}$.*

A subset $\mathbf{S} := \{\mathbf{s}_j, j \in J\}$ of vectors of $\mathbf{V}$ is called linearly independent w.r.t. $\mathbf{U}$, or - equivalently - w.r.t. $\mathbf{G}$, if $\mathbf{S}$ is linearly independent and if $\text{span}(\mathbf{S})$ and $\mathbf{U} = \text{span}(\mathbf{G})$ are independent subspaces.

A.2 Linear transformations and matrix representations

Definition A.2.1 [LINEAR TRANSFORMATION] *Let $\mathbf{V}$ and $\mathbf{W}$ be vector spaces over the field K . A map $\mathcal{A} : \mathbf{V} \rightarrow \mathbf{W} : \mathbf{v} \mapsto \mathcal{A}\mathbf{v}$ is a K -linear map or a linear transformation if*

$$\begin{aligned} \mathcal{A}(\mathbf{v} + \tilde{\mathbf{v}}) &= \mathcal{A}\mathbf{v} + \mathcal{A}\tilde{\mathbf{v}} \\ \mathcal{A}(\alpha\mathbf{v}) &= \alpha(\mathcal{A}\mathbf{v}) \end{aligned}$$

for any $\mathbf{v}$ and $\tilde{\mathbf{v}}$ in $\mathbf{V}$ and for any α in K .

It is immediate to verify that the *kernel* and the *image* of the linear transformation $\mathcal{A}$

$$\begin{aligned} \ker \mathcal{A} &:= \{\mathbf{v} \in \mathbf{V} : \mathcal{A}\mathbf{v} = \mathbf{0}\} \\ \text{Im} \mathcal{A} &:= \{\mathbf{w} \in \mathbf{W} : \mathbf{w} = \mathcal{A}\mathbf{v}, \exists \mathbf{v} \in \mathbf{V}\} \end{aligned}$$

are subspaces of $\mathbf{V}$ and $\mathbf{W}$, respectively.

The transformation is called *surjective* if $\text{Im} \mathcal{A} = \mathbf{W}$ and *injective* if, for any $\mathbf{v}$ and $\tilde{\mathbf{v}}$ in $\mathbf{V}$, $\mathcal{A}\mathbf{v} = \mathcal{A}\tilde{\mathbf{v}}$ implies $\mathbf{v} = \tilde{\mathbf{v}}$. It is an easy exercise to prove that $\mathcal{A}$ is injective if and only if $\ker \mathcal{A} = \{\mathbf{0}\}$.

A linear transformation $\mathcal{A} : \mathbf{V} \rightarrow \mathbf{W}$ is completely identified by its restriction to the vectors of a basis of $\mathbf{V}$. In fact, if $\mathbf{v}_i, i \in I$, are the vectors of the basis, indexed on the

(finite or infinite) set I , and if the values $\mathcal{A}\mathbf{v}_i$ taken by $\mathcal{A}$ on the vectors of the basis are known, each $\mathbf{v} \in \mathbf{V}$ can be expressed as $\mathbf{v} = \sum_{i \in F} \alpha_i \mathbf{v}_i$, with F a finite set, and then

$$\mathcal{A}\mathbf{v} = \sum_{i \in F} \alpha_i (\mathcal{A}\mathbf{v}_i). \quad (\text{A.8})$$

On the other hand, given a basis $\mathcal{B} = \{\mathbf{v}_i, i \in I\}$ of $\mathbf{V}$, a map arbitrarily defined on the basis $\mathcal{A} : \mathcal{B} \rightarrow \mathbf{W}$ can be extended linearly in a unique way to the whole space $\mathbf{V}$, via the (A.8).

Example A.2.1 (i) The most important linear map is probably the one induced by a matrix $M \in K^{p \times m}$:

$$\mathcal{A} : K^m \rightarrow K^p : \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_m \end{bmatrix} \mapsto M \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_m \end{bmatrix}.$$

The map is surjective if and only if the rows of M are linearly independent, namely the rank of matrix M is equal to p (*full row rank*); it is injective if and only if the columns of M are linearly independent, meaning that the rank of M is m (*full column rank*). Can you explain why?

(ii) The map $\mathcal{A} : \mathbb{R}^n \rightarrow \mathbb{R}^n$ defined by

$$\mathcal{A} : \begin{bmatrix} \alpha_1 \\ \vdots \\ \alpha_{n-1} \\ \alpha_n \end{bmatrix} \mapsto \begin{bmatrix} \alpha_2 \\ \vdots \\ \alpha_n \\ 0 \end{bmatrix}$$

is linear, it is neither surjective nor injective (what is its kernel?).

(iii) The map

$$\mathcal{A} : K[z] \rightarrow K[z] : p(z) \mapsto p(z^2)$$

is linear, injective and not surjective.

(iv) The map

$$\mathcal{A} : \mathcal{C}_{[0,1]} \rightarrow \mathcal{C}_{[0,1]}^1 : f(\cdot) \mapsto \int_0^t f(\xi) d\xi, \quad t \in [0,1]$$

is $\mathbb{R}$ -linear. Is it injective? Is it surjective?

A.2.1 Matrix representation

Let $\mathcal{A} : \mathbf{V} \rightarrow \mathbf{W}$ be a linear transformation and assume $\dim(\mathbf{V}) = m < \infty$, $\dim(\mathbf{W}) = p < \infty$. Let us choose two ordered bases in $\mathbf{V}$ and $\mathbf{W}$, $(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m)$ and $(\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_p)$ respectively, and let us represent the transformed $\mathcal{A}\mathbf{v}_j$ of the vectors of the basis of $\mathbf{V}$ as linear combinations of the vectors of the basis of $\mathbf{W}$:

$$\mathcal{A}\mathbf{v}_j = \sum_{i=1}^p \mathbf{w}_i a_{ij} = (\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_p) \begin{bmatrix} a_{1j} \\ a_{2j} \\ \vdots \\ a_{pj} \end{bmatrix}, \quad j = 1, 2, \dots, m.$$

We can represent compactly the transformed of the m basis vectors:

$$(\mathcal{A}\mathbf{v}_1, \mathcal{A}\mathbf{v}_2, \dots, \mathcal{A}\mathbf{v}_m) = (\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_p) \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1m} \\ a_{21} & a_{22} & \dots & a_{2m} \\ \dots & \dots & \dots & \dots \\ a_{p1} & a_{p2} & \dots & a_{pm} \end{bmatrix} = (\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_p)A. \quad (\text{A.9})$$

In (A.9) the j -th column of the matrix A is obtained from the components of the transformed through $\mathcal{A}$ of the j -th vector of the basis of $\mathbf{V}$, with respect to the basis of $\mathbf{W}$. Given an arbitrary vector in $\mathbf{V}$

$$\mathbf{v} = \sum_{j=1}^m \mathbf{v}_j \xi_j = (\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m) \begin{bmatrix} \xi_1 \\ \xi_2 \\ \vdots \\ \xi_m \end{bmatrix},$$

its transformed through $\mathcal{A}$ is $\mathcal{A}\mathbf{v} = \mathcal{A}(\sum_{j=1}^m \mathbf{v}_j \xi_j) = \sum_{j=1}^m (\mathcal{A}\mathbf{v}_j) \xi_j$ and, considering (A.9),

$$\mathcal{A}\mathbf{v} = (\mathcal{A}\mathbf{v}_1, \mathcal{A}\mathbf{v}_2, \dots, \mathcal{A}\mathbf{v}_m) \begin{bmatrix} \xi_1 \\ \xi_2 \\ \vdots \\ \xi_m \end{bmatrix} = (\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_p)A \begin{bmatrix} \xi_1 \\ \xi_2 \\ \vdots \\ \xi_m \end{bmatrix}. \quad (\text{A.10})$$

It follows that the p -tuple of the components of the transformed of $\mathbf{v}$ can be obtained by applying the matrix A to the m -tuple of the components of $\mathbf{v}$. We can summarize what just said in the following

Proposition A.2.2 [BASES AND REPRESENTATIONS] *If $\mathbf{V}$ and $\mathbf{W}$ have finite dimensions m and p , respectively, and if the bases in the two spaces have been chosen, then*

- the vectors of $\mathbf{V}$ and of $\mathbf{W}$ are represented isomorphically through m -tuples in K^m and through p -tuples in K^p ;
- any linear transformation $\mathcal{A} : \mathbf{V} \rightarrow \mathbf{W}$ is represented through a matrix $A \in K^{p \times m}$, such that the transformed $\mathcal{A}\mathbf{v} \in \mathbf{W}$ of a vector $\mathbf{v} \in \mathbf{V}$ is represented by the p -tuple obtained multiplying A by the m -tuple representing $\mathbf{v}$.

Let us move now to the *new* bases $(\tilde{\mathbf{v}}_1, \tilde{\mathbf{v}}_2, \dots, \tilde{\mathbf{v}}_m)$ and $(\tilde{\mathbf{w}}_1, \tilde{\mathbf{w}}_2, \dots, \tilde{\mathbf{w}}_p)$ in $\mathbf{V}$ and in $\mathbf{W}$ and let us introduce $T \in K^{m \times m}$ and $Q \in K^{p \times p}$ as the invertible matrices that realize the changes of bases:

$$\begin{aligned} (\tilde{\mathbf{v}}_1, \tilde{\mathbf{v}}_2, \dots, \tilde{\mathbf{v}}_m) &= (\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m)T \\ (\tilde{\mathbf{w}}_1, \tilde{\mathbf{w}}_2, \dots, \tilde{\mathbf{w}}_p) &= (\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_p)Q. \end{aligned} \quad (\text{A.11})$$

To determine the matrix associated to the transformation $\mathcal{A}$ with respect to the new bases, we need to find the matrix $\tilde{A}$ that satisfies the relation

$$(\mathcal{A}\tilde{\mathbf{v}}_1, \mathcal{A}\tilde{\mathbf{v}}_2, \dots, \mathcal{A}\tilde{\mathbf{v}}_m) = (\tilde{\mathbf{w}}_1, \tilde{\mathbf{w}}_2, \dots, \tilde{\mathbf{w}}_p)\tilde{A}.$$

To this aim, it is convenient to note that the transformed of the new basis vectors are related to the transformed of the old vectors via the following

$$\begin{aligned}
 (\mathcal{A}\tilde{\mathbf{v}}_1, \mathcal{A}\tilde{\mathbf{v}}_2, \dots, \mathcal{A}\tilde{\mathbf{v}}_m) &= (\mathcal{A}(\sum_{i=1}^m \mathbf{v}_i t_{i1}), \mathcal{A}(\sum_{i=1}^m \mathbf{v}_i t_{i2}), \dots, \mathcal{A}(\sum_{i=1}^m \mathbf{v}_i t_{im})) \\
 &= (\sum_{i=1}^m (\mathcal{A}\mathbf{v}_i) t_{i1}, \sum_{i=1}^m (\mathcal{A}\mathbf{v}_i) t_{i2}, \dots, \sum_{i=1}^m (\mathcal{A}\mathbf{v}_i) t_{im}) \\
 &= (\mathcal{A}\mathbf{v}_1, \mathcal{A}\mathbf{v}_2, \dots, \mathcal{A}\mathbf{v}_m)T.
 \end{aligned}$$

Based on (A.9) one gets $(\mathcal{A}\mathbf{v}_1, \mathcal{A}\mathbf{v}_2, \dots, \mathcal{A}\mathbf{v}_m) = (\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_p)A$ and, by keeping into account the second equation in (A.11), one finally finds

$$\begin{aligned}
 (\mathcal{A}\tilde{\mathbf{v}}_1, \mathcal{A}\tilde{\mathbf{v}}_2, \dots, \mathcal{A}\tilde{\mathbf{v}}_m) &= (\mathcal{A}\mathbf{v}_1, \mathcal{A}\mathbf{v}_2, \dots, \mathcal{A}\mathbf{v}_m)T \\
 &= (\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_p)AT \\
 &= (\tilde{\mathbf{w}}_1, \tilde{\mathbf{w}}_2, \dots, \tilde{\mathbf{w}}_p)Q^{-1}AT.
 \end{aligned} \tag{A.12}$$

We have thus proved the following

Proposition A.2.3 [CHANGE OF BASIS AND LINEAR TRANSFORMATIONS] *Let us consider an old basis $(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m)$ and a new basis $(\tilde{\mathbf{v}}_1, \tilde{\mathbf{v}}_2, \dots, \tilde{\mathbf{v}}_m)$ in $\mathbf{V}$, and T be the matrix that expresses the new basis with respect to the old one:*

$$(\tilde{\mathbf{v}}_1, \tilde{\mathbf{v}}_2, \dots, \tilde{\mathbf{v}}_m) = (\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m)T.$$

Similarly, let us consider an old basis $(\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_p)$ and a new one $(\tilde{\mathbf{w}}_1, \tilde{\mathbf{w}}_2, \dots, \tilde{\mathbf{w}}_p)$ in $\mathbf{W}$ and be Q the change of basis matrix:

$$(\tilde{\mathbf{w}}_1, \tilde{\mathbf{w}}_2, \dots, \tilde{\mathbf{w}}_p) = (\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_p)Q.$$

If $\mathcal{A} : \mathbf{V} \rightarrow \mathbf{W}$ is a linear transformation, the matrices A and $\tilde{A}$ that represent $\mathcal{A}$ in the old and new bases, respectively, are related each other through the following relation

$$\tilde{A} = Q^{-1}AT. \quad \blacksquare$$

If $\mathcal{A} : \mathbf{V} \rightarrow \mathbf{V}$ is a linear transformation of the vector space $\mathbf{V}$ onto itself, the matrix A is square and a change of basis in $\mathbf{V}$, characterized by an invertible matrix T , induces a *similarity* transformation on the matrix A , which becomes $\tilde{A} = T^{-1}AT$.

A.2.2 Transformation invariant subspaces

Definition A.2.4 [INVARIANT SUBSPACE] *Let $\mathcal{A} : \mathbf{V} \rightarrow \mathbf{V}$ be a linear transformation of $\mathbf{V}$ onto itself. A subspace $\mathbf{U}$ of $\mathbf{V}$ is $\mathcal{A}$ -invariant if for all $\mathbf{u} \in \mathbf{U}$ one has $\mathcal{A}\mathbf{u} \in \mathbf{U}$, that is if $\mathcal{A}\mathbf{U} \subseteq \mathbf{U}$.*

Example A.2.2 (i) The subspace $\{\mathbf{0}\}$ and the whole domain $\mathbf{V}$ of the transformation $\mathcal{A} : \mathbf{V} \rightarrow \mathbf{V}$ are trivially $\mathcal{A}$ -invariant subspaces, for any choice of $\mathcal{A}$.

(ii) For any linear transformation $\mathcal{A} : \mathbf{V} \rightarrow \mathbf{V}$, $\ker \mathcal{A}$ is $\mathcal{A}$ -invariant.

(iii) Consider the transformation $\mathcal{A} : \mathbb{R}^n \rightarrow \mathbb{R}^n$ defined in Example A.2.1 (ii): the subspace

$$\mathbf{U} = \left\{ \begin{bmatrix} \alpha_1 \\ \vdots \\ \alpha_{n-1} \\ 0 \end{bmatrix}, \alpha_i \in \mathbb{R} \right\}$$

is $\mathcal{A}$ -invariant. It is left as an exercise the description of all the $\mathcal{A}$ -invariant subspaces of $\mathbb{R}^n$.

(iv) $\mathcal{A}$ -invariant subspaces for the transformation $\mathcal{A} : K[z] \rightarrow K[z] : p(z) \mapsto p(z^2)$, considered in Example A.2.1 (iii), are the subspaces $K[z^{2^\nu}]$, $\nu = 1, 2, \dots$

If $\mathbf{V}$ has finite dimension, some basis choices induce matrices representative of the linear transformation $\mathcal{A} : \mathbf{V} \mapsto \mathbf{V}$ with a particularly simple structure.

Among the most common cases, we mention the situation in which the n -dimensional vector space $\mathbf{V}$ contains a non-trivial $\mathcal{A}$ -invariant subspace $\mathbf{U}$, of dimension m , con $0 < m < n$. In this case, a convenient choice is that of considering a basis $(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m)$ for $\mathbf{U}$ and to complete it to an ordered basis of $\mathbf{V}$ by adding $n - m$ vectors

$$\overbrace{(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m, \mathbf{v}_{m+1}, \dots, \mathbf{v}_n)}^{\mathbf{V} \text{ basis}}. \quad (\text{A.13})$$

$\mathbf{U} \text{ basis}$

Since $\mathcal{A}\mathbf{v}_1, \mathcal{A}\mathbf{v}_2, \dots, \mathcal{A}\mathbf{v}_m$ belong to $\mathbf{U}$, they have null components corresponding to the vectors $\mathbf{v}_{m+1}, \dots, \mathbf{v}_n$ and the matrix that is representative of the transformation with respect to the basis (A.13) is block-triangular

$$\begin{aligned} (\mathcal{A}\mathbf{v}_1, \dots, \mathcal{A}\mathbf{v}_m, \mathcal{A}\mathbf{v}_{m+1}, \dots, \mathcal{A}\mathbf{v}_n) &= (\mathbf{v}_1, \dots, \mathbf{v}_m, \mathbf{v}_{m+1}, \dots, \mathbf{v}_n) \begin{bmatrix} a_{11} & \dots & a_{1m} & | & a_{1,m+1} & \dots & a_{1n} \\ \dots & \dots & \dots & | & \dots & \dots & \dots \\ a_{m1} & \dots & a_{mm} & | & a_{m,m+1} & \dots & a_{mn} \\ \hline 0 & \dots & 0 & | & a_{m+1,m+1} & \dots & a_{m+1,n} \\ \dots & \dots & \dots & | & \dots & \dots & \dots \\ 0 & \dots & 0 & | & a_{n,m+1} & \dots & a_{nn} \end{bmatrix} \\ &= (\mathbf{v}_1, \dots, \mathbf{v}_m, \mathbf{v}_{m+1}, \dots, \mathbf{v}_n) \begin{bmatrix} A_{11} & A_{12} \\ 0 & A_{22} \end{bmatrix}. \end{aligned} \quad (\text{A.14})$$

The matrix A_{11} , whose size is $m \times m$, represents the *restriction* $\mathcal{A}|_{\mathbf{U}} : \mathbf{U} \rightarrow \mathbf{U}$ of the map $\mathcal{A}$ to the subspace $\mathbf{U}$, with respect to the basis $(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m)$.

Viceversa, if the matrix A , whose size is $n \times n$, represents a linear transformation $\mathcal{A}$ with respect to a basis $(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n)$ and it has upper-triangular block structure, with block A_{11} of dimension $m \times m$, then the first m basis vectors generate an $\mathcal{A}$ -invariant subspace.

Example A.2.3 (i) The vectors $\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_{n-1}$ of the canonical basis in $\mathbb{R}^n$ constitute a basis for a subspace that is $\mathbf{U}$ -invariant with respect to the transformation

$$\mathcal{A} : \begin{bmatrix} \alpha_1 \\ \vdots \\ \alpha_{n-1} \\ \alpha_n \end{bmatrix} \mapsto \begin{bmatrix} \alpha_2 \\ \vdots \\ \alpha_n \\ 0 \end{bmatrix}.$$

By adding $\mathbf{e}_n$, the basis $\mathbf{U}$ is completed to a basis in $\mathbb{R}^n$:

$$\overbrace{(\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_{n-1}, \mathbf{e}_n)}^{\text{basis for } \mathbb{R}^n}.$$

basis for $\mathbf{U}$

The corresponding matrix, representing $\mathcal{A}$, is block-triangular

$$A = \left[\begin{array}{ccccc|c} 0 & 1 & 0 & \dots & 0 & 0 \\ 0 & 0 & 1 & \dots & 0 & 0 \\ \vdots & \vdots & \ddots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \ddots & 1 & 0 \\ 0 & 0 & 0 & \dots & 0 & 1 \\ \hline 0 & 0 & 0 & \dots & 0 & 0 \end{array} \right].$$

Note that the structure of A is block-triangular even when the A_{11} block has size different from $(n-1) \times (n-1)$. This corresponds to the fact that for any $k \leq n$ the vectors $\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_k$ are the basis of an $\mathcal{A}$ -invariant subspace.

If also the subspace $\mathbf{C}$ generated by vectors $\mathbf{v}_{m+1}, \mathbf{v}_{m+2}, \dots, \mathbf{v}_n$ of the basis (A.13) is $\mathcal{A}$ -invariant, the vector space $\mathbf{V} = \mathbf{U} \oplus \mathbf{C}$ is the direct sum of two $\mathcal{A}$ -invariant subspaces and the matrix A in (A.14) takes the block diagonal form

$$A = \left[\begin{array}{c|c} \overbrace{A_{11}}^m & \overbrace{0}^{n-m} \\ \hline 0 & A_{22} \end{array} \right] \begin{matrix} m \\ n-m \end{matrix}$$

A_{22} represents the restriction $\mathcal{A}|_{\mathbf{C}} : \mathbf{C} \rightarrow \mathbf{C}$ of $\mathcal{A}$, with respect to the basis $\mathbf{v}_{m+1}, \mathbf{v}_{m+2}, \dots, \mathbf{v}_n$. In general, we can state that the determination of a basis in $\mathbf{V}$ with respect to which the transformation $\mathcal{A} : \mathbf{V} \rightarrow \mathbf{V}$ is represented by a diagonal matrix with t diagonal blocks

$$A = \text{diag}\{A_{11}, A_{22}, \dots, A_{tt}\} := \begin{bmatrix} A_{11} & & & \\ & A_{22} & & \\ & & \ddots & \\ & & & A_{tt} \end{bmatrix}$$

corresponds to the identification of t $\mathcal{A}$ -invariant subspaces, independent and finite-dimensional, whose direct sum is $\mathbf{V}$.

A.2.3 Properties of the kernel and the image of a transformation

Proposition A.2.5 [DIMENSIONS OF KERNEL AND IMAGE] *Let $\mathcal{A} : \mathbf{V} \rightarrow \mathbf{W}$ be a linear transformation and let $\mathcal{B}^{(1)} = \{\mathbf{v}_i, i \in I\}$ be a basis for $\ker \mathcal{A}$ and $\mathcal{B}^{(2)} = \{\tilde{\mathbf{v}}_j, j \in J\}$ a completion of $\mathcal{B}^{(1)}$ to a basis in $\mathbf{V}$. Then*

- i) $\{\mathcal{A}\tilde{\mathbf{v}}_j, j \in J\}$ is a basis for $\text{Im}\mathcal{A}$;
- ii) $\dim(\mathbf{V}) = \dim(\ker \mathcal{A}) + \dim(\text{Im}\mathcal{A})$.

PROOF (i) Since $\mathcal{A}\mathbf{v}_i = \mathbf{0}, \forall i \in I$, any vector $\mathbf{w}$ in the image of $\mathcal{A}$ can be represented in the form

$$\mathbf{w} = \sum_{i \in I} \alpha_i \mathcal{A}\mathbf{v}_i + \sum_{j \in J} \tilde{\alpha}_j \mathcal{A}\tilde{\mathbf{v}}_j = \sum_{j \in J} \tilde{\alpha}_j \mathcal{A}\tilde{\mathbf{v}}_j$$

where F and $\tilde{F}$ are appropriate finite subsets of I and of J , respectively.

Therefore, $\{\mathcal{A}\tilde{\mathbf{v}}_j, j \in J\}$ is a set of generators for $\text{Im}\mathcal{A}$. This set is linearly independent, since $\sum_{j \in \tilde{F}} \tilde{\alpha}_j \mathcal{A}\tilde{\mathbf{v}}_j = \mathbf{0}$ implies $\mathcal{A} \sum_{j \in \tilde{F}} \tilde{\alpha}_j \tilde{\mathbf{v}}_j = \mathbf{0}$, namely $\sum_{j \in \tilde{F}} \tilde{\alpha}_j \tilde{\mathbf{v}}_j \in \ker \mathcal{A}$, which leads to $\alpha_j = 0, \forall j \in \tilde{F}$.

(ii) is a straightforward consequence of (i). ■

If $\mathbf{V}$ is a vector space over the field K , we indicate with $\mathcal{L}(\mathbf{V}, \mathbf{V})$ the set of linear transformations of $\mathbf{V}$ onto itself. It is a vector space with respect to sum and scalar product, defined for any $\mathcal{A}$ and $\mathcal{B}$ in $\mathcal{L}(\mathbf{V}, \mathbf{V})$, for any α in K and for any $\mathbf{v}$ in $\mathbf{V}$ by

$$\begin{aligned} (\mathcal{A} + \mathcal{B})\mathbf{v} &:= \mathcal{A}\mathbf{v} + \mathcal{B}\mathbf{v} \\ (\alpha\mathcal{A})\mathbf{v} &:= \alpha(\mathcal{A}\mathbf{v}). \end{aligned} \quad (\text{A.15})$$

In $\mathcal{L}(\mathbf{V}, \mathbf{V})$ it is also possible to define a (non commutative) *product* by assuming

$$(\mathcal{A}\mathcal{B})\mathbf{v} := \mathcal{A}(\mathcal{B}\mathbf{v}) \quad (\text{A.16})$$

(hence, the product between two transformations is simply their composition). In particular, the symbol $\mathcal{A}^\nu, \nu \in \mathbb{N}$, indicates the transformation $\underbrace{\mathcal{A}\mathcal{A} \cdots \mathcal{A}}_{\nu \text{ times}}$, if $\nu > 0$, and the identity transformation $\mathcal{I}$, if $\nu = 0$. With respect to sum (A.15) and product (A.16) $\mathcal{L}(\mathbf{V}, \mathbf{V})$ is a non-commutative ring, with unit $\mathcal{I}$.

If $\mathbf{V}$ has finite dimension, once a basis $(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n)$ in $\mathbf{V}$ has been fixed, the space $\mathcal{L}(\mathbf{V}, \mathbf{V})$ is isomorphic to the space $K^{n \times n}$ of the $n \times n$ matrices over the field K . Such isomorphism is expressed by

$$(\mathcal{A}\mathbf{v}_1, \mathcal{A}\mathbf{v}_2, \dots, \mathcal{A}\mathbf{v}_n) = (\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n)A,$$

which associates the matrix A to the transformation $\mathcal{A}$: the columns of A are the components of the vectors $\mathcal{A}\mathbf{v}_1, \mathcal{A}\mathbf{v}_2, \dots, \mathcal{A}\mathbf{v}_n$ with respect to the basis $(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n)$. The space $K^{n \times n}$ (and so the space $\mathcal{L}(\mathbf{V}, \mathbf{V})$) has dimension n^2 on K . The correspondence $\mathcal{A} \rightarrow A$ described above also realizes a ring isomorphism between $\mathcal{L}(\mathbf{V}, \mathbf{V})$ and $K^{n \times n}$.

Proposition A.2.6 [CHAINS OF KERNELS AND IMAGES] *Let $\mathcal{A} : \mathbf{V} \rightarrow \mathbf{V}$ be a linear transformation.*

- i) *for any $h \in \mathbb{N}$, one has $\ker \mathcal{A}^h \subseteq \ker \mathcal{A}^{h+1}$;*
- ii) *if in the chain of subspaces*

$$\{\mathbf{0}\} = \ker \mathcal{A}^0 \subseteq \ker \mathcal{A} \subseteq \ker \mathcal{A}^2 \subseteq \ker \mathcal{A}^3 \subseteq \dots \quad (\text{A.17})$$

it occurs that $\ker \mathcal{A}^h = \ker \mathcal{A}^{h+1}$, then $\ker \mathcal{A}^{h+1} = \ker \mathcal{A}^{h+2}$ and the chain becomes stationary;

- iii) *for all $h \in \mathbb{N}$, one has $\text{Im}\mathcal{A}^h \supseteq \text{Im}\mathcal{A}^{h+1}$;*
- iv) *if in the sequence of subspaces*

$$\mathbf{V} = \text{Im}\mathcal{A}^0 \supseteq \text{Im}\mathcal{A} \supseteq \text{Im}\mathcal{A}^2 \supseteq \text{Im}\mathcal{A}^3 \supseteq \dots \quad (\text{A.18})$$

it occurs that $\text{Im}\mathcal{A}^h = \text{Im}\mathcal{A}^{h+1}$, then $\text{Im}\mathcal{A}^{h+1} = \text{Im}\mathcal{A}^{h+2}$ and the sequence becomes stationary;

v) if $\mathbf{V}$ has finite dimension n , the chains (A.17) and (A.18) are stationary at latest starting from the n -th power, namely

$$\ker \mathcal{A}^n = \ker \mathcal{A}^{n+1} = \dots \quad \text{and} \quad \text{Im} \mathcal{A}^n = \text{Im} \mathcal{A}^{n+1} = \dots \quad (\text{A.19})$$

PROOF (i) If $\mathbf{v} \in \ker \mathcal{A}^h$, then $\mathcal{A}^h \mathbf{v} = \mathbf{0}$, hence $\mathbf{0} = \mathcal{A}(\mathcal{A}^h \mathbf{v}) = \mathcal{A}^{h+1} \mathbf{v}$ and $\mathbf{v} \in \ker \mathcal{A}^{h+1}$.

(ii) If $\ker \mathcal{A}^h = \ker \mathcal{A}^{h+1}$, it is sufficient to prove that $\mathbf{v} \in \ker \mathcal{A}^{h+2}$ implies $\mathbf{v} \in \ker \mathcal{A}^{h+1}$. Indeed,

$$\mathcal{A}^{h+2} \mathbf{v} = \mathbf{0} \Rightarrow \mathcal{A}^{h+1}(\mathcal{A} \mathbf{v}) = \mathbf{0} \Rightarrow \mathcal{A}^h(\mathcal{A} \mathbf{v}) = \mathbf{0} \Rightarrow \mathcal{A}^{h+1} \mathbf{v} = \mathbf{0}.$$

(iii) If $\mathbf{v} \in \text{Im} \mathcal{A}^{h+1}$, then $\mathbf{v} = \mathcal{A}^{h+1} \mathbf{u}$, hence $\mathbf{v} = \mathcal{A}^h(\mathcal{A} \mathbf{u}) \in \text{Im} \mathcal{A}^h$.

(iv) If $\text{Im} \mathcal{A}^h = \text{Im} \mathcal{A}^{h+1}$, it is sufficient to prove that $\mathbf{v} \in \text{Im} \mathcal{A}^{h+1}$ implies $\mathbf{v} \in \text{Im} \mathcal{A}^{h+2}$. Indeed,

$$\mathbf{v} = \mathcal{A}^{h+1} \mathbf{u} \Rightarrow \mathbf{v} = \mathcal{A}(\mathcal{A}^h \mathbf{u}) \Rightarrow \mathbf{v} = \mathcal{A}(\mathcal{A}^{h+1} \mathbf{w}), \exists \mathbf{w} \Rightarrow \mathbf{v} = \mathcal{A}^{h+2} \mathbf{w}.$$

(v) If in the chain (A.17) there are no equalities before $\ker \mathcal{A}^n$, all the inclusions up to $\ker \mathcal{A}^n$ are in strict sense and each kernel has a dimension that is larger than that of the previous one. Therefore, $\ker \mathcal{A}^n = \mathbf{V}$ and the kernels that follow are consistently equal to $\mathbf{V}$. Similarly, if (A.18) is strictly decreasing to $\text{Im} \mathcal{A}^n$, that subspace is null and all the image subspaces that follow are null as well. ■

- EXERCISE A.2.1 Let $\mathbf{V}$ be of finite dimension n and let $\mathcal{A} : \mathbf{V} \rightarrow \mathbf{V}$ be a linear transformation.
 - (i) $\mathcal{A}$ is injective $\Leftrightarrow \mathcal{A}$ is surjective $\Leftrightarrow$ the chain (A.17) is stationary from the first step $\Leftrightarrow$ the sequence (A.18) is stationary from the first step;
 - (ii) $\dim(\ker \mathcal{A}^{h+1}) - \dim(\ker \mathcal{A}^h) = \dim(\text{Im} \mathcal{A}^h) - \dim(\text{Im} \mathcal{A}^{h+1})$
- ‡ Hint: remember that, for any $\nu \geq 0$, $\dim(\ker \mathcal{A}^\nu) + \dim(\text{Im} \mathcal{A}^\nu) = n$.

A.2.4 Schur complement and block matrix inversion

In this section we summarize some results that are of frequent use when we deal with square matrices with block partitions of the following kind

$$A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \quad (\text{A.20})$$

and then, in a more general sense, when the linear transformation $\mathcal{A}$ represented by A acts on a vector space which is the direct sum of two subspaces.

Definition A.2.7 [SCHUR COMPLEMENT] If in (A.20) the A_{11} block is invertible, the matrix $A_{22} - A_{21}A_{11}^{-1}A_{12}$ is called Schur complement of A_{11} and is indicated as A/A_{11} .

Proposition A.2.8 [DETERMINANT AND INVERSE OF A MATRIX]

(i) If in (A.20) the block A_{11} is invertible, then

- $\det(A)$ can be expressed as

$$\det(A) = \det A_{11} \det(A/A_{11}) \quad (\text{A.21})$$

- A is invertible if and only if the Schur complement A/A_{11} is invertible.

(ii) If in (A.20) both blocks A_{11} and A_{22} are invertible, then any of the following facts

- A is invertible,

- $A/A_{11} = A_{22} - A_{21}A_{11}^{-1}A_{12}$ is invertible
- $A/A_{22} = A_{11} - A_{12}A_{22}^{-1}A_{21}$ is invertible

implies the other two and one has

$$A^{-1} = \begin{bmatrix} (A/A_{22})^{-1} & -A_{11}^{-1}A_{12}(A/A_{11})^{-1} \\ -A_{22}^{-1}A_{21}(A/A_{22})^{-1} & (A/A_{11})^{-1} \end{bmatrix} \quad (\text{A.22})$$

$$(A/A_{11})^{-1} = A_{22}^{-1} + A_{22}^{-1}A_{12}(A/A_{22})^{-1} \quad (\text{A.23})$$

$$(A/A_{22})^{-1} = A_{11}^{-1} + A_{11}^{-1}A_{12}(A/A_{11})^{-1} \quad (\text{A.24})$$

PROOF (i) If A_{11} is invertible, the first fact follows from the identity (that can be immediately verified)

$$\begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \begin{bmatrix} I & -A_{11}^{-1}A_{12} \\ 0 & I \end{bmatrix} = \begin{bmatrix} A_{11} & 0 \\ A_{21} & A/A_{11} \end{bmatrix}$$

by equalizing the determinants of the two sides and by recalling that the determinant of the product of two square matrices coincides with the product of their determinants. From (A.21) it follows that $\det(A) \neq 0$ if and only if $\det(A/A_{11}) \neq 0$.

(ii) If A_{11} and A_{22} are both invertible, the equivalence of the invertibility of A , of A/A_{11} and of A/A_{22} follows from fact (i). If one condition (and then all three) is satisfied, to verify (A.22) it is sufficient to note that the equality

$$\begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \begin{bmatrix} (A_{11} - A_{12}A_{22}^{-1}A_{21})^{-1} & -A_{11}^{-1}A_{12}(A_{22} - A_{21}A_{11}^{-1}A_{12})^{-1} \\ -A_{22}^{-1}A_{21}(A_{11} - A_{12}A_{22}^{-1}A_{21})^{-1} & (A_{22} - A_{21}A_{11}^{-1}A_{12})^{-1} \end{bmatrix} = \begin{bmatrix} I & 0 \\ 0 & I \end{bmatrix} \quad (\text{A.25})$$

is trivially satisfied. To verify (A.24), we preliminarily notice that, if X is an $n \times m$ matrix and Y is an $m \times n$ matrix, from the equality $X(I_m + YX) = (I_n + XY)X$ it follows, when the inverse matrices exist, $(I_n + XY)^{-1}X = X(I_m + YX)^{-1}$. After these premises, we have

$$\begin{aligned} & (A_{11} - A_{12}A_{22}^{-1}A_{21})^{-1} - A_{11}^{-1} - A_{11}^{-1}A_{12}(A_{22} - A_{21}A_{11}^{-1}A_{12})^{-1}A_{21}A_{11}^{-1} \\ &= (A_{11} - A_{12}A_{22}^{-1}A_{21})^{-1} - A_{11}^{-1} - A_{11}^{-1}A_{12}(I - A_{22}^{-1}A_{21}A_{11}^{-1}A_{12})^{-1}A_{22}^{-1}A_{21}A_{11}^{-1} \\ &= (A_{11} - A_{12}A_{22}^{-1}A_{21})^{-1} - A_{11}^{-1} - (I - A_{11}^{-1}A_{12}A_{22}^{-1}A_{21})^{-1}A_{11}^{-1}A_{12}A_{22}^{-1}A_{21}A_{11}^{-1} \\ &= (A_{11} - A_{12}A_{22}^{-1}A_{21})^{-1} - A_{11}^{-1} - (A_{11} - A_{12}A_{22}^{-1}A_{21})^{-1}A_{12}A_{22}^{-1}A_{21}A_{11}^{-1} \\ &= (A_{11} - A_{12}A_{22}^{-1}A_{21})^{-1}(I - A_{12}A_{22}^{-1}A_{21}A_{11}^{-1}) - A_{11}^{-1} = 0 \end{aligned} \quad (\text{A.26})$$

(A.23) can be verified in a similar way.

- EXERCISE A.2.2 If $X \in K^{n \times m}$ and $Y \in K^{m \times n}$, verify that $\det(I_n + XY) = \det(I_m + YX)$. Then, the invertibility of one between the two matrices $(I_n + XY)$ and $(I_m + YX)$ implies the invertibility of the other.

A.3 Quotient space

Let $\mathbf{V}$ be a vector space over K and $\mathbf{U}$ one of its subspaces. Let us introduce in $\mathbf{V}$ the relation

$$\mathbf{v} \sim \tilde{\mathbf{v}} \Leftrightarrow \mathbf{v} - \tilde{\mathbf{v}} \in \mathbf{U}. \quad (\text{A.27})$$

This latter is an equivalence relation (it is reflexive, symmetric, and transitive), and hence $\mathbf{V}$ is partitioned into disjoint subsets, each of which contains all and only equivalent vectors. These subsets are called *equivalence classes of $\mathbf{V}$ modulo $\mathbf{U}$* .

If $\mathbf{v}$ is a vector of $\mathbf{V}$, we denote by $[\mathbf{v}]$ the class of vectors equivalent to $\mathbf{v}$ that is, according to the reflexive property, the class where $\mathbf{v}$ belongs.

If $\tilde{\mathbf{v}} \in [\mathbf{v}]$, since all vectors of $[\mathbf{v}]$ are equivalent to $\mathbf{v}$, it is $\mathbf{v} - \tilde{\mathbf{v}} \in \mathbf{U}$. Viceversa, if $\mathbf{v} - \tilde{\mathbf{v}} \in \mathbf{U}$, then it is $\tilde{\mathbf{v}} \sim \mathbf{v}$ and then $\tilde{\mathbf{v}} \in [\mathbf{v}]$. Therefore, the class $[\mathbf{v}]$ is also indicated in the form $\mathbf{v} + \mathbf{U}$, to highlight that its elements can be obtained by summing $\mathbf{v}$ with each of the vectors of the subspace $\mathbf{U}$. In particular, $\mathbf{U} = \mathbf{0} + \mathbf{U} = [\mathbf{0}]$ is an equivalence class. If $\tilde{\mathbf{v}} \sim \mathbf{v}$, then $[\tilde{\mathbf{v}}] = \tilde{\mathbf{v}} + \mathbf{U} = \mathbf{v} + \mathbf{U} = [\mathbf{v}]$; thus equivalent elements are representative of the same class.

The set of equivalence classes of $\mathbf{V}$ modulo $\mathbf{U}$ is indicated with $\mathbf{V}(\text{mod } \mathbf{U})$ or with $\mathbf{V}/\mathbf{U}$.

Example A.3.1 (i) If $\mathbf{V} = \mathbb{R}^2$ and $\mathbf{U} = \left\{ \begin{bmatrix} x \\ y \end{bmatrix} : x - y = 0 \right\}$, the set $\mathbf{V}/\mathbf{U}$ of the equivalence classes has elements that are the subsets of $\mathbb{R}^2$ given by the lines parallel to the bisector of the first and third quadrants.

(ii) Let us denote by (z^2) the subspace of $K[z]$ consisting of the polynomials multiple of z^2 , and let us suppose that $\mathbf{V} = K[z]$ and $\mathbf{U} = (z^2)$. To specify a class of $\mathbf{V}/\mathbf{U}$ it is sufficient to consider any element $p(z)$ of the class and add to it all polynomials of (z^2) :

$$[p(z)] = \{p(z) + z^2 q(z), q(z) \in K[z]\}.$$

By resorting to the polynomial division algorithm, it can be verified that any equivalence class contains only one polynomial of degree lower than 2. Since a class can be indicated by any element belonging to it, we can conclude that the classes of $\mathbf{V}/\mathbf{U}$ can all be only represented in the form $[a + bz]$, with a and b varying in K .

We want now to provide the set of equivalence classes $\mathbf{V}/\mathbf{U}$ with a structure of vector space over K . To this aim, we can define the sum and the product by a scalar as follows

$$(\mathbf{v} + \mathbf{U}) + (\mathbf{w} + \mathbf{U}) := (\mathbf{v} + \mathbf{w}) + \mathbf{U} \quad \forall \mathbf{v}, \mathbf{w} \in \mathbf{V} \quad (\text{A.28})$$

$$\alpha(\mathbf{v} + \mathbf{U}) := (\alpha\mathbf{v}) + \mathbf{U} \quad \forall \alpha \in K, \forall \mathbf{v} \in \mathbf{V}. \quad (\text{A.29})$$

Remark Since the same class $\mathbf{v} + \mathbf{U}$ can be indicated through different representatives $\mathbf{v}$, one may wonder whether in (A.28) and (A.29) the sum class $(\mathbf{v} + \mathbf{w}) + \mathbf{U}$ and the product class $(\alpha\mathbf{v}) + \mathbf{U}$ depend on the chosen representatives $\mathbf{v}$ and $\mathbf{w}$. If this were the case, the two operations would not be *well posed* because, although acting on the same elements (the classes) they would provide different results depending on the different representing vectors.

Actually, the operations are well defined. If $\mathbf{v} \sim \tilde{\mathbf{v}}$ and $\mathbf{w} \sim \tilde{\mathbf{w}}$, the class $\mathbf{v} + \mathbf{U}$ can be represented alternatively as $\tilde{\mathbf{v}} + \mathbf{U}$, and the class $\mathbf{w} + \mathbf{U}$ as $\tilde{\mathbf{w}} + \mathbf{U}$, and, by performing the sum of the two classes with the alternative representations, one gets

$$(\tilde{\mathbf{v}} + \mathbf{U}) + (\tilde{\mathbf{w}} + \mathbf{U}) := (\tilde{\mathbf{v}} + \tilde{\mathbf{w}}) + \mathbf{U} \quad (\text{A.30})$$

From the equivalence conditions $\mathbf{v} - \tilde{\mathbf{v}} \in \mathbf{U}$ and $\mathbf{w} - \tilde{\mathbf{w}} \in \mathbf{U}$ it follows $(\mathbf{v} + \mathbf{w}) - (\tilde{\mathbf{v}} + \tilde{\mathbf{w}}) \in \mathbf{U}$, then $(\mathbf{v} + \mathbf{w}) + \mathbf{U} = (\tilde{\mathbf{v}} + \tilde{\mathbf{w}}) + \mathbf{U}$ and the sum class obtained from (A.28) and from (A.30) is the same. It can be stated that the result of the sum of two classes does not depend on the considered representatives of the classes.

A similar validation can also be done for the scalar product.

With respect to the operations introduced in (A.28) and (A.29) and assuming as null element the class $[\mathbf{0}] = \mathbf{0} + \mathbf{U}$, the set $\mathbf{V}/\mathbf{U}$ satisfies the axioms of Definition A.1.1. Therefore, we can provide the following

Definition A.3.1 [QUOTIENT SPACE] *Let $\mathbf{V}$ be a vector space over K and let $\mathbf{U}$ be one of its subspaces.*

The set $\mathbf{V}/\mathbf{U}$ of the equivalence classes of $\mathbf{V}$ modulo $\mathbf{U}$ is a vector space over K , with respect to the sum and scalar product defined in (A.28) and (A.29), named *quotient space of $\mathbf{V}$ modulo $\mathbf{U}$* .

Proposition A.3.2 [BASIS FOR THE QUOTIENT SPACE] *If $\mathbf{V}$ has finite dimension n and if $(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m)$, $m < n$, is a basis for $\mathbf{U}$, let $(\mathbf{v}_{m+1}, \dots, \mathbf{v}_n)$ be a completion for such basis to a basis for $\mathbf{V}$. Then, the classes*

$$[\mathbf{v}_{m+1}], \dots, [\mathbf{v}_n]$$

are a basis for $\mathbf{V}/\mathbf{U}$. If $m = n$, the spaces $\mathbf{V}$ and $\mathbf{U}$ coincide and $\mathbf{V}/\mathbf{U}$ has zero dimension, since it contains only the null class.

PROOF Keeping into account (A.28)-(A.29) and being $\mathbf{v}_i \sim \mathbf{0}$, $i = 1, 2, \dots, m$, the generic vector of the quotient space can be expressed as $\mathbf{v} + \mathbf{U} = (\sum_{i=1}^n \alpha_i \mathbf{v}_i) + \mathbf{U} = \sum_{i=1}^n \alpha_i (\mathbf{v}_i + \mathbf{U}) = \sum_{i=m+1}^n \alpha_i (\mathbf{v}_i + \mathbf{U})$. This proves that $[\mathbf{v}_{m+1}], \dots, [\mathbf{v}_n]$ generate $\mathbf{V}/\mathbf{U}$. To verify that they are linearly independent, let us consider a linear combination of such vectors, which produces the null class

$$[\mathbf{0}] = \sum_{i=m+1}^n \alpha_i [\mathbf{v}_i] = \sum_{i=m+1}^n (\alpha_i \mathbf{v}_i) + \mathbf{U}. \quad (\text{A.31})$$

The vector $\sum_{i=m+1}^n \alpha_i \mathbf{v}_i$ is equivalent to $\mathbf{0}$ modulo $\mathbf{U}$, and hence it belongs to $\mathbf{U}$. On the other hand, the condition

$$\mathbf{U} \cap \text{span}(\mathbf{v}_{m+1}, \dots, \mathbf{v}_n) = \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_m) \cap \text{span}(\mathbf{v}_{m+1}, \dots, \mathbf{v}_n) = \{\mathbf{0}\}$$

implies $\sum_{i=m+1}^n \alpha_i \mathbf{v}_i = \mathbf{0}$ and from the linear independence of $\mathbf{v}_{m+1}, \dots, \mathbf{v}_n$ it follows that $\alpha_{m+1} = \dots = \alpha_n = 0$. This proves that in (A.31) the coefficients α_i are all null and $[\mathbf{v}_{m+1}], \dots, [\mathbf{v}_n]$ are independent. ■

Remark If $\mathbf{V}$ has infinite dimension, the statement of the previous proposition is still valid. Let us consider a basis $\{\mathbf{v}_a, a \in A\}$ for $\mathbf{U}$, indexed on the set A , which can have infinite cardinality, and let us complete this to a basis in $\mathbf{V}$ by adding further elements $\{\mathbf{v}_b, b \in B\}$ (this addition may have finite or infinite cardinality). By keeping into account the fact that vectors of $\mathbf{U}$ and of $\mathbf{V}$ are *finite* linear combinations of the elements of their corresponding bases, along the line of the proof of Proposition A.3.1 it can be shown that the classes $[\mathbf{v}_b], b \in B$, are a basis for the quotient space $\mathbf{V}/\mathbf{U}$.

Corollary A.3.3 [DIMENSION OF THE QUOTIENT SPACE] *If $\mathbf{U}$ is a subspace of $\mathbf{V}$, the dimensions of $\mathbf{V}$, $\mathbf{U}$ and $\mathbf{V}/\mathbf{U}$ satisfy the relation*

$$\dim(\mathbf{V}) = \dim(\mathbf{U}) + \dim(\mathbf{V}/\mathbf{U}). \quad (\text{A.32})$$

In particular, if $\mathbf{U}$ has finite dimension, it is

$$\dim(\mathbf{V}/\mathbf{U}) = \dim(\mathbf{V}) - \dim(\mathbf{U}). \quad (\text{A.33})$$

Example A.3.2 (i) In Example A.3.1 (i), one can choose the vector $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ as basis for $\mathbf{U}$ and as completion to a basis for $\mathbb{R}^2$ the vector $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$. The quotient space is generated by the class $\begin{bmatrix} 1 \\ -1 \end{bmatrix} + \mathbf{U}$ and the generic equivalence class is $\begin{bmatrix} \alpha \\ -\alpha \end{bmatrix} + \mathbf{U}$.

(ii) If $(\mathbf{v}_1, \dots, \mathbf{v}_m)$ is a basis for the subspace $\mathbf{U} \subset \mathbb{R}^n$, it is known that² the orthogonal subspace $\mathbf{U}^\perp := \{\mathbf{v} \in \mathbb{R}^n : \mathbf{v}^T \mathbf{v}_i = 0, i = 1, \dots, m\}$ has a basis made of $n - m$ elements $(\mathbf{v}_{m+1}, \dots, \mathbf{v}_n)$ and

²We refer the interested reader again to Section A.12 for inner product spaces.

one has $\mathbf{U} \oplus \mathbf{U}^\perp = \mathbf{V}$.

According to Proposition A.3.2, $([\mathbf{v}_{m+1}], \dots, [\mathbf{v}_n])$ is a basis for $\mathbf{V}/\mathbf{U}$, and there is a bijective correspondence between the vectors $\sum_{i=m+1}^n \alpha_i \mathbf{v}_i \in \mathbf{U}^\perp$ and the classes of the quotient space $\sum_{i=m+1}^n \alpha_i [\mathbf{v}_i]$.

(iii) If $\mathbf{V} = K[z]$ and $\mathbf{U} = K[z^2]$, the quotient space $K[z]/K[z^2]$ has the classes $[z], [z^3], [z^5], \dots$ as its basis. Note that in this example both $\mathbf{U}$ and $\mathbf{V}$ have infinite dimensions.

Proposition A.3.4 [FACTORIZATION OF LINEAR MAPS] *If $\mathcal{A} : \mathbf{V} \rightarrow \mathbf{W}$ is a linear map, consider the quotient space $\mathbf{V}/\ker \mathcal{A}$ and the projection map*

$$\mathcal{P} : \mathbf{V} \rightarrow \mathbf{V}/\ker \mathcal{A} : \mathbf{v} \mapsto \mathbf{v} + \ker \mathcal{A}.$$

Then, there exists a linear map $\bar{\mathcal{A}} : \mathbf{V}/\ker \mathcal{A} \rightarrow \mathbf{W}$ such that $\mathcal{A} = \bar{\mathcal{A}} \circ \mathcal{P}$, or, as it is often said, such that the following diagram commutes:

$$\begin{array}{ccc} & \mathcal{A} & \\ \mathbf{V} & \xrightarrow{\quad} & \mathbf{W} \\ \mathcal{P} \searrow & & \nearrow \bar{\mathcal{A}} \\ & \mathbf{V}/\ker \mathcal{A} & \end{array}$$

PROOF The transformation $\bar{\mathcal{A}}$ is defined by

$$\bar{\mathcal{A}}(\mathbf{v} + \ker \mathcal{A}) := \mathcal{A}\mathbf{v}. \quad (\text{A.34})$$

The right hand side of (A.34) does not depend on the representing vector $\mathbf{v}$ of the class $\mathbf{v} + \ker \mathcal{A}$.

Indeed, if $\mathbf{v} - \tilde{\mathbf{v}} \in \ker \mathcal{A}$, it follows $\bar{\mathcal{A}}(\tilde{\mathbf{v}} + \ker \mathcal{A}) = \mathcal{A}\tilde{\mathbf{v}} = \mathcal{A}\mathbf{v}$, since $\mathcal{A}\tilde{\mathbf{v}} - \mathcal{A}\mathbf{v} = \mathcal{A}(\tilde{\mathbf{v}} - \mathbf{v}) = \mathbf{0}$. ■

The map $\mathcal{P}$ is surjective, while $\bar{\mathcal{A}}$ is injective. Moreover, if in the diagram above the space $\mathbf{W}$ is replaced by the subspace $\text{Im} \mathcal{A}$, the map $\bar{\mathcal{A}}$ is also surjective and thus it is an isomorphism between the spaces $\mathbf{V}/\ker \mathcal{A}$ and $\text{Im} \mathcal{A}$. Since isomorphic spaces have the same dimensions, it always holds that $\dim(\mathbf{V}/\ker \mathcal{A}) = \dim(\text{Im} \mathcal{A})$ and, by employing (A.32), the following exercise can be solved.

- EXERCISE A.3.1 [DIMENSIONS OF KERNEL AND IMAGE] Let $\mathcal{A} : \mathbf{V} \rightarrow \mathbf{W}$ be a linear map. Use the equality $\dim(\mathbf{V}/\ker \mathcal{A}) = \dim(\text{Im} \mathcal{A})$ to provide another proof of the relation $\dim(\mathbf{V}) = \dim(\text{Im} \mathcal{A}) + \dim(\ker \mathcal{A})$, already proved in Proposition A.2.5.

Proposition A.3.5 [INDUCED TRANSFORMATION] *Let $\mathcal{A} : \mathbf{V} \rightarrow \mathbf{V}$ be a linear transformation and let $\mathbf{U}$ be an $\mathcal{A}$ -invariant subspace of $\mathbf{V}$. If*

$$\mathcal{P} : \mathbf{V} \rightarrow \mathbf{V}/\mathbf{U} : \mathbf{v} \mapsto \mathbf{v} + \mathbf{U}$$

is the canonical projection of $\mathbf{V}$ onto the quotient $\mathbf{V}/\mathbf{U}$, there exists a unique linear transformation $\bar{\mathcal{A}} : \mathbf{V}/\mathbf{U} \rightarrow \mathbf{V}/\mathbf{U}$ such that $\bar{\mathcal{A}} \circ \mathcal{P} = \mathcal{P} \circ \mathcal{A}$, namely such that the following diagram commutes

$$\begin{array}{ccccc} \mathcal{A} & \mathbf{V} & \longrightarrow & \mathbf{V} & \\ & \downarrow & & \downarrow & \\ \mathcal{P} & \mathbf{V}/\mathbf{U} & \longrightarrow & \mathbf{V}/\mathbf{U} & \bar{\mathcal{A}} \end{array} \quad (\text{A.35})$$

$\bar{\mathcal{A}}$ is called the induced linear transformation from $\mathcal{A}$ onto the quotient $\mathbf{V}/\mathbf{U}$.

PROOF Let us suppose, for simplicity's sake, that $\mathbf{V}$ has finite dimension n . Let

$$(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_r, \mathbf{v}_{r+1}, \dots, \mathbf{v}_n) \quad (\text{A.36})$$

be a basis for $\mathbf{V}$ whose first r vectors are a basis for $\mathbf{U}$. Then $[\mathbf{v}_{r+1}] = \mathbf{v}_{r+1} + \mathbf{U}, \dots, [\mathbf{v}_n] = \mathbf{v}_n + \mathbf{U}$ is a basis for $\mathbf{V}/\mathbf{U}$. To allow for the diagram to commute, $\bar{\mathcal{A}}$ should be defined so that $\bar{\mathcal{A}} \circ \mathcal{P} = \mathcal{P} \circ \mathcal{A}$ is verified on the vectors of the basis of $\mathbf{V}$. Therefore, it will hold that $\bar{\mathcal{A}}(\mathcal{P}\mathbf{v}_i) = \mathcal{P}(\mathcal{A}\mathbf{v}_i)$, namely

$$\bar{\mathcal{A}}[\mathbf{v}_i] = [\mathcal{A}\mathbf{v}_i], \quad i = 1, 2, \dots, n. \quad (\text{A.37})$$

For $i = 1, 2, \dots, r$, (A.37) establishes only that $\bar{\mathcal{A}}$ is zero corresponding to the null vector: indeed $\mathbf{v}_i$ and $\mathcal{A}\mathbf{v}_i$ are both in $\mathbf{U}$ and thus $[\mathbf{v}_i] = [\mathcal{A}\mathbf{v}_i] = [\mathbf{0}]$.

For $i = r+1, \dots, n$, the classes $[\mathbf{v}_i]$ are a basis for the quotient space, so that (A.37) assigns the values of the transformation $\bar{\mathcal{A}}$ on a basis of $\mathbf{V}/\mathbf{U}$. The assignment does not depend on the representative $\mathbf{v}_i$ of the class $[\mathbf{v}_i]$: indeed, any vector $\hat{\mathbf{v}}_i \in [\mathbf{v}_i]$ satisfies $\hat{\mathbf{v}}_i - \mathbf{v}_i \in \mathbf{U}$, then $\mathcal{A}\hat{\mathbf{v}}_i - \mathcal{A}\mathbf{v}_i \in \mathbf{U}$, and hence $[\mathcal{A}\hat{\mathbf{v}}_i] = [\mathcal{A}\mathbf{v}_i]$. The map $\bar{\mathcal{A}}$, defined on the basis of $\mathbf{V}/\mathbf{U}$, can be linearly extended to the whole space $\mathbf{V}/\mathbf{U}$: it is sufficient to choose $\bar{\mathcal{A}}(\sum_{i=r+1}^n \alpha_i [\mathbf{v}_i]) := \sum_{i=r+1}^n \alpha_i \bar{\mathcal{A}}[\mathbf{v}_i]$. In this way, the diagram (A.35) commutes because, for any $\mathbf{v} = \sum_{i=1}^n \alpha_i \mathbf{v}_i \in \mathbf{V}$ it holds:

$$\bar{\mathcal{A}}\mathcal{P}\mathbf{v} = \bar{\mathcal{A}}\mathcal{P}\left(\sum_{i=1}^r \alpha_i \mathbf{v}_i\right) + \bar{\mathcal{A}}\mathcal{P}\left(\sum_{i=r+1}^n \alpha_i \mathbf{v}_i\right) = \bar{\mathcal{A}}\left(\sum_{i=r+1}^n \alpha_i [\mathbf{v}_i]\right) = \sum_{i=r+1}^n \alpha_i (\bar{\mathcal{A}}[\mathbf{v}_i])$$

which, considering (A.37), coincides with

$$\mathcal{P}\mathcal{A}\mathbf{v} = \mathcal{P}\mathcal{A}\left(\sum_{i=1}^n \alpha_i \mathbf{v}_i\right) = \mathcal{P}\mathcal{A}\left(\sum_{i=r+1}^n \alpha_i \mathbf{v}_i\right) = \sum_{i=r+1}^n \alpha_i \mathcal{P}(\mathcal{A}\mathbf{v}_i) = \sum_{i=r+1}^n \alpha_i [\mathcal{A}\mathbf{v}_i]. \quad \blacksquare$$

If

$$A = \left[\begin{array}{c|c} \overbrace{A_{11}}^r & \overbrace{A_{12}}^{n-r} \\ \hline 0 & A_{22} \end{array} \right] \begin{matrix} r \\ n-r \end{matrix}$$

is the block triangular matrix representing $\mathcal{A} : \mathbf{V} \rightarrow \mathbf{V}$ with respect to the basis (A.36), we have seen that the block A_{11} represents the restriction $\mathcal{A}_{\mathbf{U}}$ of $\mathcal{A}$ to the invariant subspace $\mathbf{U}$.

Instead, the block A_{22} represents the linear transformation $\bar{\mathcal{A}}$ induced on the quotient space $\mathbf{V}/\mathbf{U}$, with respect to the basis $([\mathbf{v}_{r+1}], \dots, [\mathbf{v}_n])$. Indeed, one has, for $j = r+1, r+2, \dots, n$,

$$\bar{\mathcal{A}}[\mathbf{v}_j] = [\mathcal{A}\mathbf{v}_j] = \left[\sum_{i=1}^n \mathbf{v}_i a_{ij} \right] = \sum_{i=r+1}^n [\mathbf{v}_i] a_{ij} = ([\mathbf{v}_{r+1}], \dots, [\mathbf{v}_n]) \begin{bmatrix} a_{r+1,j} \\ \vdots \\ a_{n,j} \end{bmatrix}.$$

Therefore, the columns of matrix A_{22} provides the components of the transformed vectors according to $\bar{\mathcal{A}}$ of the basis vectors, with respect to the quotient space basis vectors $([\mathbf{v}_{r+1}], \dots, [\mathbf{v}_n])$.

A.4 Eigenvalues and eigenvectors

To simplify the discussion, we will restrict ourselves to finite dimensional spaces. The field k will be the field of real numbers or, when there will be the need to refer to an algebraically closed field, we will consider the complex field.

Definition A.4.1 [EIGENVALUES AND EIGENVECTORS] Let $\mathbf{V}$ be a vector space of finite dimension n over the field K and let $\mathcal{A} : \mathbf{V} \rightarrow \mathbf{V}$ be a linear transformation. A vector $\mathbf{v} \in \mathbf{V}$ is an *eigenvector* of $\mathcal{A}$ if $\mathbf{v} \neq \mathbf{0}$ and if there exist a number in $\lambda \in K$ such that

$$\mathcal{A}\mathbf{v} = \lambda\mathbf{v}.$$

λ is only identified by $\mathbf{v}$ and is called an *eigenvalue* of $\mathcal{A}$. The set $\Lambda(\mathcal{A})$ of the eigenvalues of $\mathcal{A}$ is called *spectrum* of $\mathcal{A}$.

Please note that:

1. an eigenvector $\mathbf{v}$ is transformed by $\mathcal{A}$ into a vector that is parallel to $\mathbf{v}$;
2. the whole subspace $\text{span}(\mathbf{v})$, except the null vector, consists of eigenvectors that correspond to the same eigenvalue λ ;
3. the subspace $\text{span}(\mathbf{v})$ is $\mathcal{A}$ -invariant;
4. if $\lambda \neq 0$, the restriction of $\mathcal{A}$ to $\text{span}(\mathbf{v})$ is an isomorphism.

Definition A.4.2 [EIGENSPACE] If λ is an eigenvalue of $\mathcal{A} : \mathbf{V} \rightarrow \mathbf{V}$, the subspace of $\mathbf{V}$ generated by the eigenvectors of $\mathcal{A}$ corresponding to λ

$$\mathbf{U}_\lambda := \text{span}\{\mathbf{v} : \mathcal{A}\mathbf{v} = \lambda\mathbf{v}, \mathbf{v} \neq \mathbf{0}\}$$

is called *eigenspace* of the eigenvalue λ and the dimension of $\mathbf{U}_\lambda$ is called *geometric multiplicity* of the eigenvalue λ .

The eigenspace $\mathbf{U}_\lambda$ has the following properties:

1. Every non-null vector $\mathbf{u}$ in $\mathbf{U}_\lambda$ is an eigenvector of $\mathcal{A}$ related to λ . Indeed $\mathbf{u}$ is a linear combination of eigenvectors $\mathbf{v}_j$ of $\mathcal{A}$ related to λ , that is $\mathbf{u} = \sum_j \alpha_j \mathbf{v}_j$; it therefore results

$$\mathcal{A}\mathbf{u} = \mathcal{A} \sum_j \alpha_j \mathbf{v}_j = \sum_j \alpha_j \mathcal{A}\mathbf{v}_j = \sum_j \alpha_j \lambda \mathbf{v}_j = \lambda \mathbf{u}.$$

2. If $\mathcal{I} : \mathbf{V} \rightarrow \mathbf{V} : \mathbf{v} \mapsto \mathbf{v}$ denotes the identity transformation, $\mathbf{U}_\lambda$ is the kernel of the linear transformation $\mathcal{A} - \lambda\mathcal{I}$.
3. $\mathbf{U}_\lambda$ and all its subspaces are $\mathcal{A}$ -invariant.

Example A.4.1 (i) For any vector space $\mathbf{V}$, the identity transformation $\mathcal{I} : \mathbf{V} \rightarrow \mathbf{V}$ has only one eigenvalue $\lambda = 1$ and the corresponding eigenspace $\mathbf{U}_1$ coincides with $\mathbf{V}$.

Let $\mathbf{V}$ be the space of the real n -tuples, represented as row vectors.

(ii) If $\mathcal{A} : \mathbf{V} \rightarrow \mathbf{V} : [\xi_1 \ \xi_2 \ \dots \ \xi_n] \mapsto [\lambda_1 \xi_1 \ \lambda_2 \xi_2 \ \dots \ \lambda_n \xi_n]$, with $\lambda_1, \lambda_2, \dots, \lambda_n$ distinct real numbers, then λ_i is an eigenvalue of $\mathcal{A}$ and $\mathbf{U}_{\lambda_i} = \text{span}(\mathbf{e}_i^T)$.

(iii) If $\mathcal{A} : \mathbf{V} \rightarrow \mathbf{V} : [\xi_1 \ \xi_2 \ \dots \ \xi_n] \mapsto [\xi_2 \ \xi_3 \ \dots \ \xi_1]$ induces a cyclic permutation on the components of the n -tuples, in order to have the n -tuple $[\xi_1 \ \xi_2 \ \dots \ \xi_n] \neq \mathbf{0}^T$ as an eigenvector of $\mathcal{A}$, there must exist $\lambda \in \mathbb{R}$ such that it holds

$$\xi_2 = \lambda \xi_1, \ \xi_3 = \lambda \xi_2, \ \dots, \ \xi_n = \lambda \xi_{n-1}, \ \xi_1 = \lambda \xi_n. \quad (\text{A.38})$$

Since at least one of the components ξ_i is non null, from (A.38) it follows $\lambda^n \xi_i = \xi_i$, then

$$\lambda^n = 1. \quad (\text{A.39})$$

If n is odd, (A.39) has only one real solution $\lambda = 1$, the corresponding eigenvectors are the n -tuples constant and non-null and $\mathbf{U}_1 = \text{span}([1 \ 1 \ \dots \ 1])$ has dimension equal to 1. If n is even, (A.39) has the additional solution equal to -1 , corresponding to eigenvectors with components of the same absolute value and alternate sign, and an eigenspace $\mathbf{U}_{-1} = \text{span}([1 \ -1 \ 1 \ \dots \ -1])$ of dimension equal to 1.

Proposition A.4.3 [INDEPENDENCE OF EIGENSPACES] *If $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_t$ are eigenvectors of $\mathcal{A}$ corresponding to distinct eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_t$, then $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_t$ are linearly independent.*

PROOF By induction on t . If $t = 1$, the thesis is trivially true. Let us assume that the proposition is valid for $t - 1$ eigenvectors corresponding to $t - 1$ distinct eigenvalues ($t > 1$) and let us suppose that

$$\sum_{i=1}^t \alpha_i \mathbf{v}_i = \mathbf{0} \quad (\text{A.40})$$

with $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_t$ eigenvectors corresponding to the distinct eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_t$. By subtracting from each side the equalities

$$\begin{aligned} \mathbf{0} &= \mathcal{A} \sum_{i=1}^t \alpha_i \mathbf{v}_i = \sum_{i=1}^t \alpha_i \lambda_i \mathbf{v}_i \\ \mathbf{0} &= \lambda_1 \sum_{i=1}^t \alpha_i \mathbf{v}_i = \sum_{i=1}^t \alpha_i \lambda_1 \mathbf{v}_i \end{aligned}$$

one finds $\mathbf{0} = \sum_{i=2}^t \alpha_i (\lambda_1 - \lambda_i) \mathbf{v}_i$. According to the inductive hypothesis, $\mathbf{v}_2, \mathbf{v}_3, \dots, \mathbf{v}_t$ are linearly independent, and from $\lambda_1 - \lambda_i \neq 0$, $i = 2, 3, \dots, t$, it follows $\alpha_2 = \alpha_3 = \dots = \alpha_t = 0$. Then, by (A.40), also α_1 is null and the eigenvectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_t$ are linearly independent. ■

As a consequence of the proposition above, the eigenspaces corresponding to the distinct eigenvalues of $\mathcal{A}$ are independent, therefore the number of distinct eigenvalues cannot exceed the dimension of $\mathbf{V}$ and the sum $\mathbf{U}$ of the eigenspaces is a direct sum:

$$\mathbf{U} := \text{span}_{\lambda \in \Lambda(\mathcal{A})}(\mathbf{U}_\lambda) = \bigoplus_{\lambda \in \Lambda(\mathcal{A})} \mathbf{U}_\lambda.$$

The direct sum of the eigenspaces of $\mathcal{A}$ does not always coincide with $\mathbf{V}$. When this situation occurs, $\mathbf{V}$ has a basis consisting of eigenvectors of $\mathcal{A}$ and the A matrix, which represents $\mathcal{A}$ with respect to such basis, is diagonal.

Viceversa, if the A matrix that represents the transformation $\mathcal{A}$ with respect to a basis $(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n)$ is diagonalizable by similarity, namely $T^{-1}AT = \text{diag}\{\lambda_1, \lambda_2, \dots, \lambda_n\}$, then $(\tilde{\mathbf{v}}_1, \tilde{\mathbf{v}}_2, \dots, \tilde{\mathbf{v}}_n) = (\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n)T$ is a basis of eigenvectors for $\mathcal{A}$.

A.5 Polynomials associated to a linear transformation

Let $(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n)$ be a basis for the space $\mathbf{V}$ and let A be the matrix associated to the linear transformation $\mathcal{A} : \mathbf{V} \rightarrow \mathbf{V}$:

$$(\mathcal{A}\mathbf{v}_1, \mathcal{A}\mathbf{v}_2, \dots, \mathcal{A}\mathbf{v}_n) = (\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n)A.$$

A non-null vector $\mathbf{v} = (\mathbf{v}_1, \dots, \mathbf{v}_n) \begin{bmatrix} \xi_1 \\ \vdots \\ \xi_n \end{bmatrix} \in \mathbf{V}$ is an eigenvector of $\mathcal{A}$ corresponding to the eigenvalue λ if and only if

$$\mathbf{0} = \mathcal{A}\mathbf{v} - \lambda\mathbf{v} = (\mathbf{v}_1, \dots, \mathbf{v}_n)A \begin{bmatrix} \xi_1 \\ \vdots \\ \xi_n \end{bmatrix} - (\mathbf{v}_1, \dots, \mathbf{v}_n)\lambda I_n \begin{bmatrix} \xi_1 \\ \vdots \\ \xi_n \end{bmatrix} = (\mathbf{v}_1, \dots, \mathbf{v}_n)(A - \lambda I_n) \begin{bmatrix} \xi_1 \\ \vdots \\ \xi_n \end{bmatrix},$$

that is if and only if $\lambda I_n - A$ is singular and the n -tuple $\begin{bmatrix} \xi_1 \\ \vdots \\ \xi_n \end{bmatrix}$ belongs to its kernel.

We can conclude with the following

Proposition A.5.1 [CHARACTERISTIC POLYNOMIAL] *Let $\mathcal{A} : \mathbf{V} \rightarrow \mathbf{V}$ be a linear transformation and let A be the $n \times n$ matrix that represents $\mathcal{A}$ with respect to a basis of $\mathbf{V}$.*

(i) *The eigenvalues of $\mathcal{A}$ are the solutions of the n -th order polynomial in the variable s*

$$\det(sI_n - A). \quad (\text{A.41})$$

(ii) *A change of basis in $\mathbf{V}$ induces a similarity relation on the matrix representing $\mathcal{A}$, but it leaves unchanged the polynomial (A.41), which is called characteristic polynomial of the transformation (and of any of the representing matrices) and is indicated either with $\Delta_{\mathcal{A}}(s)$ or with $\Delta_A(s)$.*

The invariance of the polynomial (A.41) with respect to similarity transformations on A is an implication of the following equalities

$$\begin{aligned} \det(sI_n - T^{-1}AT) &= \det(T^{-1}(sI_n)T - T^{-1}AT) = \det(T^{-1}) \det(sI_n - A) \det(T) \\ &= \det(sI_n - A). \end{aligned}$$

- EXERCISE A.5.1 [SYMMETRIC MATRICES] If the matrix $A \in \mathbb{R}^{n \times n}$ is symmetric (that is $A = A^T$), all its eigenvalues are real.

‡ Hint: by denoting the conjugation operation with $\bar{\cdot}$, from $A\mathbf{x} = \lambda\mathbf{x}$ it follows $\bar{\mathbf{x}}^T A\mathbf{x} = \lambda\bar{\mathbf{x}}^T \mathbf{x}$. But $\bar{\mathbf{x}}^T A\mathbf{x}$ and $\bar{\mathbf{x}}^T \mathbf{x}$ are real and $\bar{\mathbf{x}}^T \mathbf{x} \neq 0$.

Definition A.5.2 [EIGENVALUES MULTIPLICITY] *The multiplicity of λ as a solution of the characteristic polynomial $\Delta_{\mathcal{A}}(s)$ is called algebraic multiplicity of the eigenvalue λ . In general, it does not coincide with the geometric multiplicity of the eigenvalue λ .*

- EXERCISE A.5.2 [ALGEBRAIC MULTIPLICITY AND GEOMETRIC MULTIPLICITY] If λ is an eigenvalue of the transformation $\mathcal{A} : \mathbf{V} \rightarrow \mathbf{V}$, its algebraic multiplicity is not lower than the geometric multiplicity.

‡ Hint: choose a basis in $\mathbf{U}_{\lambda}$ and complete it to a basis in $\mathbf{V}$. Evaluate the characteristic polynomial $\Delta_{\mathcal{A}}(s)$ of the matrix

$$\begin{bmatrix} \lambda & & & & \\ & \ddots & & & \\ & & \lambda & & \\ - & - & - & - & - \\ & & & & \\ & 0 & & & \star \end{bmatrix}$$

representing the transformation.

If λ is an eigenvalue, the non-null solutions of the system of equations with unknowns $\xi_1, \dots, \xi_n$

$$(A - \lambda I_n) \begin{bmatrix} \xi_1 \\ \vdots \\ \xi_n \end{bmatrix} = 0$$

provide the representations, with respect to the chosen basis in $\mathbf{V}$, of the eigenvectors corresponding to the eigenvalue λ (and hence of the non-null vectors of $\mathbf{U}_\lambda$).

Proposition A.5.3 [EXISTENCE OF EIGENVALUES AND EIGENVECTORS] *If the transformation $\mathcal{A} : \mathbf{V} \rightarrow \mathbf{V}$ is given on the complex field,*

- (i) *its spectrum cannot be empty;*
- (ii) *if $\mathbf{U}$ is an $\mathcal{A}$ -invariant subspace of $\mathbf{V}$, it contains at least one eigenvector of $\mathcal{A}$.*

PROOF (i) the characteristic polynomial $\det(sI_n - A)$ has at least one zero, since the complex field is algebraically closed.

(ii) According to (i), the restriction $\mathcal{A}|_{\mathbf{U}}$ of $\mathcal{A}$ to the subspace $\mathbf{U}$ has at least one eigenvalue and thus one eigenvector. They remain such also for the transformation $\mathcal{A}$, acting over all the space $\mathbf{V}$ ■

Both the statements of the previous Proposition are not valid on the real field, as can be shown by the following example.

Example A.5.1 The matrix

$$A = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}$$

has no real eigenvalues if $\theta \neq k\pi, k = 0, \pm 1, \pm 2, \dots$. This result has an immediate geometric interpretation: the matrix A , when applied to the vectors of $\mathbb{R}^2$, induces a counterclockwise rotation of θ radians. If θ is not an integer multiple of π , no $\mathbb{R}^2$ vector is mapped into a parallel vector.

If $\mathcal{A} : \mathbf{V} \rightarrow \mathbf{V}$ is a linear transformation and $\mathbf{V}$ has finite dimension n , let us consider the subspace $\text{span}(\mathcal{I}, \mathcal{A}, \mathcal{A}^2, \dots)$ generated by the powers of $\mathcal{A}$. Since $\mathcal{L}(\mathbf{V}, \mathbf{V})$ has finite dimension, there exists a *minimum* positive integer m such that $\mathcal{A}^m$ is linearly dependent on $\mathcal{I}, \mathcal{A}, \dots, \mathcal{A}^{m-1}$, that is

$$\mathcal{A}^m + \beta_{m-1}\mathcal{A}^{m-1} + \dots + \beta_1\mathcal{A} + \beta_0\mathcal{I} = 0 \quad (\text{A.42})$$

Then, $\mathcal{A}^{m+k}$ is linearly dependent on $\mathcal{I}, \mathcal{A}, \dots, \mathcal{A}^{m-1}$ for any $k \geq 0$ and from

$$\text{span}(\mathcal{I}, \mathcal{A}, \mathcal{A}^2, \dots) = \text{span}(\mathcal{I}, \mathcal{A}, \dots, \mathcal{A}^{m-1})$$

it follows that m is the dimension of the space generated by the powers of $\mathcal{A}$.

Definition A.5.4 [ANNIHILATOR POLYNOMIALS] *A polynomial $p(s) = \gamma_0 + \gamma_1 s + \dots + \gamma_t s^t$ in $K[s]$ is called annihilator of the linear transformation $\mathcal{A}$ ($\mathcal{A}$ -annihilator) if*

$$p(\mathcal{A}) := \gamma_0\mathcal{I} + \gamma_1\mathcal{A} + \dots + \gamma_t\mathcal{A}^t = 0 \quad (\text{A.43})$$

namely if, by combining the powers of $\mathcal{A}$ with the coefficients of the monomials of corresponding degree in $p(s)$, the null transformation is obtained.

Once a basis in $\mathbf{V}$ has been fixed and $\mathcal{A}$ is represented by the matrix A with respect to that basis, it can be easily verified that $p(\mathcal{A}) = 0$ if and only if $p(A) = 0$. Therefore, an $\mathcal{A}$ -annihilator polynomial is said to be also annihilator of the matrix A and, clearly, of any other matrix similar to A .

Example A.5.2 (i) Let us consider the matrix

$$A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$

Matrices $A^0 = I_2$ and A are linearly independent, while $A^2 = 0_2$ is linearly dependent on I_2 and A . It is immediate to verify that $A^2 + 0A + 0I_2$ is the null matrix and hence that $p(s) = s^2$ is an annihilator polynomial for A .

(ii) Let us consider the matrix

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Clearly

$$A^0 = I_3, \quad A, \quad A^2 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

are linearly independent, while $A^3 - A^2 + 0A + 0I_3 = 0$. It follows that $p(s) = s^3 - s^2$ is an annihilator polynomial for A .

In both the above examples, the set of annihilator polynomials for A (and of the transformation $\mathcal{A}$ that A is representing) does not include only the null polynomial. This is a general fact: indeed, from (A.42), it follows that the non-null polynomial $\beta_0 + \beta_1 s + \beta_2 s^2 + \dots + s^m$ is an annihilator polynomial. Moreover, it is clear that

1. if $p_1(s)$ and $p_2(s)$ are annihilators for $\mathcal{A}$, also $p_1(s) + p_2(s)$ is an annihilator;
2. if $p(s)$ is an annihilator for $\mathcal{A}$, all polynomials multiple of $p(s)$ are annihilators.

Proposition A.5.5 [MINIMUM POLYNOMIAL] *If $\mathcal{A} \in \mathcal{L}(\mathbf{V}, \mathbf{V})$, there exists a unique monic polynomial of minimum degree that is an annihilator of $\mathcal{A}$, and all annihilator polynomials of $\mathcal{A}$ are all and only its multiples.*

This is called minimal (annihilating) polynomial of $\mathcal{A}$ (and of any matrix A representing $\mathcal{A}$), it is written as $\psi_{\mathcal{A}}(s)$ (or $\psi_A(s)$), and coincides with the polynomial $\beta_0 + \beta_1 s + \beta_2 s^2 + \dots + s^m$ associated to the combination (A.42).

PROOF Given how the combination (A.42) is generated, the polynomial $\psi_{\mathcal{A}}(s) = \beta_0 + \beta_1 s + \beta_2 s^2 + \dots + s^m$ is a monic annihilator of minimum degree. It remains to be proven that every annihilator $p(s)$ of $\mathcal{A}$ is multiple of $\psi_{\mathcal{A}}(s)$. Indeed, by applying Euclid's algorithm it follows that $p(s) = \psi_{\mathcal{A}}(s)q(s) + r(s)$, where $r(s)$ is either the null polynomial or it has a lower degree than that of $\psi_{\mathcal{A}}(s)$. Since $\psi_{\mathcal{A}}(s)q(s)$ is an annihilator of $\mathcal{A}$, also $r(s) = p(s) - \psi_{\mathcal{A}}(s)q(s)$ is such. The fact that $\psi_{\mathcal{A}}(s)$ has minimum degree implies $r(s) = 0$ and $p(s)$ is multiple of $\psi_{\mathcal{A}}(s)$. ■

- EXERCISE A.5.3 [ANNIHILATOR POLYNOMIALS] If $q(s)$ is any polynomial,
 - (i) Is it true that the annihilators of $\mathcal{A}$ are all annihilators of the transformation $\mathcal{B} = q(\mathcal{A})$?
 ‡ Hint: consider the matrix $-I_2$, annihilated by polynomial $p(s) = s + 1$, and polynomial $q(s) = s^2$.
 - (ii) Verify that the polynomial $\psi_{\mathcal{B}}(s)$ has degree that is not higher than that of $\psi_{\mathcal{A}}(s)$;
 ‡ Hint: the subspace of $\mathcal{L}(\mathbf{V}, \mathbf{V})$ generated by the powers of $\mathcal{B}$ is included in the one generated by the powers of $\mathcal{A}$.
 - (iii) Is it true that $\psi_{\mathcal{B}}(s)$ divides $\psi_{\mathcal{A}}(s)$?

Teorema A.5.6 [CAYLEY-HAMILTON] *Let $\mathbf{V}$ be a vector space of finite dimension n on the field K and $\mathcal{A} : \mathbf{V} \rightarrow \mathbf{V}$ a linear transformation. The characteristic polynomial $\Delta_{\mathcal{A}}(s) = \alpha_0 + \alpha_1 s + \dots + s^n$ is an annihilator for $\mathcal{A}$, that is*

$$\Delta_{\mathcal{A}}(\mathcal{A}) = \alpha_0 \mathcal{I} + \alpha_1 \mathcal{A} + \dots + \mathcal{A}^n = 0.$$

PROOF It will be sufficient to verify that, if the matrix $A \in K^{n \times n}$ represents the transformation $\mathcal{A}$ with respect to some basis in $\mathbf{V}$, then

$$\Delta_A(A) = \alpha_0 I_n + \alpha_1 A + \dots + A^n \quad (\text{A.44})$$

is the null matrix. To this aim, we will use the algebraic tool of polynomial matrices, namely matrices whose entries are polynomials in $K[s]$, proving the result that follows, which will be useful also later on:

Lemma A.5.7 *The polynomial $g(s) = \gamma_0 + \gamma_1 s + \dots + \gamma_t s^t$ is an annihilator of $A \in K^{n \times n}$ if and only if there exists a polynomial matrix $M(s) \in K^{n \times n}$ for which it holds*

$$g(s)I_n = M(s)(sI_n - A). \quad (\text{A.45})$$

PROOF OF LEMMA A.5.7 For any $h \geq 1$ it results

$$s^h I_n - A^h = (s^{h-1} I_n + s^{h-2} A + \dots + s A^{h-2} + A^{h-1})(sI_n - A) =: L_h(s)(sI_n - A)$$

and, with $L_0(s) := 0$, for any $h \in \mathbb{N}$ we can write A^h as

$$A^h = s^h I_n - L_h(s)(sI_n - A). \quad (\text{A.46})$$

If (A.45) holds, from the two identities

$$\begin{aligned} M(s)(sI_n - A) &= \gamma_0 I_n + \gamma_1 s I_n + \dots + \gamma_t s^t I_n \\ g(A) &= \gamma_0 I_n + \gamma_1 A + \dots + \gamma_t A^t \end{aligned}$$

we obtain

$$M(s)(sI_n - A) - g(A) = \gamma_1(sI_n - A) + \gamma_2(s^2 I_n - A^2) + \dots + \gamma_t(s^t I_n - A^t) = \sum_{i=1}^t \gamma_i L_i(s)(sI_n - A)$$

and then

$$g(A) = \left(M(s) - \sum_{i=1}^t \gamma_i L_i(s) \right) (sI_n - A) =: N(s)(sI_n - A) \quad (\text{A.47})$$

Since either $g(A)$ has zero degree or it is null, se $N(s)$ were not null (A.47) could not be verifies. Hence, it is $N(s) = 0$ and, consequently, $g(A) = 0$.

Viceversa, if $g(A) = 0$ holds, by taking into account (A.46), it follows

$$\begin{aligned} 0 = g(A) &= \gamma_0 I_n + \gamma_1 A + \dots + \gamma_t A^t \\ &= \gamma_0 I_n + \gamma_1 (sI_n - L_1(s)(sI_n - A)) + \dots + \gamma_t (s^t I_n - L_t(s)(sI_n - A)) \\ &= g(s)I_n - \left(\sum_{i=1}^t L_i(s) \right) (sI_n - A). \end{aligned}$$

Therefore, (A.45) holds, and the lemma is proved. ■

The following identity holds:

$$I_n \Delta_A(s) = \text{adj}(sI_n - A)(sI_n - A), \quad (\text{A.48})$$

which comes from the Laplace expansion of the determinant of a matrix according to the entries of one row. As a consequence of Lemma A.5.7 and of (A.48), Cayley-Hamilton's theorem is immediately derived. ■

- EXERCISE A.5.4 [CAYLEY-HAMILTON'S THEOREM] An alternative proof of Cayley-Hamilton's theorem is based on the employment of formal power series:

(i) $(I_n - A\xi)^{-1}$ is represented through the series $I_n + A\xi + A^2\xi^2 + \dots$

(ii) The identity

$$\begin{aligned} (\alpha_0 \xi^n + \alpha_1 \xi^{n-1} + \dots + \alpha_{n-1} \xi + 1)(I_n - A\xi)^{-1} &= \text{adj}(\xi^{-1}I_n - A)\xi^{n-1} \\ &= M_{n-1}\xi^{n-1} + M_{n-2}\xi^{n-2} + \dots + M_0 \end{aligned}$$

is considered

(iii) In the previous equality factor $(I_n - A\xi)^{-1}$ is replaced by the series and the nullity of the coefficient of ξ^n is imposed at the left hand side, as at the right hand side.

- EXERCISE A.5.5 [ADJOINT MATRIX] Let us assume $\text{adj}(sI_n - A) = M_{n-1} + M_{n-2}s + \dots + M_1s^{n-2} + M_0s^{n-1}$. By adopting (A.48), verify that
 - (i) $M_{n-1} = I_n$ and, recursively, $M_{n-h} = \alpha_{n-h}I_n + AM_{n-h+1}$, $h = 2, \dots, n-1$
 - (ii) Then $M_{n-h} = \alpha_{n-h}I_n + \alpha_{n-h+1}A + \dots + \alpha_{n-1}A^{h-1} + A^h$, $h = 1, 2, \dots, n-1$

A first, immediate consequence of the fact that $\Delta_{\mathcal{A}}(s)$ is an annihilator polynomial is that it is multiple of the minimal polynomial $\psi_{\mathcal{A}}(s)$. This implies that the zeros of $\psi_{\mathcal{A}}(s)$ are among those of $\Delta_{\mathcal{A}}(s)$ (of not higher multiplicity). The following proposition states, viceversa, that any zero of the characteristic polynomial, that is any eigenvalue of $\mathcal{A}$, is a zero of the minimal polynomial, even if of a lower multiplicity.

Proposition A.5.8 [ZEROS OF THE MINIMAL POLYNOMIAL] *If λ is a zero of $\Delta_{\mathcal{A}}(s)$, it is also a zero of $\psi_{\mathcal{A}}(s)$.*

PROOF If $\mathbf{v} \neq \mathbf{0}$ is an eigenvector of $\mathcal{A}$ corresponding to the eigenvalue λ , for any integer $k > 0$ one has $\mathcal{A}^k \mathbf{v} = \lambda^k \mathbf{v}$. If the minimal polynomial is $\psi_{\mathcal{A}}(s) = \beta_0 + \beta_1 s + \beta_2 s^2 + \dots + s^m$, from $\psi_{\mathcal{A}}(\mathcal{A}) = 0$ it follows that

$$\mathbf{0} = \psi_{\mathcal{A}}(\mathcal{A})\mathbf{v} = (\beta_0 \mathcal{I} + \beta_1 \mathcal{A} + \beta_2 \mathcal{A}^2 + \dots + \mathcal{A}^m)\mathbf{v} = (\beta_0 + \beta_1 \lambda + \beta_2 \lambda^2 + \dots + \lambda^m)\mathbf{v} = \psi_{\mathcal{A}}(\lambda)\mathbf{v}.$$

Then, $\psi_{\mathcal{A}}(\lambda) = 0$. ■

The multiplicity of λ as a zero of $\psi_{\mathcal{A}}(s)$ is called “degree” of the λ eigenvalue.

The characteristic polynomial $\Delta_{\mathcal{A}}(s)$, being an annihilator of the linear transformation $\mathcal{A}$, is a multiple of the minimal polynomial $\psi_{\mathcal{A}}(s)$. Therefore there exists a polynomial $b(s)$ satisfying the identity $b(s)\psi_{\mathcal{A}}(s) = \Delta_{\mathcal{A}}(s)$. The meaning of the factor $b(s)$ is detailed by the following.

Proposition A.5.9 [FACTORIZATION OF THE CHARACTERISTIC POLYNOMIAL] *The minimal polynomial $\psi_{\mathcal{A}}(s)$ and the characteristic polynomial $\Delta_{\mathcal{A}}(s)$ satisfy the relation*

$$b(s)\psi_{\mathcal{A}}(s) = \Delta_{\mathcal{A}}(s)$$

where $b(s)$ is the (monic) greatest common divisor (GCD) of the elements of the matrix $\text{adj}(sI_n - A)$.

PROOF We exploit once again the identity

$$I_n \Delta_{\mathcal{A}}(s) = \text{adj}(sI_n - A)(sI_n - A). \quad (\text{A.49})$$

If $b(s)$ is the monic GCD of the elements of $\text{adj}(sI_n - A)$, (A.49) can be rewritten in the form

$$I_n \Delta_{\mathcal{A}}(s) = b(s)M(s)(sI_n - A),$$

whence it appears that $\Delta_{\mathcal{A}}(s)$ can be divided by $b(s)$, namely

$$\Delta(s) = b(s)\phi(s), \quad (\text{A.50})$$

with monic $\phi(s)$. By deleting from both sides in (A.49) the common factor $b(s)$, it can be obtained

$$I_n \phi(s) = M(s)(sI_n - A), \quad (\text{A.51})$$

where the GCD of the elements of $M(s)$ is 1.

By Lemma A.5.7, from (A.51) it follows that $\phi(s)$ is an annihilator polynomial of A , and then a multiple of the minimal polynomial: $\phi(s) = \psi_A(s)q(s)$. Moreover, according to Lemma A.5.7, it is $I_n \psi_A(s) = N(s)(sI_n - A)$ and, by multiplying both sides by $q(s)$, it holds

$$I_n \phi(s) = q(s)N(s)(sI_n - A). \quad (\text{A.52})$$

By subtracting (A.52) from (A.51), one gets $0 = (M(s) - q(s)N(s))(sI_n - A)$ and the invertibility of $sI_n - A$ as a rational matrix implies $M(s) = q(s)N(s)$, hence $q(s) = 1$, since the factors that are common to all elements in $M(s)$ are only the constant values. It can be concluded that in (A.50) $\phi(s) = \psi_A(s)$. ■

By Propositions A.5.8 and A.5.9, if the characteristic polynomial factorizes on the complex field as

$$\Delta_A(s) = (s - \lambda_1)^{n_1} (s - \lambda_2)^{n_2} \dots (s - \lambda_t)^{n_t}, \quad (\text{A.53})$$

where $n_1, n_2, \dots, n_t$ are the algebraic multiplicities of the eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_t$, the factorization of the minimal polynomial is

$$\psi_A(s) = (s - \lambda_1)^{m_1} (s - \lambda_2)^{m_2} \dots (s - \lambda_t)^{m_t}, \quad (\text{A.54})$$

with

$$1 \leq m_1 \leq n_1, \quad 1 \leq m_2 \leq n_2, \quad \dots, \quad 1 \leq m_t \leq n_t,$$

while the GCD of the elements of $\text{adj}(sI_n - A)$ is given by

$$b(s) = (s - \lambda_1)^{n_1 - m_1} (s - \lambda_2)^{n_2 - m_2} \dots (s - \lambda_t)^{n_t - m_t}.$$

A.6 Polynomials associated to a vector

Let $\mathbf{V}$ be a space of finite dimension n , $\mathbf{v}$ a non-null vector in $\mathbf{V}$ and $\mathcal{A} : \mathbf{V} \rightarrow \mathbf{V}$ a linear transformation.

Definition A.6.1 [ANNIHILATOR POLYNOMIAL OF A VECTOR] A polynomial $p(s) = p_0 + p_1 s + \dots + p_h s^h$ is called *annihilator of the vector $\mathbf{v}$ (with respect to the transformation $\mathcal{A}$)* if

$$p_0 \mathbf{v} + p_1 \mathcal{A} \mathbf{v} + \dots + p_h \mathcal{A}^h \mathbf{v} = p(\mathcal{A}) \mathbf{v} = \mathbf{0} \quad (\text{A.55})$$

that is if $\mathbf{v} \in \ker p(\mathcal{A})$.

Any annihilator polynomial of $\mathcal{A}$ is an annihilator of $\mathbf{v}$. Generally speaking, though, the set of the annihilator polynomials of $\mathbf{v}$ strictly contains that of the annihilators of $\mathcal{A}$.

If $p(s)$ is an annihilator of $\mathbf{v}$, also every multiple $p(s)q(s)$ is an annihilator; if $p_1(s)$ and $p_2(s)$ are annihilators of $\mathbf{v}$, also $p_1(s) + p_2(s)$ is an annihilator.

- EXERCISE A.6.1 If $p_1(s)$ and $p_2(s)$ are annihilators of $\mathbf{v}$, also their GCD is an annihilator.

‡ *Solution:* Set $c(s) = \text{GCD}(p_1(s), p_2(s))$. Then $p_i(s) = c(s)h_i(s)$, $i = 1, 2$, with $\text{GCD}(h_1(s), h_2(s)) = 1$ and there exist two additional polynomials $q_i(s)$, $i = 1, 2$, such that $q_1(s)h_1(s) + q_2(s)h_2(s) = 1$. Then $c(s) = q_1(s)p_1(s) + q_2(s)p_2(s)$ is the sum of two annihilators of $\mathbf{v}$.

Similarly to the proof of Proposition A.5.5, it can be proved also the following.

Proposition A.6.2 [MINIMUM POLYNOMIAL OF A VECTOR] *If $\mathcal{A} \in \mathcal{L}(\mathbf{V}, \mathbf{V})$ and $\mathbf{v} \in \mathbf{V}$, there exists a unique monic polynomial of minimum degree annihilator of $\mathbf{v}$, and the annihilator polynomials of $\mathbf{v}$ are all and only its multiple. This is called the minimal polynomial of $\mathbf{v}$, it is indicated with $\psi_{\mathbf{v}}(s)$, and is a divisor of $\psi_{\mathcal{A}}(s)$, the minimal polynomial of the transformation.*

- EXERCISE A.6.2 Let $\mathcal{A}$ be a linear transformation and let $\psi_{\mathcal{A}} = (s - \lambda_1)^{m_1}(s - \lambda_2)^{m_2} \dots (s - \lambda_t)^{m_t}$ be its minimal polynomial.
 - (i) $\mathbf{v} \neq \mathbf{0}$ is eigenvector of $\mathcal{A}$ corresponding to the eigenvalue λ_1 if and only if its minimal polynomial is $s - \lambda_1$.
 - (ii) If $k > m_1$, it does not exist any vector whose minimal polynomial is $(s - \lambda_1)^k$.
 - (iii) There exists a vector $\mathbf{v}$ whose minimal polynomial is $(s - \lambda_1)^{m_1}$.
- ‡ Hint: (ii) If $\mathbf{v}$ had minimal polynomial equal to $(s - \lambda_1)^k$, the vector $\mathbf{w} := (\mathcal{A} - \lambda_1 \mathcal{I})^{m_1} \mathbf{v}$ would have both $(s - \lambda_2)^{m_2} \dots (s - \lambda_t)^{m_t}$ and $(s - \lambda_1)^{k-m_1}$ as annihilators, which are coprime. Therefore, ...
- (iii) Consider a vector $\mathbf{w}$ that does not belong to the kernel of $(\mathcal{A} - \lambda_1 \mathcal{I})^{m_1-1}(\mathcal{A} - \lambda_2 \mathcal{I})^{m_2} \dots (\mathcal{A} - \lambda_t \mathcal{I})^{m_t}$ and be $\mathbf{v} := (\mathcal{A} - \lambda_2 \mathcal{I})^{m_2} \dots (\mathcal{A} - \lambda_t \mathcal{I})^{m_t} \mathbf{w}$.

An alternative characterization of the minimal polynomial of a vector $\mathbf{v}$ can be obtained by considering the smallest positive integer t in correspondence to which vector $\mathcal{A}^t \mathbf{v}$ is linearly dependent on $\mathbf{v}, \mathcal{A}\mathbf{v}, \dots, \mathcal{A}^{t-1}\mathbf{v}$, that is

$$\eta_0 \mathbf{v} + \eta_1 \mathcal{A}\mathbf{v} + \dots + \eta_{t-1} \mathcal{A}^{t-1} \mathbf{v} + \mathcal{A}^t \mathbf{v} = \mathbf{0}. \quad (\text{A.56})$$

It clearly results $\psi_{\mathbf{v}}(s) = \eta_0 + \eta_1 s + \dots + \eta_{t-1} s^{t-1} + s^t$.

Corollary A.6.3 [CYCLIC SPACE] *The subspace of $\mathbf{V}$ defined by*

$$< \mathcal{A} | \mathbf{v} > := \text{span}\{\mathbf{v}, \mathcal{A}\mathbf{v}, \mathcal{A}^2 \mathbf{v}, \dots\} \quad (\text{A.57})$$

is the smallest $\mathcal{A}$ -invariant subspace including $\mathbf{v}$. It is called cyclic space with respect to the transformation $\mathcal{A}$ generated by vector $\mathbf{v}$ and, according to (A.56), it follows that

$$< \mathcal{A} | \mathbf{v} > := \text{span}\{\mathbf{v}, \mathcal{A}\mathbf{v}, \dots, \mathcal{A}^{t-1} \mathbf{v}\}. \quad (\text{A.58})$$

Vectors $\mathbf{v}, \mathcal{A}\mathbf{v}, \dots, \mathcal{A}^{t-1} \mathbf{v}$ make a cyclic basis for $< \mathcal{A} | \mathbf{v} >$. ■

Proposition A.6.4 [MINIMUM POLYNOMIAL OF THE SUM OF VECTORS] *If the minimal polynomials $\psi_{\mathbf{v}_1}(s), \psi_{\mathbf{v}_2}(s), \dots, \psi_{\mathbf{v}_k}(s)$ of vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k$ (with respect to the same linear transformation $\mathcal{A}$) are pairwise coprime, the sum $\mathbf{v} = \mathbf{v}_1 + \mathbf{v}_2 + \dots + \mathbf{v}_k$ has minimal polynomial equal to the product of the minimal polynomials of the summands, namely $\psi_{\mathbf{v}} = \psi_{\mathbf{v}_1}(s) \psi_{\mathbf{v}_2}(s) \dots \psi_{\mathbf{v}_k}(s)$.*

PROOF The product $\psi_{\mathbf{v}_1}(s) \psi_{\mathbf{v}_2}(s)$ is an annihilator of $\mathbf{v}_1 + \mathbf{v}_2$ and hence it is multiple of $\psi_{\mathbf{v}_1 + \mathbf{v}_2}(s)$. The product $\psi_{\mathbf{v}_1 + \mathbf{v}_2}(s) \psi_{\mathbf{v}_2}(s)$ is an annihilator of $\mathbf{v}_2$ and of $\mathbf{v}_1 + \mathbf{v}_2$, hence of $\mathbf{v}_1 = (\mathbf{v}_1 + \mathbf{v}_2) - \mathbf{v}_2$. This implies that $\psi_{\mathbf{v}_1}(s)$ divides $\psi_{\mathbf{v}_1 + \mathbf{v}_2}(s) \psi_{\mathbf{v}_2}(s)$ and being $\psi_{\mathbf{v}_1}(s)$ and $\psi_{\mathbf{v}_2}(s)$ relatively coprime, $\psi_{\mathbf{v}_1}(s)$ divides $\psi_{\mathbf{v}_1 + \mathbf{v}_2}(s)$.

Similarly it can be verified that $\psi_{\mathbf{v}_2}(s)$ divides $\psi_{\mathbf{v}_1 + \mathbf{v}_2}(s)$. Then the product $\psi_{\mathbf{v}_1}(s) \psi_{\mathbf{v}_2}(s)$ is at the same time a factor and a multiple of $\psi_{\mathbf{v}_1 + \mathbf{v}_2}(s)$, it coincides with itself and the proposition is verified if $k = 2$. If the thesis is proved inductively for $k-1$ vectors, since $\psi_{\mathbf{v}_1 + \mathbf{v}_2}(s)$ is coprime with respect to $\psi_{\mathbf{v}_3}(s), \dots, \psi_{\mathbf{v}_k}(s)$ then $\psi_{\mathbf{v}}(s) = \psi_{\mathbf{v}_1 + \mathbf{v}_2}(s) \psi_{\mathbf{v}_3}(s) \dots \psi_{\mathbf{v}_k}(s) = \psi_{\mathbf{v}_1}(s) \psi_{\mathbf{v}_2}(s) \psi_{\mathbf{v}_3}(s) \dots \psi_{\mathbf{v}_k}(s)$. ■

Proposition A.6.5 [VECTORS $\mathbf{v}$ WITH $\psi_{\mathbf{v}} = \psi_{\mathcal{A}}$] *For any linear transformation $\mathcal{A} : \mathbf{V} \rightarrow \mathbf{V}$ there exist vectors $\mathbf{v} \in \mathbf{V}$ whose minimal polynomial is the minimal polynomial of $\mathcal{A}$.*

PROOF Let $\mathbf{w} \neq \mathbf{0}$ be a vector whose minimal polynomial $\psi_{\mathbf{w}}(s)$ has lower degree than that of $\psi_{\mathcal{A}}(s)$. Since $\psi_{\mathbf{w}}(\mathcal{A}) \neq 0$, there exists a vector $\mathbf{u}$ such that $\psi_{\mathbf{w}}(\mathcal{A})\mathbf{u} \neq \mathbf{0}$ and thus $\psi_{\mathbf{w}}(s)$ is not multiple of $\psi_{\mathbf{u}}(s)$. Then the least common multiple $g(s)$ of $\psi_{\mathbf{w}}(s)$ and $\psi_{\mathbf{u}}(s)$ has degree strictly greater than the degree of $\psi_{\mathbf{w}}(s)$, and its factorization into irreducible factors can be expressed as

$$g(s) = p_1(s)^{\nu_1 \vee \mu_1} p_2(s)^{\nu_2 \vee \mu_2} \cdots p_k(s)^{\nu_k \vee \mu_k}$$

where ν_i and μ_i are the multiplicities of $p_i(s)$, in $\psi_{\mathbf{w}}(s)$ and $\psi_{\mathbf{u}}(s)$, respectively. If it is

$$\begin{aligned} \mathbf{v}_1 &:= p_2(\mathcal{A})^{\nu_2 \vee \mu_2} p_3(\mathcal{A})^{\nu_3 \vee \mu_3} \cdots p_k(\mathcal{A})^{\nu_k \vee \mu_k} (\mathbf{u} + \mathbf{w}) \\ \mathbf{v}_2 &:= p_1(\mathcal{A})^{\nu_1 \vee \mu_1} p_3(\mathcal{A})^{\nu_3 \vee \mu_3} \cdots p_k(\mathcal{A})^{\nu_k \vee \mu_k} (\mathbf{u} + \mathbf{w}) \\ &\vdots \\ \mathbf{v}_k &:= p_1(\mathcal{A})^{\nu_1 \vee \mu_1} p_2(\mathcal{A})^{\nu_2 \vee \mu_2} \cdots p_{k-1}(\mathcal{A})^{\nu_{k-1} \vee \mu_{k-1}} (\mathbf{u} + \mathbf{w}) \end{aligned}$$

the minimal polynomials of $\mathbf{v}_i$ are $p_i(s)^{\nu_i \vee \mu_i}$, $i = 1, 2, \dots, k$, and hence they are pairwise coprime. By the proposition before, $g(s)$ is the minimal polynomial of $\mathbf{v} := \sum_{i=1}^k \mathbf{v}_i$.

It follows that a vector $\mathbf{v}$ is found, whose minimal polynomial has higher degree than that of $\psi_{\mathbf{w}}(s)$. This reasoning can be repeated until a vector is obtained whose minimal polynomial coincides with $\psi_{\mathcal{A}}(s)$. ■

- EXERCISE A.6.3 (i) If $\mathbf{v}_1$ and $\mathbf{v}_2$ have coprime minimal polynomials, then the cyclic space generated by $\mathbf{v} := \mathbf{v}_1 + \mathbf{v}_2$ has a dimension given by the sum of the dimensions of the cyclic spaces generated by $\mathbf{v}_1$ and $\mathbf{v}_2$.
(ii) What is the dimension of the cyclic space generated by $\mathbf{v}_1 + \mathbf{v}_2 + \cdots + \mathbf{v}_k$, where $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k$ are eigenvectors corresponding to distinct eigenvalues?
(iii) If any non-null vector of $\mathbf{V}$ generates a cyclic space of dimension equal to 1, then $\mathcal{A}$ has a single eigenvalue λ and $\mathbf{V} = \mathbf{U}_{\lambda}$.
- EXERCISE A.6.4 The restriction $\mathcal{A}|_{\mathbf{C}}$ of the transformation $\mathcal{A}$ to a cyclic subspace $\mathbf{C}$ has the same minimal and characteristic polynomials.
‡ Hint: there exists a vector $\mathbf{v}$ such that $\psi_{\mathbf{v}}(s)$ has degree equal to that of $\Delta_{\mathcal{A}|_{\mathbf{C}}}(s)$.
- EXERCISE A.6.5* Two eigenvectors $\mathbf{w}_1$ and $\mathbf{w}_2$, corresponding to the same eigenvalue λ of $\mathcal{A}$ and belonging to the same cyclic subspace of $\mathbf{V}$, are linearly dependent.
‡ Solution: if $\mathbf{v}$ generates the cyclic subspace, $\mathbf{w}_i = p_i(\mathcal{A})\mathbf{v}$ for some $p_i(s)$, $i = 1, 2$.
If $\psi_{\mathbf{v}}(s)$ is the minimal polynomial of $\mathbf{v}$, it is $\psi_{\mathbf{v}}(\mathcal{A})\mathbf{w}_1 = \psi_{\mathbf{v}}(\mathcal{A})p_1(\mathcal{A})\mathbf{v} = \mathbf{0}$, then $\psi_{\mathbf{v}}(s)$ is multiple of $s - \lambda$, the minimal polynomial of the eigenvector $\mathbf{w}_1$.
It the factorization $\psi_{\mathbf{v}}(s) = (s - \lambda)g(s)$, $g(s)$ is a polynomial of minimum degree such that $g(\mathcal{A})\mathbf{v}$ is eigenvector corresponding to the eigenvalue λ and, thanks to this, one gets $p_i(s) = h_i(s)g(s)$, $i = 1, 2$.
Being $\mathbf{w}_i = p_i(\mathcal{A})\mathbf{v} = h_i(\mathcal{A})[g(\mathcal{A})\mathbf{v}] = h_i(\lambda)[g(\mathcal{A})\mathbf{v}]$, $i = 1, 2$, the eigenvectors $\mathbf{w}_1$ and $\mathbf{w}_2$ are linearly dependent.

A.7 Generalized eigenspaces and Jordan chains

In this and in the following paragraphs we will suppose that the field K is the complex field, that $\mathbf{V}$ has dimension n on $\mathbb{C}$ and that the linear transformation $\mathcal{A}$ has characteristic and minimal polynomials factorized as in (A.53) and (A.54).

Proposition A.7.1 [GENERALIZED EIGENSPACES] *The subspaces*

$$\mathbf{N}_{\lambda_i} := \ker(\mathcal{A} - \lambda_i \mathcal{I})^{m_i}, \quad i = 1, 2, \dots, t, \quad (\text{A.59})$$

with m_i multiplicity of the eigenvalue λ_i in the minimal polynomial $\psi_{\mathcal{A}}(s)$, are called *generalized eigenspaces* corresponding to the eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_t$. They have the following properties:

- i) they are $\mathcal{A}$ -invariant;
 ii) their sum is a direct sum and it is

$$\mathbf{V} = \mathbf{N}_{\lambda_1} \oplus \mathbf{N}_{\lambda_2} \oplus \cdots \oplus \mathbf{N}_{\lambda_t}; \quad (\text{A.60})$$

- iii) the chain of subspaces $\ker(\mathcal{A} - \lambda_i \mathcal{I})^h$, $h = 1, 2, \dots$, is strictly increasing up to the power m_i , and it is stationary for higher powers

$$\underbrace{\ker(\mathcal{A} - \lambda_i \mathcal{I})}_{\mathbf{U}_{\lambda_i}} \subset \ker(\mathcal{A} - \lambda_i \mathcal{I})^2 \subset \cdots \subset \underbrace{\ker(\mathcal{A} - \lambda_i \mathcal{I})^{m_i}}_{\mathbf{N}_{\lambda_i}} = \cdots \quad (\text{A.61})$$

- iv) the dimension of $\mathbf{N}_{\lambda_i}$ coincides with the algebraic multiplicity n_i of the eigenvalue λ_i .

PROOF (i) If the vector $\mathbf{v}$ is in $\mathbf{N}_{\lambda_i}$, then $(\mathcal{A} - \lambda_i \mathcal{I})^{m_i} \mathbf{v} = \mathbf{0}$. Therefore $(\mathcal{A} - \lambda_i \mathcal{I})^{m_i} (\mathcal{A} \mathbf{v}) = \mathcal{A} (\mathcal{A} - \lambda_i \mathcal{I})^{m_i} \mathbf{v} = \mathbf{0}$ and also the vector $\mathcal{A} \mathbf{v}$ is in $\mathbf{N}_{\lambda_i}$.

(ii) The GCD of polynomials $p_i(s) := \psi_{\mathcal{A}}(s) / (s - \lambda_i)^{m_i}$, $i = 1, 2, \dots, t$, is unitary; then, there exist polynomials $g_i(s)$, $i = 1, 2, \dots, t$, that satisfy the identity

$$g_1(s)p_1(s) + g_2(s)p_2(s) + \cdots + g_t(s)p_t(s) = 1. \quad (\text{A.62})$$

If $\mathbf{v}$ is any vector in $\mathbf{V}$, the vectors $\mathbf{v}_i := g_i(\mathcal{A})p_i(\mathcal{A})\mathbf{v}$, $i = 1, 2, \dots, t$, satisfy

$$\mathbf{v}_i \in \mathbf{N}_{\lambda_i}, \quad i = 1, 2, \dots, t,$$

because $(\mathcal{A} - \lambda_i \mathcal{I})^{m_i} \mathbf{v}_i = g_i(\mathcal{A})\psi_{\mathcal{A}}(\mathcal{A})\mathbf{v} = \mathbf{0}$. Moreover, taking into account (A.62), one gets

$$\mathbf{v} = \sum_{i=1}^t g_i(\mathcal{A})p_i(\mathcal{A})\mathbf{v} = \sum_{i=1}^t \mathbf{v}_i, \quad (\text{A.63})$$

and hence the sum of the generalized eigenspaces coincides with $\mathbf{V}$.

To prove that their sum is a direct sum, let us suppose, by contradiction, that a non-null vector $\mathbf{v}$ belongs both to a generalized eigenspace, for example to $\mathbf{N}_{\lambda_1}$, and to the sum of the others, namely $\sum_{i \neq 1} \mathbf{N}_{\lambda_i}$. Then both polynomials $(s - \lambda_1)^{m_1}$ and $\prod_{i=2}^t (s - \lambda_i)^{m_i}$ would be annihilators of $\mathbf{v}$, resulting in

$$\mathbf{0} = (\mathcal{A} - \lambda_1 \mathcal{I})^{m_1} \mathbf{v} = (\mathcal{A} - \lambda_2 \mathcal{I})^{m_2} \cdots (\mathcal{A} - \lambda_t \mathcal{I})^{m_t} \mathbf{v},$$

and the polynomial 1, their GCD, would be the annihilator of $\mathbf{v}$ (see Exercise A.6.1).

(iii) Similarly to what we have seen in section A.2.4, if in the chain (A.61) two consecutive spaces coincide, starting from the first of them the chain becomes stationary. If (A.61) became stationary at step $k < m_i$, it would be

$$\ker(\mathcal{A} - \lambda_i \mathcal{I})^k = \ker(\mathcal{A} - \lambda_i \mathcal{I})^{m_i}$$

and, thanks to (A.63), any vector $\mathbf{v}$ in $\mathbf{V}$ would belong to the kernel of the transformation

$$(\mathcal{A} - \lambda_1 \mathcal{I})^{m_1} \cdots (\mathcal{A} - \lambda_i \mathcal{I})^k \cdots (\mathcal{A} - \lambda_t \mathcal{I})^{m_t}.$$

In such a case, $p(s) = (s - \lambda_1)^{m_1} \cdots (s - \lambda_k)^k (s - \lambda_t)^{m_t}$ would be an annihilator polynomial of $\mathcal{A}$ of degree lower than the one of the minimal polynomial, which is a contradiction. On the other hand, if for $k > m_i$ it were

$$\ker(\mathcal{A} - \lambda_i \mathcal{I})^k \supset \ker(\mathcal{A} - \lambda_i \mathcal{I})^{m_i},$$

in the proof of step (ii) we could replace $\mathbf{N}_{\lambda_i}$ with $\ker(\mathcal{A} - \lambda_i \mathcal{I})^k$, getting

$$\mathbf{V} = \mathbf{N}_{\lambda_1} \oplus \cdots \oplus \mathbf{N}_{\lambda_{i-1}} \oplus \ker(\mathcal{A} - \lambda_i \mathcal{I})^k \oplus \mathbf{N}_{\lambda_{i+1}} \oplus \cdots \oplus \mathbf{N}_{\lambda_t}. \quad (\text{A.64})$$

The dimension of $\mathbf{V}$ would be different depending on whether the decomposition (A.60) or the decomposition (A.64) is considered.

(iv) The restriction of the transformation $\mathcal{A}$ to the subspace $\mathbf{N}_{\lambda_i}$ has only the eigenvalue λ_i . Indeed, if $\mathbf{v} \in \mathbf{N}_{\lambda_i}$ is an eigenvector corresponding to the eigenvalue μ , it must be

$$\mathbf{0} = (\mathcal{A} - \lambda_i \mathcal{I})^{m_i} \mathbf{v} = \sum_{h=0}^{m_i} \binom{m_i}{h} (-1)^h \lambda_i^h \mathcal{A}^{m_i-h} \mathbf{v} = \sum_{h=0}^{m_i} \binom{m_i}{h} (-1)^h \lambda_i^h \mu^{m_i-h} \mathbf{v} = (\mu - \lambda_i)^{m_i} \mathbf{v}$$

which leads to $\lambda_i = \mu$. The characteristic polynomial of the transformation $\mathcal{A}$ restricted to $\mathbf{N}_{\lambda_i}$ is then $(s - \lambda_i)^{\dim \mathbf{N}_{\lambda_i}}$ and that of $\mathcal{A} : \mathbf{V} \rightarrow \mathbf{V}$ is

$$\Delta_{\mathcal{A}}(s) = (s - \lambda_1)^{\dim \mathbf{N}_{\lambda_1}} (s - \lambda_2)^{\dim \mathbf{N}_{\lambda_2}} \dots (s - \lambda_t)^{\dim \mathbf{N}_{\lambda_t}}, \quad (\text{A.65})$$

being the generalized eigenspaces summand in (A.60) $\mathcal{A}$ -invariant. By comparing expressions (A.53) and (A.65) of the characteristic polynomial, it follows that $n_i = \dim(\mathbf{N}_{\lambda_i})$, $i = 1, 2, \dots, t$. ■

Note that, as a follow up of the previous proposition, the multiplicity of an eigenvalue λ_i in the minimal polynomial can be obtained by computing the first power of transformation $(\mathcal{A} - \lambda_i \mathcal{I})$ starting from which the chain (A.61) becomes stationary.

Example A.7.1 Consider the transformation

$$\mathcal{A} : \mathbb{R}^5 \rightarrow \mathbb{R}^5 : \begin{bmatrix} \xi_1 \\ \xi_2 \\ \xi_3 \\ \xi_4 \\ \xi_5 \end{bmatrix} \mapsto \begin{bmatrix} 0 & 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \xi_1 \\ \xi_2 \\ \xi_3 \\ \xi_4 \\ \xi_5 \end{bmatrix} = A \begin{bmatrix} \xi_1 \\ \xi_2 \\ \xi_3 \\ \xi_4 \\ \xi_5 \end{bmatrix}.$$

The characteristic polynomial of $\mathcal{A}$ is $s^5 - s^4$, and therefore the eigenvalues are $\lambda_1 = 1$, with algebraic multiplicity $n_1 = 1$, and $\lambda_2 = 0$, with algebraic multiplicity $n_2 = 4$.

The eigenvalue λ_1 has clearly unitary multiplicity also in the minimal polynomial, and the eigenspace $\mathbf{U}_{\lambda_1}$ coincides with the generalized eigenspace $\mathbf{N}_{\lambda_1}$, yielding

$$\mathbf{U}_{\lambda_1} = \ker(A - \lambda_1 I_5) = \ker \begin{bmatrix} -1 & 0 & 0 & 1 & 1 \\ 1 & -1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 1 \\ 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} = \text{span} \left(\begin{bmatrix} 1 \\ 1 \\ 1 \\ 0 \\ 1 \end{bmatrix} \right)$$

The eigenvalue $\lambda_2 = 0$ is associated to the eigenspace

$$\mathbf{U}_{\lambda_2} = \ker(A - \lambda_2 I_5) = \ker \begin{bmatrix} 0 & 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} = \text{span} \left(\begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} \right) = \text{span}(\mathbf{e}_2, \mathbf{e}_3)$$

and the kernels that follow

$$\begin{aligned} \ker(A - \lambda_2 I_5)^2 &= \ker \begin{bmatrix} 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} = \text{span} \left(\begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} \right) = \text{span}(\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3) \\ \ker(A - \lambda_2 I_5)^3 &= \ker \begin{bmatrix} 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} = \text{span} \left(\begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} \right) = \text{span}(\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3, \mathbf{e}_4) \end{aligned}$$

$$\ker(A - \lambda_2 I_5)^4 = \ker(A - \lambda_2 I_5)^3.$$

Since $\mathbf{N}_{\lambda_2} = \ker(A - \lambda_2 I_5)^{m_2} = \ker(A - \lambda_2 I_5)^3 = \text{span}(\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3, \mathbf{e}_4)$, it can be stated that the multiplicity of λ_2 in the minimal polynomial is 3, while its algebraic multiplicity is 4 (coinciding with the dimension of $\mathbf{N}_{\lambda_2}$).

- EXERCISE A.7.1 [SYMMETRIC MATRICES] (i) The eigenvalues of a symmetric matrix $A \in \mathbb{R}^{n \times n}$ are simple roots of the minimal polynomial.
(ii) Employ the statement above to show that any real symmetric matrix is diagonalizable by similarity.
‡ *Hint: (i) The eigenvalues of A are real (Exercise A.5.1) and from $(A - \lambda I_n)^2 \mathbf{x} = \mathbf{0}$ it follows that $\mathbf{x}^T (A - \lambda I_n)^T (A - \lambda I_n) \mathbf{x} = \mathbf{0}$. Hence, $(A - \lambda I_n) \mathbf{x} = \mathbf{0}$.*

Taking in $\mathbf{V}$ a basis obtained by juxtaposing an arbitrary basis for $\mathbf{N}_{\lambda_1}$, an arbitrary basis for $\mathbf{N}_{\lambda_2}$, $\dots$, an arbitrary basis for $\mathbf{N}_{\lambda_t}$, and by taking into account the $\mathcal{A}$ -invariance of the subspaces $\mathbf{N}_{\lambda_i}$, the matrix representing $\mathcal{A}$ takes a block diagonal structure, with i -th diagonal block of dimension $n_i \times n_i$.

The procedure we describe aims at building in each generalized eigenspace $\mathbf{N}_{\lambda_i}$ - and then in $\mathbf{V}$ - a particular basis, obtained by juxtaposition of specific “Jordan chains”, according to the following scheme

$$\underbrace{\underbrace{\begin{array}{c} \star \dots \star \\ \text{chain} \end{array}}_{\text{basis of } \mathbf{N}_{\lambda_1}} \underbrace{\begin{array}{c} \star \dots \star \\ \text{chain} \end{array}}_{\text{basis of } \mathbf{N}_{\lambda_2}} \dots \underbrace{\begin{array}{c} \star \dots \star \\ \text{chain} \end{array}}_{\text{basis of } \mathbf{N}_{\lambda_t}}}_{\text{basis of } \mathbf{V}} \quad (\text{A.66})$$

Each chain gives rise to an $\mathcal{A}$ -invariant subspace, which in the matrix representation of $\mathcal{A}$ is associated to a diagonal block, called “Jordan mini-block”.

We will limit the discussion to a single generalized eigenspace $\mathbf{N}_{\lambda}$, which can be obtained as the first stationary element of the subspace chain

$$\ker(\mathcal{A} - \lambda \mathcal{I}) \subset \ker(\mathcal{A} - \lambda \mathcal{I})^2 \subset \dots \subset \ker(\mathcal{A} - \lambda \mathcal{I})^{\tilde{m}} = \dots \quad (\text{A.67})$$

The exponent $\tilde{m}$ represents the multiplicity of λ as a zero of the minimal polynomial $\psi_{\mathcal{A}}(s)$.

Definition A.7.2 [GENERALIZED EIGENVECTOR] *A non-null vector $\mathbf{v} \in \mathbf{N}_{\lambda}$ is said to be a generalized eigenvector of order k corresponding to the eigenvalue λ if*

$$\mathbf{v} \in \ker(\mathcal{A} - \lambda \mathcal{I})^k \quad \text{but} \quad \mathbf{v} \notin \ker(\mathcal{A} - \lambda \mathcal{I})^{k-1},$$

namely if the smallest subspace of the sequence (A.67) containing $\mathbf{v}$ is $\ker(\mathcal{A} - \lambda \mathcal{I})^k$.

For each generalized eigenvector $\mathbf{v}^{(k)}$ of order k corresponding to the eigenvalue λ , it can be found in $\mathbf{N}_\lambda$ a chain of generalized eigenvectors of decreasing order, called *Jordan chain*, by choosing

$$\begin{aligned} \mathbf{v}^{(k)} & \\ \mathbf{v}^{(k-1)} &:= (\mathcal{A} - \lambda\mathcal{I})\mathbf{v}^{(k)} \\ \mathbf{v}^{(k-2)} &:= (\mathcal{A} - \lambda\mathcal{I})\mathbf{v}^{(k-1)} = (\mathcal{A} - \lambda\mathcal{I})^2\mathbf{v}^{(k)} \\ &\vdots \\ \mathbf{v}^{(1)} &:= (\mathcal{A} - \lambda\mathcal{I})\mathbf{v}^{(2)} = (\mathcal{A} - \lambda\mathcal{I})^{k-1}\mathbf{v}^{(k)} \end{aligned} \quad (\text{A.68})$$

- EXERCISE A.7.2 (i) The minimal polynomial of a generalized eigenvector of order h corresponding to the eigenvalue λ is $(s - \lambda)^h$.
(ii) If $\mathbf{v}^{(k)}$ is a generalized eigenvector of order k and $\mathbf{w}^{(h)}$ one of order h , corresponding to the same eigenvalue but not necessarily to the same chain, $\mathbf{v}^{(k)} + \mathbf{w}^{(h)}$ is again a generalized eigenvector. Of which order?

Proposition A.7.3 [JORDAN CHAIN] *Let $\mathbf{v}^{(k)}, \mathbf{v}^{(k-1)}, \dots, \mathbf{v}^{(1)}$ be a Jordan chain corresponding to the eigenvalue λ of the transformation $\mathcal{A} : \mathbf{V} \rightarrow \mathbf{V}$. Then*

- (i) *the vectors $\mathbf{v}^{(k)}, \mathbf{v}^{(k-1)}, \dots, \mathbf{v}^{(1)}$ are linearly independent;*
- (ii) *the subspace $\mathbf{C} := \text{span}\{\mathbf{v}^{(k)}, \mathbf{v}^{(k-1)}, \dots, \mathbf{v}^{(1)}\} \subseteq \mathbf{N}_\lambda \subseteq \mathbf{V}$ is $\mathcal{A}$ -invariant;*
- (iii) *if in $\mathbf{C}$ we take the ordered basis $(\mathbf{v}^{(1)}, \mathbf{v}^{(2)}, \dots, \mathbf{v}^{(k)})$, the restriction of $\mathcal{A}$ to $\mathbf{C}$ is represented by the $k \times k$ Jordan mini-block*

$$\begin{bmatrix} \lambda & 1 & & & \\ & \lambda & 1 & & \\ & & \ddots & \ddots & \\ & & & \lambda & 1 \\ & & & & \lambda \end{bmatrix}. \quad (\text{A.69})$$

PROOF (i) Set

$$\alpha_k \mathbf{v}^{(k)} + \alpha_{k-1} \mathbf{v}^{(k-1)} + \dots + \alpha_1 \mathbf{v}^{(1)} = \mathbf{0}. \quad (\text{A.70})$$

By applying the linear transformation $(\mathcal{A} - \lambda\mathcal{I})^{k-1}$ to both sides of (A.70) one gets $\alpha_k \mathbf{v}^{(1)} = \mathbf{0}$ and so α_k is null, being the vector $\mathbf{v}^{(1)}$ non null. By applying then to (A.70), respectively $(\mathcal{A} - \lambda\mathcal{I})^{k-2}, (\mathcal{A} - \lambda\mathcal{I})^{k-3}, \dots$ it follows that also $\alpha_{k-1}, \alpha_{k-2}, \dots, \alpha_1$ are null.

(ii) From (A.68) it follows

$$\mathcal{A}\mathbf{v}^{(1)} = \lambda\mathbf{v}^{(1)}; \quad \mathcal{A}\mathbf{v}^{(h)} = \lambda\mathbf{v}^{(h)} + \mathbf{v}^{(h-1)}, \quad h = 2, 3, \dots, k. \quad (\text{A.71})$$

Since the $\mathcal{A}$ -transform of each vector of the Jordan chain belongs to the space generated by the chain, this latter is $\mathcal{A}$ -invariant.

(iii) follows straightforwardly from (A.71). ■

Proposition A.7.4 [INDEPENDENCE OF JORDAN CHAINS] *If the generalized eigenvectors of order k corresponding to the eigenvalue λ , $\mathbf{v}_\nu^{(k)}$, $\nu = 1, 2, \dots, N$, are linearly independent³ with respect to the subspace $\ker(\mathcal{A} - \lambda\mathcal{I})^{k-1}$, then the generalized eigenvectors*

$$\mathbf{v}_\nu^{(k-1)} := (\mathcal{A} - \lambda\mathcal{I})\mathbf{v}_\nu^{(k)}, \quad \nu = 1, 2, \dots, N$$

are linearly independent with respect to the subspace $\ker(\mathcal{A} - \lambda\mathcal{I})^{k-2}$.

³Equivalently, according to Definition A.1.7, no non-null linear combination of vectors $\mathbf{v}_\nu^{(k)}$, $\nu = 1, 2, \dots, N$, belongs to $\ker(\mathcal{A} - \lambda\mathcal{I})^{k-1}$, namely the vector space generated by $\ker(\mathcal{A} - \lambda\mathcal{I})^{k-1}$ and by $\mathbf{v}_\nu^{(k)}$, $\nu = 1, 2, \dots, N$, has dimension $N + \dim \ker(\mathcal{A} - \lambda\mathcal{I})^{k-1}$.

PROOF If the linear combination

$$\alpha_1 \mathbf{v}_1^{(k-1)} + \dots + \alpha_N \mathbf{v}_N^{(k-1)} = \alpha_1 (\mathcal{A} - \lambda \mathcal{I}) \mathbf{v}_1^{(k)} + \dots + \alpha_N (\mathcal{A} - \lambda \mathcal{I}) \mathbf{v}_N^{(k)} \quad (\text{A.72})$$

belongs to $\ker(\mathcal{A} - \lambda \mathcal{I})^{k-2}$, then

$$\mathbf{0} = (\mathcal{A} - \lambda \mathcal{I})^{k-2} \left(\alpha_1 (\mathcal{A} - \lambda \mathcal{I}) \mathbf{v}_1^{(k)} + \dots + \alpha_N (\mathcal{A} - \lambda \mathcal{I}) \mathbf{v}_N^{(k)} \right) = (\mathcal{A} - \lambda \mathcal{I})^{k-1} \left(\alpha_1 \mathbf{v}_1^{(k)} + \dots + \alpha_N \mathbf{v}_N^{(k)} \right)$$

implies that the vector $\alpha_1 \mathbf{v}_1^{(k)} + \dots + \alpha_N \mathbf{v}_N^{(k)}$ belongs to $\ker(\mathcal{A} - \lambda \mathcal{I})^{k-1}$. Since $\mathbf{v}_\nu^{(k)}$, $\nu = 1, 2, \dots, N$, are independent with respect to subspace $\ker(\mathcal{A} - \lambda \mathcal{I})^{k-1}$, the combinators $\alpha_1, \dots, \alpha_N$ are all null. ■

As a follow-up of the proposition before, it is always true that

$$\dim \ker(\mathcal{A} - \lambda \mathcal{I})^k - \dim \ker(\mathcal{A} - \lambda \mathcal{I})^{k-1} \geq \dim \ker(\mathcal{A} - \lambda \mathcal{I})^{k+1} - \dim \ker(\mathcal{A} - \lambda \mathcal{I})^k,$$

namely the increase of dimension along the subspace sequence (A.67) is not growing as the exponent increases.

A.8 Basis and Jordan normal form

We can now describe the construction of a *Jordan basis* for the generalized eigenspace $\mathbf{N}_\lambda$ given by (A.67). A basis for $\mathbf{V}$ will be obtained by juxtaposition of the bases of the generalized eigenspaces, according to the scheme (A.66).

- If $d_{\tilde{m}}$ is the difference between the dimensions of $\ker(\mathcal{A} - \lambda \mathcal{I})^{\tilde{m}}$ and $\ker(\mathcal{A} - \lambda \mathcal{I})^{\tilde{m}-1}$, the first step consists in searching for a set of $d_{\tilde{m}}$ generalized eigenvectors of order $\tilde{m}$, independent with respect to $\ker(\mathcal{A} - \lambda \mathcal{I})^{\tilde{m}-1}$

$$\mathbf{v}_1^{(\tilde{m})}, \dots, \mathbf{v}_{d_{\tilde{m}}}^{(\tilde{m})}. \quad (\text{A.73})$$

Note that the union of the set (A.73) with a basis $\mathcal{B}^{(\tilde{m}-1)}$ for $\ker(\mathcal{A} - \lambda \mathcal{I})^{\tilde{m}-1}$ provides a basis for $\ker(\mathcal{A} - \lambda \mathcal{I})^{\tilde{m}}$.

- As a second step, the generalized eigenvectors of order $\tilde{m} - 1$ are considered

$$\mathbf{v}_1^{(\tilde{m}-1)} := (\mathcal{A} - \lambda \mathcal{I}) \mathbf{v}_1^{(\tilde{m})}, \quad \dots, \quad \mathbf{v}_{d_{\tilde{m}}}^{(\tilde{m}-1)} := (\mathcal{A} - \lambda \mathcal{I}) \mathbf{v}_{d_{\tilde{m}}}^{(\tilde{m})}.$$

By Proposition A.7.4 they are independent with respect to $\ker(\mathcal{A} - \lambda \mathcal{I})^{\tilde{m}-2}$. If the difference $d_{(\tilde{m}-1)}$ between the dimensions of $\ker(\mathcal{A} - \lambda \mathcal{I})^{\tilde{m}-1}$ and $\ker(\mathcal{A} - \lambda \mathcal{I})^{\tilde{m}-2}$ is higher than $d_{\tilde{m}}$, $d_{(\tilde{m}-1)} - d_{\tilde{m}}$ additional generalized eigenvectors are found in $\ker(\mathcal{A} - \lambda \mathcal{I})^{\tilde{m}-1}$, $\mathbf{v}_{d_{\tilde{m}}+1}^{(\tilde{m}-1)}, \dots, \mathbf{v}_{d_{(\tilde{m}-1)}}^{(\tilde{m}-1)}$ of order $\tilde{m} - 1$, such that

$$\mathbf{v}_1^{(\tilde{m}-1)}, \dots, \mathbf{v}_{d_{\tilde{m}}}^{(\tilde{m}-1)}, \mathbf{v}_{d_{\tilde{m}}+1}^{(\tilde{m}-1)}, \dots, \mathbf{v}_{d_{(\tilde{m}-1)}}^{(\tilde{m}-1)} \quad (\text{A.74})$$

are a maximal set of generalized eigenvectors of order $\tilde{m} - 1$, independent with respect to $\ker(\mathcal{A} - \lambda \mathcal{I})^{\tilde{m}-2}$.

By adding the vectors of (A.74) to a basis $\mathcal{B}^{(\tilde{m}-2)}$ for $\ker(\mathcal{A} - \lambda \mathcal{I})^{\tilde{m}-2}$, a basis for $\ker(\mathcal{A} - \lambda \mathcal{I})^{\tilde{m}-1}$ is obtained.

- If $d_{(\tilde{m}-2)}$ is the difference between the dimensions of $\ker(\mathcal{A} - \lambda \mathcal{I})^{\tilde{m}-2}$ and $\ker(\mathcal{A} - \lambda \mathcal{I})^{\tilde{m}-3}$, the $d_{(\tilde{m}-1)}$ generalized eigenvectors of order $\tilde{m} - 2$

$$\mathbf{v}_1^{(\tilde{m}-2)} := (\mathcal{A} - \lambda \mathcal{I}) \mathbf{v}_1^{(\tilde{m}-1)}, \quad \dots, \quad \mathbf{v}_{d_{(\tilde{m}-1)}}^{(\tilde{m}-2)} := (\mathcal{A} - \lambda \mathcal{I}) \mathbf{v}_{d_{(\tilde{m}-1)}}^{(\tilde{m}-1)}$$

are complemented with other $d_{(\tilde{m}-2)} - d_{(\tilde{m}-1)}$ generalized eigenvectors of the same order, such that a set is obtained

$$\mathbf{v}_1^{(\tilde{m}-2)}, \dots, \mathbf{v}_{d_{(\tilde{m}-1)}}^{(\tilde{m}-2)}, \mathbf{v}_{d_{(\tilde{m}-1)}+1}^{(\tilde{m}-2)}, \dots, \mathbf{v}_{d_{(\tilde{m}-2)}}^{(\tilde{m}-2)} \quad (\text{A.75})$$

independent with respect to $\ker(\mathcal{A} - \lambda\mathcal{I})^{\tilde{m}-3}$.

By adding the vectors of (A.75) to a basis $\mathcal{B}^{(\tilde{m}-3)}$ for $\ker(\mathcal{A} - \lambda\mathcal{I})^{\tilde{m}-3}$, a basis for $\ker(\mathcal{A} - \lambda\mathcal{I})^{\tilde{m}-2}$ is obtained.

- At the $\tilde{m}$ -th step of the procedure, by adding to the eigenvectors

$$\mathbf{v}_1^{(1)} := (\mathcal{A} - \lambda\mathcal{I})\mathbf{v}_1^{(2)}, \dots, \mathbf{v}_{\tilde{d}_2}^{(1)} := (\mathcal{A} - \lambda\mathcal{I})\mathbf{v}_{\tilde{d}_2}^{(2)}$$

other eigenvectors $\mathbf{v}_{\tilde{d}_2+1}^{(1)}, \dots, \mathbf{v}_{\dim \mathbf{U}_\lambda}^{(1)}$ a basis is obtained

$$\mathbf{v}_1^{(1)}, \dots, \mathbf{v}_{\tilde{d}_2}^{(1)}, \mathbf{v}_{\tilde{d}_2+1}^{(1)}, \dots, \mathbf{v}_{\dim \mathbf{U}_\lambda}^{(1)} \quad (\text{A.76})$$

for the eigenspace $\mathbf{U}_\lambda = \ker(\mathcal{A} - \lambda\mathcal{I})$.

The set of the generalized eigenvectors (A.73), (A.74), (A.75), $\dots$, (A.76), of order $\tilde{m}, \tilde{m}-1, \tilde{m}-2, \dots, 1$, built in the subsequent steps of the procedure, provides a basis for $\mathbf{N}_\lambda$. The following diagram highlights the construction of the basis in $\mathbf{N}_\lambda$ just described: along the first row there are the generalized eigenvectors of maximum order, chosen at the first step, along the second row those of immediately lower order, obtained by applying the transformation $\mathcal{A} - \lambda\mathcal{I}$ to the previous eigenvectors and completing the list with $d_{(\tilde{m}-1)} - d_{\tilde{m}}$ other vectors, and so on.

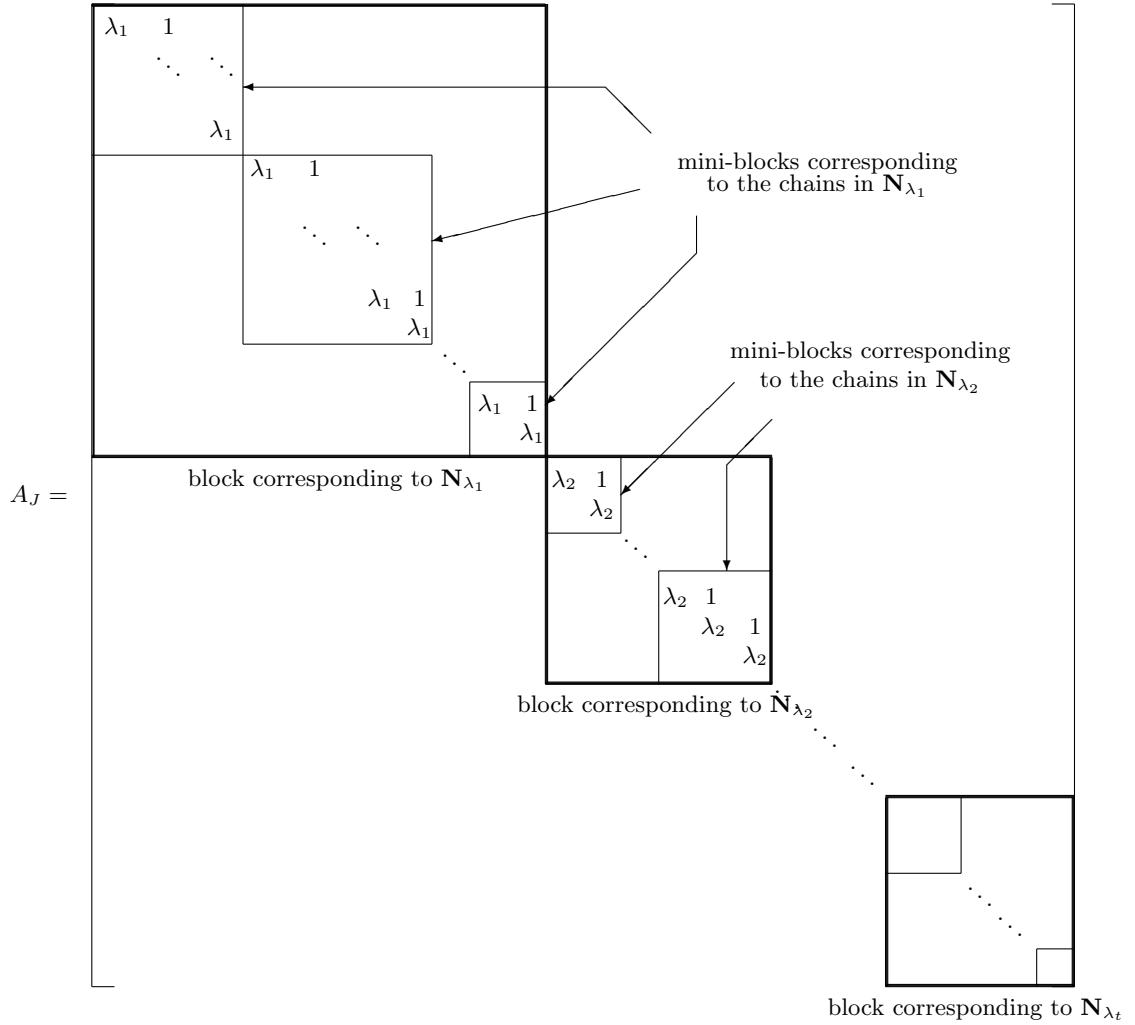
The columns of the diagram are *Jordan chains*, as many as the dimension of $\mathbf{U}_\lambda$, and each of them is as long as the order of the generalized eigenvector that initialized it.

$$\begin{array}{cccccccc}
 & \mathbf{v}_1^{(\tilde{m})} & \dots & \mathbf{v}_{d_{\tilde{m}}}^{(\tilde{m})} & & & & \\
 \mathcal{A} - \lambda\mathcal{I} & \downarrow & & \downarrow & & & & \\
 & \mathbf{v}_1^{(\tilde{m}-1)} & \dots & \mathbf{v}_{d_{\tilde{m}}}^{(\tilde{m}-1)} & \mathbf{v}_{d_{\tilde{m}}+1}^{(\tilde{m}-1)} & \dots & \mathbf{v}_{d_{(\tilde{m}-1)}}^{(\tilde{m}-1)} & \\
 \mathcal{A} - \lambda\mathcal{I} & \downarrow & & \downarrow & \downarrow & & \downarrow & \\
 & \mathbf{v}_1^{(\tilde{m}-2)} & \dots & \mathbf{v}_{d_{\tilde{m}}}^{(\tilde{m}-2)} & \mathbf{v}_{d_{\tilde{m}}+1}^{(\tilde{m}-2)} & \dots & \mathbf{v}_{d_{(\tilde{m}-1)}}^{(\tilde{m}-2)} & \mathbf{v}_{d_{(\tilde{m}-1)}+1}^{(\tilde{m}-2)} \dots \mathbf{v}_{d_{(\tilde{m}-2)}}^{(\tilde{m}-2)} \\
 \mathcal{A} - \lambda\mathcal{I} & \downarrow & & \downarrow & \downarrow & & \downarrow & \downarrow \\
 & \vdots & & \vdots & \vdots & & \vdots & \vdots \\
 \mathcal{A} - \lambda\mathcal{I} & \downarrow & & \downarrow & \downarrow & & \downarrow & \downarrow \\
 & \mathbf{v}_1^{(1)} & \dots & \mathbf{v}_{d_{\tilde{m}}}^{(1)} & \mathbf{v}_{d_{\tilde{m}}+1}^{(1)} & \dots & \mathbf{v}_{d_{(\tilde{m}-1)}}^{(1)} & \mathbf{v}_{d_{(\tilde{m}-1)}+1}^{(1)} \dots \mathbf{v}_{d_{(\tilde{m}-2)}}^{(1)} & \mathbf{v}_{d_{(\tilde{m}-2)}+1}^{(1)} \dots \mathbf{v}_{\dim \mathbf{U}_\lambda}^{(1)}
 \end{array}$$

To get to a matrix representation of $\mathcal{A}$, we need to adopt an order in the basis of $\mathbf{N}_\lambda$. The *Jordan basis* for $\mathbf{N}_\lambda$ can be obtained by juxtaposition of the chains that we have obtained, sorting each of them starting from the proper eigenvector and ending with the generalized eigenvector (following the inverse order with respect to the one according to which the vectors have been obtained).

Since the subspace generated by each Jordan chain is $\mathcal{A}$ -invariant, the restriction of $\mathcal{A}$ to $\mathbf{N}_\lambda$ is represented, in the Jordan basis, by a block diagonal matrix (*Jordan mini-blocks*) that contains as many mini-blocks as the chains are. Each mini-block has dimension equal to the length of the chain it corresponds to and has the structure (A.69).

The matrix A_J associated to the transformation $\mathcal{A}$ in the Jordan basis can be partitioned into diagonal blocks according to two levels:



- at a raw level, it is made by diagonal blocks, one for each eigenvalue λ_i of $\mathcal{A}$, of dimension n_i equal to the algebraic multiplicity of λ_i ;
- at a finer level, each diagonal block obtained in this way is the direct sum of Jordan

mini-blocks⁴, one for each chain in the Jordan basis.

Remark The dimension of the mini-block of maximum order corresponding to the eigenvalue λ_i coincides with the length of the longest Jordan chain corresponding to the eigenvalue λ_i and hence with the multiplicity m_i of λ_i as a root of the minimal polynomial. The number of mini-blocks associated with λ_i is the number of chains that belong to the Jordan basis, hence it coincides with the dimension of the eigenspace $\mathbf{U}_{\lambda_i}$.

The structure of the matrix A_J representing $\mathcal{A}$ with respect to a Jordan basis is unique, up to a permutation of the mini-blocks, meaning that it does not depend on the specific Jordan basis that has been chosen. Indeed, for each eigenvalue λ_i , the set of integers that represent the chain lengths and thus the set of mini-blocks of A_J is uniquely determined. The mini-blocks may only appear in different diagonal locations inside the matrix, according to the order after which the chains are listed in the selected basis.

Therefore the matrix A_J is called the *Jordan normal form* and on the complex field any square matrix is similar to one and only Jordan normal form (up to a mini-block permutation).

- EXERCISE A.8.1 (i) Given a linear transformation $\mathcal{A} : \mathbf{V} \rightarrow \mathbf{V}$, the following facts are equivalent:
 - the Jordan normal form A_J has a single mini-block for each eigenvalue;
 - every Jordan basis has a single chain for each eigenvalue;
 - in all the sequences (A.61) the dimension increase from one kernel to the following one is unitary;
 - $\psi_{\mathcal{A}}(s) = \Delta_{\mathcal{A}}(s)$;
 - there exists a vector $\mathbf{v}$ (*cyclic* vector) such that the vectors $\mathcal{A}^i \mathbf{v}$, $i = 0, 1, 2, \dots$, generate the space $\mathbf{V}$. In these conditions the transformation $\mathcal{A}$ and the matrices that represent it are called *cyclic* (see also paragraph A.10).
- (ii) The following facts are equivalent:
 - the Jordan normal form A_J has only mini-blocks of unitary dimension;
 - $\mathbf{U}_{\lambda_i} = \mathbf{N}_{\lambda_i}$, $\forall i$;
 - on the complex field the minimal polynomial $\psi_{\mathcal{A}}(s)$ has only simple zeros.

In these conditions the transformation $\mathcal{A}$ and the matrices that represent it are said to have *simple structure* (for more details, see also Paragraph A.9).

- **Example A.8.1** Let us consider again Example A.7.1. We have

$$\begin{aligned} \mathbf{U}_{\lambda_1} &= \text{span} \left(\begin{bmatrix} 1 \\ 1 \\ 1 \\ 0 \\ 1 \end{bmatrix} \right) = \text{span}(\mathbf{w}^{(1)}) = \mathbf{N}_{\lambda_1} \\ \mathbf{U}_{\lambda_2} &= \ker(A - \lambda_2 I_5) = \text{span}(\mathbf{e}_2, \mathbf{e}_3) \\ &\subset \ker(A - \lambda_2 I_5)^2 = \text{span}(\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3) \end{aligned}$$

⁴If the elements of a Jordan chain are taken in the reverse order with respect to the one here considered, namely starting from the maximum order generalized eigenvector to end with the proper eigenvector, the restriction of $\mathcal{A}$ to the space generated by the chain is represented by the mini-block

$$\begin{bmatrix} \lambda & & & & \\ 1 & \lambda & & & \\ & \ddots & \ddots & & \\ & & 1 & \lambda & \\ & & & 1 & \lambda \end{bmatrix}$$

$$\subset \ker(A - \lambda_2 I_5)^3 = \text{span}(\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3, \mathbf{e}_4) = \mathbf{N}_{\lambda_2}.$$

1. The difference between the dimensions of $\ker(A - \lambda_2 I_5)^3$ and of $\ker(A - \lambda_2 I_5)^2$ is 1 and the vector $\mathbf{v}_1^{(3)} := \mathbf{e}_4 \in \ker(A - \lambda_2 I_5)^3$ is independent of $\ker(A - \lambda_2 I_5)^2$. Then, by adding $\mathbf{v}_1^{(3)}$ to a basis for $\ker(A - \lambda_2 I_5)^2$, a basis for $\ker(A - \lambda_2 I_5)^3$ is obtained.
2. Let us have $\mathbf{v}_1^{(2)} = (A - \lambda_2 I_5)\mathbf{v}_1^{(3)} = \mathbf{e}_1$. Since the difference between the dimensions of $\ker(A - \lambda_2 I_5)^2$ and of $\ker(A - \lambda_2 I_5)$ is 1, by adding $\mathbf{v}_1^{(2)}$ to a basis for $\ker(A - \lambda_2 I_5)$ a basis for $\ker(A - \lambda_2 I_5)^2$ is obtained.
3. Let us have $\mathbf{v}_1^{(1)} = (A - \lambda_2 I_5)\mathbf{v}_1^{(2)} = \mathbf{e}_2$. Since the dimension of $\ker(A - \lambda_2 I_5)$ is 2, with $\mathbf{v}_2^{(1)} := \mathbf{e}_3$ we have that $\{\mathbf{v}_1^{(1)}, \mathbf{v}_2^{(1)}\}$ is a basis for the eigenspace $\ker(A - \lambda_2 I_5)$.
4. A Jordan basis is therefore

$$\left(\underbrace{\underbrace{\mathbf{w}^{(1)}}_{\text{chain associated with } \lambda_1}}_{\text{basis for } \mathbf{N}_{\lambda_1}}, \underbrace{\underbrace{\mathbf{v}_1^{(1)}, \mathbf{v}_1^{(2)}, \mathbf{v}_1^{(3)}}_{\text{chain associated with } \lambda_2}, \underbrace{\mathbf{v}_2^{(1)}}_{\text{chain associated with } \lambda_2}}_{\text{basis for } \mathbf{N}_{\lambda_2}} \right), \quad (\text{A.77})$$

and with respect to such basis the linear transformation is represented by the matrix

$$A_J = \left[\begin{array}{c|ccc|c} 1 & & 0 & 0 & 0 & 0 \\ \hline 0 & & 0 & 1 & 0 & 0 \\ 0 & & 0 & 0 & 1 & 0 \\ 0 & & 0 & 0 & 0 & 0 \\ \hline 0 & & 0 & 0 & 0 & 0 \end{array} \right].$$

5. The transformation matrix from the “old” canonical basis in $\mathbb{R}^5$ to the “new” Jordan basis has columns composed by the (components of the) vectors $\mathbf{w}^{(1)}, \mathbf{v}_1^{(1)}, \mathbf{v}_1^{(2)}, \mathbf{v}_1^{(3)}, \mathbf{v}_2^{(1)}$ and hence it is

$$T = [\mathbf{w}^{(1)} \quad \mathbf{e}_2 \quad \mathbf{e}_1 \quad \mathbf{e}_4 \quad \mathbf{e}_3] = \begin{bmatrix} 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 0 \end{bmatrix}$$

and the similarity transformation on the matrix A

$$T^{-1}AT = \begin{bmatrix} 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & -1 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 0 \end{bmatrix}$$

yields the Jordan form A_J .

A.9 Real Jordan form

Definition A.9.1 [SIMPLE STRUCTURE TRANSFORMATIONS] *Let $\mathbf{V}$ be a vector space over $\mathbb{C}$. A linear transformation $\mathcal{A} : \mathbf{V} \rightarrow \mathbf{V}$ is said to have simple structure if $\mathbf{V}$ has a basis made of eigenvectors of $\mathcal{A}$.*

$\mathcal{A}$ is of a simple structure if and only if the matrix that represents it is diagonalizable by similarity. Notice that, even if the matrix A that represents the transformation $\mathcal{A}$ is real, diagonalization may lead to complex elements (if some eigenvalues are complex) and the matrix T that induces the similarity transformation can have complex values. In these

cases, instead of diagonalizing a matrix by acting on the complex field, it can be useful to get a block diagonal real matrix, with 2-dimensional blocks, by resorting to a similarity transformation induced by a real matrix.

Let A be the real matrix representing the linear transformation of simple structure $\mathcal{A}$ with respect to a basis $(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n)$ in $\mathbf{V}$. The characteristic polynomial of $\mathcal{A}$ has real coefficients and if $\lambda = \sigma + j\omega$ is a complex eigenvalue of $\mathcal{A}$, with algebraic multiplicity n_λ , also the complex conjugate $\check{\lambda} = \sigma - j\omega$ is an eigenvalue, with the same multiplicity.

If the eigenvector $\mathbf{v}$, corresponding to the eigenvalue λ , is represented with respect to the

basis by the complex n -tuple $\mathbf{x} = \begin{bmatrix} \xi_1 \\ \vdots \\ \xi_n \end{bmatrix}$, the conjugate n -tuple $\check{\mathbf{x}} = \begin{bmatrix} \check{\xi}_1 \\ \vdots \\ \check{\xi}_n \end{bmatrix}$ represents an

eigenvector, which will be denoted with $\check{\mathbf{v}}$, corresponding to the conjugate eigenvalue $\check{\lambda}$.

From the relations

$$\begin{aligned} A\mathbf{x} &= (\sigma + j\omega)\mathbf{x} \\ A\check{\mathbf{x}} &= (\sigma - j\omega)\check{\mathbf{x}}, \end{aligned} \quad (\text{A.78})$$

by setting

$$\mathbf{r} := \Re(\mathbf{x}) = \frac{\mathbf{x} + \check{\mathbf{x}}}{2}, \quad \mathbf{c} := \Im(\mathbf{x}) = \frac{\mathbf{x} - \check{\mathbf{x}}}{2j} \quad (\text{A.79})$$

it immediately follows

$$\begin{aligned} A\mathbf{r} &= A \frac{\mathbf{x} + \check{\mathbf{x}}}{2} = \frac{\sigma + j\omega}{2}\mathbf{x} + \frac{\sigma - j\omega}{2}\check{\mathbf{x}} = \sigma\mathbf{r} - \omega\mathbf{c} \\ A\mathbf{c} &= A \frac{\mathbf{x} - \check{\mathbf{x}}}{2j} = \frac{\sigma + j\omega}{2j}\mathbf{x} - \frac{\sigma - j\omega}{2j}\check{\mathbf{x}} = \omega\mathbf{r} + \sigma\mathbf{c} \end{aligned} \quad (\text{A.80})$$

The transformed through A of the real vectors $\mathbf{r}$ and $\mathbf{c}$ can be both expressed as combinations with real coefficients of $\mathbf{r}$ and $\mathbf{c}$ (note that neither $\mathbf{r}$, nor $\mathbf{c}$ are eigenvectors of A).

According to (A.80), $A\mathbf{r}$ and $A\mathbf{c}$ belong to the subspace $\text{span}\{\mathbf{r}, \mathbf{c}\}$, which is then A -invariant. As it is a subspace on $\mathbb{C}$, it coincides with the one generated by $\mathbf{x}$ and $\check{\mathbf{x}}$, being valid at the same time both (A.79) and $\mathbf{x} = \mathbf{r} + j\mathbf{c}$, $\check{\mathbf{x}} = \mathbf{r} - j\mathbf{c}$.

Proposition A.9.2 [REAL JORDAN BASIS AND FORM] *Suppose that the linear transformation $\mathcal{A} : \mathbf{V} \rightarrow \mathbf{V}$ is of simple structure and is represented by a real matrix A with respect to a basis in $\mathbf{V}$. Let $\lambda_1, \check{\lambda}_1, \dots, \lambda_k, \check{\lambda}_k$ be the complex eigenvalues of $\mathcal{A}$ and $\lambda_{2k+1}, \dots, \lambda_n$ the real ones, each of them counted as many times as its algebraic multiplicity, and let*

$$\mathbf{x}_1, \check{\mathbf{x}}_1, \dots, \mathbf{x}_k, \check{\mathbf{x}}_k, \mathbf{x}_{2k+1}, \dots, \mathbf{x}_n \quad (\text{A.81})$$

be the corresponding eigenvectors, represented as complex n -tuples with respect to the basis in $\mathbf{V}$.

- (i) *The vectors (A.81) are a basis with respect to which $\mathcal{A}$ is represented by a diagonal matrix;*
- (ii) *if we set $\mathbf{r}_i := (\mathbf{x}_i + \check{\mathbf{x}}_i)/2$ and $\mathbf{c}_i := (\mathbf{x}_i - \check{\mathbf{x}}_i)/2j$, $i = 1, \dots, k$, the n -uples*

$$\mathbf{r}_1, \mathbf{c}_1, \dots, \mathbf{r}_k, \mathbf{c}_k, \mathbf{x}_{2k+1}, \dots, \mathbf{x}_n \quad (\text{A.82})$$

are real and linearly independent over the real field (and over the complex field);

(iii) the (A.82) provide (the components with respect to the original basis of) a new basis, called *real characteristic basis*, with respect to which the transformation $\mathcal{A}$ is represented by the block diagonal matrix, called *real Jordan form*,

$$A_R = \text{diag}\left\{\begin{bmatrix} \sigma_1 & \omega_1 \\ -\omega_1 & \sigma_1 \end{bmatrix}, \dots, \begin{bmatrix} \sigma_k & \omega_k \\ -\omega_k & \sigma_k \end{bmatrix}, \lambda_{2k+1}, \dots, \lambda_n\right\}; \quad (\text{A.83})$$

(iv) The change of basis matrix is

$$T = [\mathbf{r}_1 \quad \mathbf{c}_1 \quad \dots \quad \mathbf{r}_k \quad \mathbf{c}_k \quad \mathbf{x}_{2k+1} \quad \dots \quad \mathbf{x}_n]$$

and one has that $A_R = T^{-1}AT$.

PROOF Only point (ii) needs to be verified. From $\mathbf{0} = \sum_{i=1}^k [\rho_i \mathbf{r}_i + \gamma_i \mathbf{c}_i] + \sum_{i=2k+1}^n \alpha_i \mathbf{x}_i$, $\rho_i, \gamma_i, \alpha_i \in \mathbb{R}$, it follows

$$\mathbf{0} = \sum_{i=1}^k \left[\rho_i \frac{\mathbf{x}_i + \check{\mathbf{x}}_i}{2} + \gamma_i \frac{\mathbf{x}_i - \check{\mathbf{x}}_i}{2j} \right] + \sum_{i=2k+1}^n \alpha_i \mathbf{x}_i = \sum_{i=1}^k \left[\frac{\rho_i - j\gamma_i}{2} \mathbf{x}_i + \frac{\rho_i + j\gamma_i}{2} \check{\mathbf{x}}_i \right] + \sum_{i=2k+1}^n \alpha_i \mathbf{x}_i$$

which implies $\rho_i = \gamma_i = \alpha_i = 0, \forall i$. ■

Example A.9.1 Let us consider the matrix

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 2 & 1 & 1 \\ 1 & -1 & 2 \end{bmatrix}$$

The characteristic polynomial is $\Delta_A(s) = s^3 - 4s^2 + 5s$ and the eigenvalues are $\lambda_1 = 2 + j$, $\lambda_2 = 2 - j$, $\lambda_3 = 0$. Hence, A has simple structure and can be transformed by similarity to the diagonal matrix

$$A_C = \begin{bmatrix} 2+j & 0 & 0 \\ 0 & 2-j & 0 \\ 0 & 0 & 0 \end{bmatrix}. \quad (\text{A.84})$$

To obtain the change of basis matrix, let us compute the eigenvectors corresponding to $\lambda_1, \lambda_2, \lambda_3$ by solving the following equations:

$$\left(A - (2+j)I_3 \right) \begin{bmatrix} \xi_1 \\ \xi_2 \\ \xi_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}; \quad \left(A - (2-j)I_3 \right) \begin{bmatrix} \xi_1 \\ \xi_2 \\ \xi_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}; \quad A \begin{bmatrix} \xi_1 \\ \xi_2 \\ \xi_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad (\text{A.85})$$

The eigenvectors result, up to a multiplicative constant

$$\mathbf{x}_1 = \begin{bmatrix} 1 \\ 2-j \\ 1+j \end{bmatrix}, \quad \check{\mathbf{x}}_1 = \begin{bmatrix} 1 \\ 2+j \\ 1-j \end{bmatrix}, \quad \mathbf{x}_3 = \begin{bmatrix} 1 \\ -1 \\ -1 \end{bmatrix},$$

corresponding to $\lambda_1, \lambda_2 = \bar{\lambda}_1, \lambda_3$.

It then results $A_C = T_C^{-1}AT_C$ when the change of basis matrix is chosen as

$$T_C = [\mathbf{x}_1 \quad \check{\mathbf{x}}_1 \quad \mathbf{x}_3] = \begin{bmatrix} 1 & 1 & 1 \\ 2+j & 2-j & -1 \\ 1-j & 1+j & -1 \end{bmatrix}.$$

Note that both the matrix A_C and the change of basis matrix T_C have complex entries.

To transform, by similarity, the matrix A into the real Jordan form

$$A_R = \left[\begin{array}{cc|c} 2 & 1 & 0 \\ -1 & 2 & 0 \\ \hline 0 & 0 & 0 \end{array} \right]$$

we can resort to the characteristic real basis $(\mathbf{r}, \mathbf{c}, \mathbf{x}_3)$, where $\mathbf{r}$ and $\mathbf{c}$ are solutions to the system

$$\begin{cases} A\mathbf{r} = \sigma\mathbf{r} - \omega\mathbf{c} = 2\mathbf{r} - \mathbf{c} \\ A\mathbf{c} = \omega\mathbf{r} + \sigma\mathbf{c} = \mathbf{r} + 2\mathbf{c} \end{cases} \quad (\text{A.86})$$

By setting $\mathbf{r}^T = [\rho_1 \ \rho_2 \ \rho_3]$ and $\mathbf{c}^T = [\gamma_1 \ \gamma_2 \ \gamma_3]$ the system (A.86) becomes

$$\begin{aligned} \begin{bmatrix} 1 & 0 & 1 \\ 2 & 1 & 1 \\ 1 & -1 & 2 \end{bmatrix} \begin{bmatrix} \rho_1 \\ \rho_2 \\ \rho_3 \end{bmatrix} &= 2 \begin{bmatrix} \rho_1 \\ \rho_2 \\ \rho_3 \end{bmatrix} - \begin{bmatrix} \gamma_1 \\ \gamma_2 \\ \gamma_3 \end{bmatrix} \\ \begin{bmatrix} 1 & 0 & 1 \\ 2 & 1 & 1 \\ 1 & -1 & 2 \end{bmatrix} \begin{bmatrix} \gamma_1 \\ \gamma_2 \\ \gamma_3 \end{bmatrix} &= \begin{bmatrix} \rho_1 \\ \rho_2 \\ \rho_3 \end{bmatrix} + 2 \begin{bmatrix} \gamma_1 \\ \gamma_2 \\ \gamma_3 \end{bmatrix} \end{aligned}$$

Taking into account only the first, the second, the forth, and the fifth equations, the system is derived

$$\begin{cases} \rho_1 + \rho_3 = 2\rho_1 - \gamma_1 \\ 2\rho_1 + \rho_2 + \rho_3 = 2\rho_2 - \gamma_2 \\ \gamma_1 + \gamma_3 = \rho_1 + 2\gamma_1 \\ 2\gamma_1 + \gamma_2 + \gamma_3 = \rho_2 + 2\gamma_2 \end{cases}$$

whence the triplets that follow are obtained

$$\mathbf{r} = \begin{bmatrix} \frac{1}{2}(\rho_3 + \gamma_3) \\ \frac{1}{2}(\rho_3 + 3\gamma_3) \\ \rho_3 \end{bmatrix}, \quad \mathbf{c} = \begin{bmatrix} \frac{1}{2}(-\rho_3 + \gamma_3) \\ \frac{1}{2}(-3\rho_3 + \gamma_3) \\ \gamma_3 \end{bmatrix},$$

which solve also the third and sixth equations. By finally setting $\rho_3 = \gamma_3 = 1$, one obtains the vectors of the characteristic real basis

$$\mathbf{r} = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \quad \mathbf{c} = \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}$$

and the transformation real matrix

$$T_R = [\mathbf{r} \ \mathbf{c} \ \mathbf{x}_3] = \begin{bmatrix} 1 & 0 & 1 \\ 2 & -1 & -1 \\ 1 & 1 & -1 \end{bmatrix}$$

which leads to the similarity relation $A_R = T_R^{-1}AT_R$.

To conclude, we mention the possibility to obtain a real basis and a real Jordan form for a linear transformation $\mathcal{A} : \mathbf{V} \rightarrow \mathbf{V}$ that is not of simple structure and is not represented, with respect to a specific basis, by a real matrix A .

By limiting ourselves to the case in which $\mathcal{A}$ has a single pair of complex conjugate eigenvalues λ and $\bar{\lambda}$ and a single Jordan chain for each eigenvalue, the Jordan basis is obtained by juxtaposing the chain associated to the eigenvalue λ and the chain (equally long) corresponding to the eigenvalue $\bar{\lambda}$. With respect to the chosen basis in $\mathbf{V}$, the elements of the chains will be represented by vectors of $\mathbb{C}^n$

$$\underbrace{\mathbf{x}^{(1)}, \mathbf{x}^{(2)}, \dots, \mathbf{x}^{(k)}}_{\text{chain related to } \lambda}, \underbrace{\check{\mathbf{x}}^{(1)}, \check{\mathbf{x}}^{(2)}, \dots, \check{\mathbf{x}}^{(k)}}_{\text{chain related to } \bar{\lambda}}$$

and it will be sufficient to set

$$\mathbf{r}^{(i)} := \Re(\mathbf{x}^{(i)}) = \frac{\mathbf{x}^{(i)} + \check{\mathbf{x}}^{(i)}}{2}, \quad \mathbf{c}^{(i)} := \Im(\mathbf{x}^{(i)}) = \frac{\mathbf{x}^{(i)} - \check{\mathbf{x}}^{(i)}}{2j}, \quad i = 1, 2, \dots, k \quad (\text{A.87})$$

to obtain (the components with respect to the old basis of) the *real Jordan basis*

$$\mathbf{r}^{(1)}, \mathbf{c}^{(1)}, \mathbf{r}^{(2)}, \mathbf{c}^{(2)}, \dots, \mathbf{r}^{(k)}, \mathbf{c}^{(k)}. \quad (\text{A.88})$$

With respect to this, the linear transformation is represented by the 2×2 block matrix

$$\begin{bmatrix} M & I_2 & & & \\ & M & I_2 & & \\ & & \ddots & \ddots & \\ & & & M & I_2 \\ & & & & M \end{bmatrix}, \quad \text{with } M := \begin{bmatrix} \sigma & \omega \\ -\omega & \sigma \end{bmatrix}. \quad (\text{A.89})$$

A.10 Rational canonical form

Definition A.10.1 [CYCLIC LINEAR TRANSFORMATIONS] *Let $\mathbf{V}$ be a vector space of dimension n . A linear transformation $\mathcal{A} : \mathbf{V} \rightarrow \mathbf{V}$ is called *cyclic* if there exists a vector $\mathbf{g}$, called *cyclic generator*, such that the cyclic space $\langle \mathcal{A} | \mathbf{g} \rangle$ coincides with $\mathbf{V}$, namely if*

$$\mathbf{g}, \mathcal{A}\mathbf{g}, \dots, \mathcal{A}^{n-1}\mathbf{g} \quad (\text{A.90})$$

is a basis for $\mathbf{V}$.

Exercise A.8.1 has showed some equivalent definitions for the transformation cyclicity (minimal and characteristic polynomials coincide, Jordan form with a number of mini-blocks equal to the number of distinct eigenvalues). A further characterization is here given, based on the possibility to represent $\mathcal{A}$ by means of a companion matrix.

Proposition A.10.2 [COMPANION MATRIX] *Given the monic polynomial*

$$f(s) = \phi_0 + \phi_1 s + \phi_2 s^2 + \dots + s^n,$$

the matrix

$$C_f := \begin{bmatrix} 0 & 1 & & & \\ & & 1 & & \\ & & & \ddots & \\ & & & & 1 \\ -\phi_0 & -\phi_1 & -\phi_2 & \cdots & -\phi_{n-1} \end{bmatrix}$$

*has $f(s)$ as its characteristic polynomial. This matrix is called *companion matrix* (of $f(s)$).*

PROOF The proposition is true for $n = 1$. Let us assume, by inductive reasoning, that the proposition is true for matrices up to the order $n - 1$ and let us compute the characteristic polynomial of C_f

$$\Delta_{C_f}(s) = \det(sI_n - C_f) = \det \begin{bmatrix} s & -1 & & & \\ & s & -1 & & \\ & & \ddots & \ddots & \\ & & & s & -1 \\ \phi_0 & \phi_1 & \phi_2 & \cdots & (s + \phi_{n-1}) \end{bmatrix}$$

$$\begin{aligned}
&= s \det \begin{bmatrix} s & -1 & & \\ & \ddots & \ddots & \\ \phi_1 & \phi_2 & \cdots & (s + \phi_{n-1}) \end{bmatrix} + (-1)^{n+1} \phi_0 \det \begin{bmatrix} -1 & & \\ s & -1 & \\ & \ddots & \\ & s & -1 \end{bmatrix} \\
&= s(\phi_1 + \phi_2 s + \dots + s^{n-1}) + (-1)^{n+1} \phi_0 (-1)^{n-1} = f(s) \quad \blacksquare
\end{aligned}$$

If $\mathcal{A}$ has characteristic polynomial equal to $\Delta_{\mathcal{A}}(s) = \alpha_0 + \alpha_1 s + \dots + s^n$, by resorting to the Cayley Hamilton's theorem, one finds, for any $\mathbf{v} \in \mathbf{V}$, $\mathcal{A}^n \mathbf{v} = -\alpha_0 \mathbf{v} - \alpha_1 \mathcal{A} \mathbf{v} - \dots - \alpha_{n-1} \mathcal{A}^{n-1} \mathbf{v}$.

Hence, if $\mathcal{A}$ is cyclic and if the basis (A.90) is assumed in $\mathbf{V}$, it results

$$(\mathcal{A}\mathbf{g}, \mathcal{A}^2\mathbf{g}, \dots, \mathcal{A}^n\mathbf{g}) = (\mathbf{g}, \mathcal{A}\mathbf{g}, \dots, \mathcal{A}^{n-1}\mathbf{g}) \begin{bmatrix} 0 & & -\alpha_0 \\ 1 & & -\alpha_1 \\ & \ddots & \vdots \\ & & 1 & -\alpha_{n-1} \end{bmatrix},$$

namely $\mathcal{A}$, with respect to the cyclic basis, is represented by the transpose of the companion matrix of the characteristic polynomial of $\mathcal{A}$.

If instead of the cyclic basis it is considered the basis $\tilde{\mathbf{v}}_1, \tilde{\mathbf{v}}_2, \dots, \tilde{\mathbf{v}}_n$ given by

$$\begin{aligned}
\tilde{\mathbf{v}}_n &= \mathbf{g} \\
\tilde{\mathbf{v}}_{n-1} &= \mathcal{A}\mathbf{g} + \alpha_{n-1}\mathbf{g} \\
&\dots \\
\tilde{\mathbf{v}}_2 &= \mathcal{A}^{n-2}\mathbf{g} + \alpha_{n-1}\mathcal{A}^{n-3}\mathbf{g} + \dots + \alpha_2\mathbf{g} \\
\tilde{\mathbf{v}}_1 &= \mathcal{A}^{n-1}\mathbf{g} + \alpha_{n-1}\mathcal{A}^{n-2}\mathbf{g} + \dots + \alpha_2\mathcal{A}\mathbf{g} + \alpha_1\mathbf{g}
\end{aligned} \tag{A.91}$$

it results

$$\begin{cases} \mathcal{A}\tilde{\mathbf{v}}_i = \tilde{\mathbf{v}}_{i-1} - \alpha_{i-1}\tilde{\mathbf{v}}_n & \text{if } i > 1 \\ \mathcal{A}\tilde{\mathbf{v}}_1 = \mathcal{A}^n\tilde{\mathbf{v}}_n + \alpha_{n-1}\mathcal{A}^{n-1}\tilde{\mathbf{v}}_n + \dots + \alpha_1\mathcal{A}\tilde{\mathbf{v}}_n = -\alpha_0\tilde{\mathbf{v}}_n & \text{if } i = 1 \end{cases}$$

and then

$$(\mathcal{A}\tilde{\mathbf{v}}_1, \mathcal{A}\tilde{\mathbf{v}}_2, \mathcal{A}\tilde{\mathbf{v}}_3, \dots, \mathcal{A}\tilde{\mathbf{v}}_n) = (\tilde{\mathbf{v}}_1, \tilde{\mathbf{v}}_2, \tilde{\mathbf{v}}_3, \dots, \tilde{\mathbf{v}}_n) \begin{bmatrix} 0 & 1 & & & \\ & & 1 & & \\ & & & \ddots & \\ & & & & 1 \\ -\alpha_0 & -\alpha_1 & -\alpha_2 & \cdots & -\alpha_{n-1} \end{bmatrix}.$$

We have thus proved the first part of the following

Proposition A.10.3 [REPRESENTATION OF CYCLIC TRANSFORMATIONS] *Let $\mathbf{V}$ be a vector space of finite dimension n .*

If $\mathcal{A} : \mathbf{V} \rightarrow \mathbf{V}$ is a cyclic transformation, there exists a basis with respect to which the transformation is represented by the companion matrix of the characteristic polynomial of $\mathcal{A}$.

Viceversa, if $\mathcal{A}$ is represented by a companion matrix, then $\mathcal{A}$ is cyclic.

PROOF (of the second part) It is sufficient to verify that $\mathbf{e}_n$ is a cyclic vector for the companion matrix. ■

Let now $\mathcal{A} : \mathbf{V} \rightarrow \mathbf{V}$ be a generic linear transformation (not necessarily cyclic). According to Corollary A.6.3, a subspace $\mathbf{U} \subseteq \mathbf{V}$ is said to be *cyclic* (with respect to $\mathcal{A}$) if there exists a vector $\mathbf{u}$ such that $\text{span}(\mathbf{u}, \mathcal{A}\mathbf{u}, \mathcal{A}^2\mathbf{u}, \dots) = \mathbf{U}$. A cyclic subspace is $\mathcal{A}$ -invariant and the restriction $\mathcal{A}|_{\mathbf{U}}$ of $\mathcal{A}$ to $\mathbf{U}$ is a cyclic transformation.

Even when the transformation $\mathcal{A}$ is not cyclic, $\mathbf{V}$ can always be decomposed into a direct sum of cyclic subspaces (with respect to $\mathcal{A}$): it is sufficient to think, for example, that the subspaces generated by the Jordan chains are cyclic.

In the proposition that follows we consider the problems of

1. studying what the smallest number of independent cyclic subspaces is, whose direct sum provides a decomposition of $\mathbf{V}$;
2. finding conditions on the cyclic spaces $\mathbf{C}_i$ of the decomposition, which guarantee the uniqueness of the characteristic (and minimal) polynomials of the restrictions $\mathcal{A}|_{\mathbf{C}_i}$
3. describing the matrix representing $\mathcal{A}$ when the basis is the juxtaposition of suitable bases for the spaces $\mathbf{C}_i$ and these latter have been chosen such that they satisfy 1. and 2.

Proposition A.10.4 [DECOMPOSITION THEOREM AND RATIONAL CANONICAL FORM] *Let $\mathcal{A} : \mathbf{V} \rightarrow \mathbf{V}$ be a linear transformation over the real field, let $\lambda_1, \dots, \lambda_t$ be the distinct eigenvalues (real or complex) of $\mathcal{A}$ and let*

$$\begin{array}{ll} \nu_{1,1} \geq \nu_{1,2} \geq \dots \nu_{1,c_1} & \text{the lengths of the Jordan chains corresponding to } \lambda_1 \\ \nu_{2,1} \geq \nu_{2,2} \geq \dots \nu_{2,c_2} & \text{the lengths of the Jordan chains corresponding to } \lambda_2 \\ \dots & \\ \nu_{t,1} \geq \nu_{t,2} \geq \dots \nu_{t,c_t} & \text{the lengths of the Jordan chains corresponding to } \lambda_t \end{array} \quad (\text{A.92})$$

Denote by $c := \max_i \{c_i, i = 1, \dots, t\}$ the maximum among the dimensions of the eigenspaces $\mathbf{U}_{\lambda_i}$. If $c_i < c$, we also set, by definition, $\nu_{i,c_i+1} = \dots = \nu_{i,c} = 0$.

- (i) $\mathbf{V}$ can be decomposed into the direct sum of c cyclic subspaces $\tilde{\mathbf{C}}_j$, with dimensions $r_1 = \sum_{i=1}^t \nu_{i,1}$, $r_2 = \sum_{i=1}^t \nu_{i,2}$, $\dots$, $r_c = \sum_{i=1}^t \nu_{i,c}$.

If $\mathbf{V} = \oplus_{j=1}^c \tilde{\mathbf{C}}_j$ is such decomposition and if in each of the cyclic subspace $\tilde{\mathbf{C}}_j$ a basis (A.91) is assumed, with respect to which $\mathcal{A}|_{\tilde{\mathbf{C}}_j}$ is represented by a companion matrix, with respect to the basis $\mathbf{V}$ obtained by juxtaposition of the aforementioned bases $\mathcal{A}$ is represented by a matrix $A \in \mathbb{R}^{n \times n}$, block diagonal with companion diagonal blocks, which is called *rational canonical form*

$$A = \text{diag} \left\{ \begin{bmatrix} 0 & 1 & & \\ & & \ddots & \\ & & & 1 \\ -\alpha_0 & -\alpha_1 & \dots & -\alpha_{r_1-1} \end{bmatrix}, \begin{bmatrix} 0 & 1 & & \\ & & \ddots & \\ & & & 1 \\ -\alpha_0 & -\alpha_1 & \dots & -\alpha_{r_2-1} \end{bmatrix}, \dots, \begin{bmatrix} 0 & 1 & & \\ & & \ddots & \\ & & & 1 \\ -\alpha_0 & -\alpha_1 & \dots & -\alpha_{r_c-1} \end{bmatrix} \right\} \quad (\text{A.93})$$

The diagonal blocks are companion to the monic polynomials with real coefficients

$$\begin{aligned}\psi_1(s) &= (s - \lambda_1)^{\nu_{1,1}}(s - \lambda_2)^{\nu_{2,1}} \cdots (s - \lambda_t)^{\nu_{t,1}} = \psi_{\mathcal{A}}(s) \\ \psi_2(s) &= (s - \lambda_1)^{\nu_{1,2}}(s - \lambda_2)^{\nu_{2,2}} \cdots (s - \lambda_t)^{\nu_{t,2}} \\ &\vdots \\ \psi_c(s) &= (s - \lambda_1)^{\nu_{1,c}}(s - \lambda_2)^{\nu_{2,c}} \cdots (s - \lambda_t)^{\nu_{t,c}},\end{aligned}$$

then $\psi_i(s)$ is a divisor of $\psi_{i-1}(s)$, for $i = 2, 3, \dots, c$. The polynomials $\psi_i(s)$ are called invariant polynomials and c is the ciclicity index of the transformation.

ii) Any decomposition $\mathbf{V} = \oplus_{\mu} \mathbf{C}_{\mu}$ into $\mathcal{A}$ -cyclic subspaces contains at least c summands.

PROOF (i) We can sort the Jordan basis by juxtaposing

- firstly, the t chains of maximum length $\nu_{1,1}, \nu_{2,1}, \dots, \nu_{t,1}$ corresponding to t eigenvalues;
- then, the shorter chains $\nu_{1,2}, \nu_{2,2}, \dots, \nu_{t,2}$;
-
- finally, the chains of length $\nu_{1,c}, \nu_{2,c}, \dots, \nu_{t,c}$.

$$\begin{array}{ccccccc} \underbrace{\star \star \star \dots \star \star \star \dots \star}_{\lambda_1 \text{ chain}} & \underbrace{\star \star \star \dots \star \star \star \dots \star}_{\lambda_2 \text{ chain}} & \dots & \underbrace{\star \star \star \dots \star \star \star \dots \star}_{\lambda_t \text{ chain}} & \underbrace{\star \star \star \dots \star \star \star \dots \star}_{\lambda_1} & \underbrace{\star \star \star \dots \star \star \star \dots \star}_{\lambda_2} & \dots & \underbrace{\star \star \star \dots \star \star \star \dots \star}_{\lambda_t} \\ \text{length } \nu_{1,1} & \text{length } \nu_{2,1} & & \text{length } \nu_{t,1} & \nu_{1,2} & \nu_{2,2} & & \nu_{t,2} \\ \hline \underbrace{\hspace{10em}}_{\text{basis of } \mathbf{C}_1} & \underbrace{\hspace{10em}}_{\text{basis of } \mathbf{C}_2} & & \underbrace{\hspace{10em}}_{\text{basis di } \mathbf{C}_c} \\ \hline \underbrace{\hspace{15em}}_{\text{basis di } \mathbf{V}} \end{array} \quad (\text{A.94})$$

The subspace $\tilde{\mathbf{C}}_1$ generated by the first t chains is cyclic. Indeed, the generalized eigenvectors of maximum order $\mathbf{v}_i^{(\nu_{i,1})}$, $i = 1, 2, \dots, t$, of each chain have minimal polynomials $(s - \lambda_i)^{\nu_{i,1}}$ that are coprime, hence $\mathbf{v} := \sum_{i=1}^t \mathbf{v}_i^{(\nu_{i,1})}$ has minimal polynomial

$$\psi_1(s) := (s - \lambda_1)^{\nu_{1,1}}(s - \lambda_2)^{\nu_{2,1}} \cdots (s - \lambda_t)^{\nu_{t,1}} \quad (\text{A.95})$$

of degree equal to the dimension of $\tilde{\mathbf{C}}_1$ and is a cyclic vector for $\tilde{\mathbf{C}}_1$. Note that (A.126) coincides with the minimal polynomial $\psi_{\mathcal{A}}(s)$ of the transformation.

The subspace $\tilde{\mathbf{C}}_2$, generated by the chains that follow, of length $\nu_{i,2}$, $i = 1, 2, \dots, t$, is also cyclic and the minimal polynomial of one of its cyclic generators is

$$\psi_2(s) = (s - \lambda_1)^{\nu_{1,2}}(s - \lambda_2)^{\nu_{2,2}} \cdots (s - \lambda_t)^{\nu_{t,2}},$$

hence it is a divisor of $\psi_1(s)$.

A similar reasoning can be done for the cyclic space $\tilde{\mathbf{C}}_3$, generated by the third group of chains, and for the minimal polynomial $\psi_3(s)$ of one of its generators, divisor of $\psi_2(s) \dots$. Therefore, it is possible to decompose $\mathbf{V}$ into a sum of c cyclic subspaces. It is then clear that, with respect to the basis described in the proposition statement, the transformation is represented by a matrix with structure (A.93).

(ii) It is impossible to decompose the vector space $\mathbf{V}$ into the direct sum of less than c cyclic subspaces. This comes from the fact that one eigenvalue λ of the transformation $\mathcal{A}$ has an eigenspace containing c linearly independent eigenvectors.

If $c = 2$, in Exercise A.6.5 it has been shown that two eigenvectors corresponding to the same eigenvalue λ and contained in the same cyclic space $\mathbf{C}$ are linearly dependent.

Let us suppose, by induction, to have proved that if k eigenvectors corresponding to the same eigenvalue λ are contained in the direct sum of $k - 1$ cyclic subspaces, then they must be linearly dependent. Let then $\mathbf{C}_i$, $i = 1, 2, \dots, k$, be independent cyclic subspaces and suppose that $\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_{k+1}$ are eigenvectors corresponding to the same eigenvalue λ . If all the eigenvectors belong to the direct sum $\oplus_{j=1}^k \mathbf{C}_j$, let us

verify that the eigenvectors are linearly dependent.

To this aim, let us consider the quotient space $\bar{\mathbf{V}} = \mathbf{V}/\mathbf{C}_1$ and the linear transformation on it⁵

$$\bar{\mathcal{A}} : \mathbf{V}/\mathbf{C}_1 \rightarrow \mathbf{V}/\mathbf{C}_1$$

induced by $\mathcal{A}$. In $\bar{\mathbf{V}}$ the subspaces

$$\bar{\mathbf{C}}_j := \mathbf{C}_j/\mathbf{C}_1 = \{\mathbf{v} + \mathbf{C}_1, \mathbf{v} \in \mathbf{C}_j\}, \quad j = 2, \dots, k,$$

- are cyclic, because generated by the images, modulo $\mathbf{C}_1$, of the cyclic vectors of $\mathbf{C}_j$, $j = 2, \dots, k$;
- have a direct sum $\oplus_{j=2}^k \bar{\mathbf{C}}_j$, as a consequence of Proposition A.3.2.

The quotient classes $[\mathbf{w}_1], [\mathbf{w}_2], \dots, [\mathbf{w}_{k+1}]$, when they are not null, are eigenvectors of $\bar{\mathcal{A}}$ corresponding to the eigenvalue λ and belong to $\oplus_{j=2}^k \bar{\mathbf{C}}_j$ and, according to the inductive hypothesis, they generate a subspace of $\bar{\mathbf{V}}$ of dimension not greater than $k - 1$. If, for instance, the last two classes $[\mathbf{w}_k]$ and $[\mathbf{w}_{k+1}]$ depend on the previous one, for suitable β_i and α_i coefficients one gets

$$\mathbf{u}_1 := \mathbf{w}_k - \sum_{i=1}^{k-1} \beta_i \mathbf{w}_i \in \mathbf{C}_1, \quad \mathbf{u}_2 := \mathbf{w}_{k+1} - \sum_{i=1}^{k-1} \alpha_i \mathbf{w}_i \in \mathbf{C}_1.$$

But $\mathbf{u}_1$ and $\mathbf{u}_2$, if not null, are eigenvectors corresponding to the eigenvalue λ contained in the cyclic space $\mathbf{C}_1$. Then, they are linearly dependent and this is sufficient to induce a linear dependence relation among $\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_{k+1}$. ■

Remark The Jordan normal form contains all the information needed to build the rational canonical form:

- the number of companion blocks in the rational canonical form, that is the cyclic index, coincides with the maximum number of mini-blocks corresponding to the same eigenvalue, that is the maximum dimension of the eigenspaces $\mathbf{U}_\lambda$;
- the characteristic (and minimal) polynomials of the companion blocks, that is the invariant polynomials of $\mathcal{A}$, are the products of the characteristic (and minimal) polynomials of direct sums of Jordan mini-blocks selected according to the criterion described at step (i) of the proof.

A.11 Norms of vectors, matrices, and operators

Definition A.11.1 [VECTOR NORM] *If V is a vector space over the real (complex) field, a norm on $\mathbf{V}$ is a function from $\mathbf{V}$ to the non-negative real numbers*

$$\|\cdot\| : \mathbf{V} \rightarrow \mathbb{R}_+ : \mathbf{v} \mapsto \|\mathbf{v}\|$$

satisfying the following axioms

- (i) $\|\mathbf{v}\| = 0$ implies $\mathbf{v} = \mathbf{0}$;
- (ii) $\|\alpha\mathbf{v}\| = |\alpha| \|\mathbf{v}\|$, $\forall \alpha \in \mathbb{R} (\mathbb{C})$, $\forall \mathbf{v} \in \mathbf{V}$;
- (iii) $\|\mathbf{v}_1 + \mathbf{v}_2\| \leq \|\mathbf{v}_1\| + \|\mathbf{v}_2\|$, $\forall \mathbf{v}_1, \mathbf{v}_2 \in \mathbf{V}$ (triangular inequality).

A space $\mathbf{V}$ where a norm $\|\cdot\|$ has been introduced is called *normed space* and is denoted by $(\mathbf{V}, \|\cdot\|)$.

⁵See Proposition A.3.5.

Example A.11.1 [NORMS IN $\mathbb{R}^n$] Some types of norms⁶ for a vector $\mathbf{v} = \begin{bmatrix} v_1 \\ \vdots \\ v_n \end{bmatrix} \in \mathbb{R}^n$ are:

- (i) [ℓ_1 NORM] $\|\mathbf{v}\|_1 = \sum_{i=1}^n |v_i|$.
- (ii) [ℓ_2 NORM, OR EUCLIDEAN NORM] $\|\mathbf{v}\|_2 = (\sum_{i=1}^n v_i^2)^{1/2} = (\mathbf{v}^T \mathbf{v})^{1/2}$;
- (iii) [ℓ_∞ NORM, OR “SUP NORM”] $\|\mathbf{v}\|_\infty = \max_i |v_i|$;

In $\mathbb{C}^n$ the only difference concerns the ℓ_2 norm, and it is $\|\mathbf{v}\|_2 = (\sum_{i=1}^n |v_i|^2)^{1/2} = (\bar{\mathbf{v}}^T \mathbf{v})^{1/2}$.

Example A.11.2 [COMPLEX SEQUENCE NORMS] Some examples of norms in infinite-dimension vector spaces are:

- (i) [ℓ_2 NORM] Let $\mathbf{H}$ be the (Hilbert) space of sequences of complex numbers $x = (\xi_1, \xi_2, \dots)$ satisfying condition $\sum_{j=1}^\infty |\xi_j|^2 < \infty$. The ℓ_2 norm of the sequence x is defined as

$$\|x\|_2 = \left(\sum_{j=1}^\infty |\xi_j|^2 \right)^{\frac{1}{2}}.$$

- (ii) [ℓ_∞ NORM] Let $\mathbf{B}$ be the space of bounded sequences of complex numbers, namely of the sequences $x = (\xi_1, \xi_2, \dots)$ with $|\xi_j| < c_x, \forall j$, where c_x is a positive real number that can depend on x , but not on j . The ℓ_∞ norm of the sequence x is defined as

$$\|x\|_\infty = \sup_j |\xi_j|.$$

- EXERCISE A.11.1 [ℓ_1 NORM] Under what condition the sequences of complex numbers are such that for them an ℓ_1 norm can be defined, similar to the one of Example A.11.1? If $\mathbf{S}$ is the space of such sequences, discuss the inclusion relations among the spaces $\mathbf{H}, \mathbf{B}$, and $\mathbf{S}$.
- EXERCISE A.11.2 If $A = [a_{ij}] \in \mathbb{R}^{n \times n}$, $|a_{ij}| \leq M, \forall i, j$ and $\|\mathbf{x}\|_2 = 1$, verify that $\|A\mathbf{x}\|_2 \leq Mn$.

‡ *Hint:* $\|A\mathbf{x}\|_2 = \left\| \begin{bmatrix} \sum_j a_{1j}x_j \\ \vdots \\ \sum_j a_{nj}x_j \end{bmatrix} \right\|_2 \leq M \left\| \begin{bmatrix} \sum_j |x_j| \\ \vdots \\ \sum_j |x_j| \end{bmatrix} \right\|_2 = M \|\mathbf{x}\|_1 \left\| \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix} \right\|_2 \leq Mn$. Note that, if

$\|\mathbf{x}\|_2 = 1$, the norm $\|\mathbf{x}\|_1$ reaches the maximum value when all components of $\mathbf{x}$ are equal to $\pm\sqrt{n^{-1}}$ and this corresponds to $\|\mathbf{x}\|_1 = |x_1| + |x_2| + \dots + |x_n| = \sqrt{n}$.

- EXERCISE A.11.3 [TRIANGULAR DIFFERENCE] In a normed space $(V, \|\cdot\|)$, for any $\mathbf{v}_1, \mathbf{v}_2 \in V$ it holds

$$\|\mathbf{v}_1 + \mathbf{v}_2\| \geq \left| \|\mathbf{v}_1\| - \|\mathbf{v}_2\| \right| \quad (\text{A.96})$$

(in a triangle, the length of one side is not smaller than the difference -in absolute value- between the lengths of the other sides).

‡ *Solution:* $\|\mathbf{v}_2\| = \|-\mathbf{v}_1 + (\mathbf{v}_2 + \mathbf{v}_1)\| \leq \|\mathbf{v}_1\| + \|\mathbf{v}_2 + \mathbf{v}_1\| \Rightarrow \|\mathbf{v}_1 + \mathbf{v}_2\| \geq \|\mathbf{v}_2\| - \|\mathbf{v}_1\|$. By exchanging $\mathbf{v}_1$ and $\mathbf{v}_2$ it follows that $\|\mathbf{v}_1 + \mathbf{v}_2\| \geq \|\mathbf{v}_1\| - \|\mathbf{v}_2\|$, whence the inequality with the absolute value is derived.

- EXERCISE A.11.4 Let $\|\cdot\|$ be a generic norm in $\mathbb{R}^n$, $n \geq 2$, and consider the vector $\begin{bmatrix} \mathbf{a} \\ \mathbf{b} \end{bmatrix} \in \mathbb{R}^n$, where $\mathbf{a}$ and $\mathbf{b}$ have both dimensions smaller than n .

- (i) Verify that $\left\| \begin{bmatrix} \mathbf{a} \\ \mathbf{0} \end{bmatrix} \right\| \leq \max \left\{ \left\| \begin{bmatrix} \mathbf{a} \\ \mathbf{b} \end{bmatrix} \right\|, \left\| \begin{bmatrix} \mathbf{a} \\ -\mathbf{b} \end{bmatrix} \right\| \right\}$.

- (ii) If $\mathbf{b} \neq \mathbf{0}$, it is not necessarily true that $\left\| \begin{bmatrix} \mathbf{a} \\ \mathbf{0} \end{bmatrix} \right\| \leq \left\| \begin{bmatrix} \mathbf{a} \\ \mathbf{b} \end{bmatrix} \right\|$.

‡ *Hint:* (i) $2 \left\| \begin{bmatrix} \mathbf{a} \\ \mathbf{0} \end{bmatrix} \right\| = \left\| \begin{bmatrix} 2\mathbf{a} \\ \mathbf{0} \end{bmatrix} \right\| = \left\| \begin{bmatrix} \mathbf{a} \\ \mathbf{b} \end{bmatrix} + \begin{bmatrix} \mathbf{a} \\ -\mathbf{b} \end{bmatrix} \right\| \leq \left\| \begin{bmatrix} \mathbf{a} \\ \mathbf{b} \end{bmatrix} \right\| + \left\| \begin{bmatrix} \mathbf{a} \\ -\mathbf{b} \end{bmatrix} \right\|$;

- (ii) Consider the positive definite matrix $P = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$ and introduce a norm in $\mathbb{R}^2$ by stating

$$\left\| \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} \right\|^2 = [v_1 \ v_2] P \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = 2v_1^2 + 2v_1v_2 + v_2^2. \text{ Compare the norms of the vectors } \begin{bmatrix} 2 \\ -1 \end{bmatrix} \text{ and } \begin{bmatrix} 2 \\ 0 \end{bmatrix}.$$

⁶We'll resort to the norm symbol without any subscript when it is not relevant to specify the type of employed norm.

Definition A.11.2 [EQUIVALENT NORMS] *Two norms $\|\cdot\|$ and $\|\cdot\|_*$ on a vector space $\mathbf{V}$ are called equivalent if there exist two positive constants α and β such that, for any $\mathbf{v} \in \mathbf{V}$, one has*

$$\|\mathbf{v}\|_* \leq \alpha \|\mathbf{v}\|, \quad \|\mathbf{v}\| \leq \beta \|\mathbf{v}\|_* \quad (\text{A.97})$$

The equivalence between norms is an equivalence relation, as clearly suggested by the name. In finite dimensional spaces all norms belong to the same equivalence class.

Proposition A.11.3 [EQUIVALENT NORMS IN A FINITE DIMENSIONAL SPACE] *In a vector space $\mathbf{V}$ of finite dimension n , all norms are equivalent.*

PROOF It is sufficient to prove that a generic norm $\|\cdot\|$ is equivalent to the norm $\|\cdot\|_1$. Let $(\mathbf{v}_1, \dots, \mathbf{v}_n)$ be a basis in $\mathbf{V}$. If $\mathbf{x} = \xi_1 \mathbf{v}_1 + \dots + \xi_n \mathbf{v}_n$ is a vector of $\mathbf{V}$ and hence $\|\mathbf{x}\|_1 := \sum_{j=1}^n |\xi_j|$, the triangular inequality applied to $\|\cdot\|$ provides $\|\mathbf{x}\| \leq \sum_{j=1}^n |\xi_j| \|\mathbf{v}_j\|$. By setting $\alpha := \max_j \{\|\mathbf{v}_j\|\}$ one gets

$$\|\mathbf{x}\| \leq \alpha \sum_{j=1}^n |\xi_j| = \alpha \|\mathbf{x}\|_1, \quad \forall \mathbf{x} \in \mathbf{V}. \quad (\text{A.98})$$

Note that (A.98) implies that the function $\|\cdot\| : \mathbf{V} \rightarrow \mathbb{R} : \mathbf{x} \mapsto \|\mathbf{x}\|$ is continuous in the normed space $(\mathbf{V}, \|\cdot\|_1)$. The proof is a simple exercise.

To obtain the inverse inequality, let us suppose by contradiction that it does not exist any positive constant β such that $\|\mathbf{x}\|_1 \leq \beta \|\mathbf{x}\|$ for any $\mathbf{x} \in \mathbf{V}$ and thus the upper limit on the set $\{\|\mathbf{x}\|_1 / \|\mathbf{x}\| : \mathbf{x} \in \mathbf{V}\}$ is infinite. It then exists a sequence of vectors $(\mathbf{x}^{(k)})_{k \in \mathbb{N}}$ with $\|\mathbf{x}^{(k)}\|_1 = 1$ satisfying $\lim_{k \rightarrow +\infty} \|\mathbf{x}^{(k)}\| = 0$. Since the sphere $\{\mathbf{x} \in \mathbf{V} : \|\mathbf{x}\|_1 = 1\}$ is a compact set of the normed space $(\mathbf{V}, \|\cdot\|_1)$, we can assume, possibly by extracting a subsequence, that $(\mathbf{x}^{(k)})_{k \in \mathbb{N}}$ converges to a vector $\bar{\mathbf{x}}$ with $\|\bar{\mathbf{x}}\|_1 = 1$, being this not null. But, as already said, $\|\cdot\|$ is a continuous function in the space $(\mathbf{V}, \|\cdot\|_1)$, and therefore $0 = \lim_{k \rightarrow +\infty} \|\mathbf{x}^{(k)}\| = \|\lim_{k \rightarrow +\infty} \mathbf{x}^{(k)}\| = \|\bar{\mathbf{x}}\|$, which is clearly not possible since $\bar{\mathbf{x}}$ is not null. ■

• EXERCISE A.11.5 [EQUIVALENCE OF NORMS AND APPLICATIONS]

- (i) In the space $\mathbb{R}^n$, verify that $\max_{\|\mathbf{x}\|_2=1} \|\mathbf{x}\|_1 = \sqrt{n}$ (see hint in Exercise A.11.2) and $\min_{\|\mathbf{x}\|_2=1} \|\mathbf{x}\|_1 = 1$. Compute the constants α and β of inequalities (A.97) for $\|\cdot\|_2$ and $\|\cdot\|_1$.
- (ii) Let $\|\cdot\|$ be an arbitrary norm in $\mathbb{R}^n$ and be $\mathbf{v}_1, \mathbf{v}_2, \dots$ a sequence of vectors in $\mathbb{R}^n$. By taking the ℓ_∞ norm as a second norm in $\mathbb{R}^n$, verify that
 - the sequence of vectors converges to zero with respect to the norm $\|\cdot\|$ (namely it converges to zero the sequence $\|\mathbf{v}_1\|, \|\mathbf{v}_2\|, \dots$) if and only if all the components of vector $\mathbf{v}_k$ go to zero, for $k \rightarrow \infty$ (namely if the sequence $\|\mathbf{v}_1\|_1, \|\mathbf{v}_2\|_1, \dots$ converges to zero).
 - the following statements are equivalent: (a) for any positive real number M it exists an index K such that, for any $k > K$, vectors $\mathbf{v}_k$ of the sequence have some components (not necessarily the same for every k) whose absolute value is higher than M , (b) the sequence $\|\mathbf{v}_1\|_1, \|\mathbf{v}_2\|_1, \dots$ is divergent, (c) the sequence $\|\mathbf{v}_1\|, \|\mathbf{v}_2\|, \dots$ is divergent.

Often, when considering spaces of matrices, it is required that the norms satisfy an additional condition with respect to those of Definition A.11.1, namely the submultiplicativity:

Definition A.11.4 [SUBMULTIPLICATIVITY⁷] *The norms $\|\cdot\|_{p \times m}$, $\|\cdot\|_{m \times n}$, $\|\cdot\|_{p \times n}$ in $\mathbb{C}^{p \times m}$, $\mathbb{C}^{m \times n}$, $\mathbb{C}^{p \times n}$ are submultiplicative if, for any $A \in \mathbb{C}^{p \times m}$ and any $B \in \mathbb{C}^{m \times n}$*

$$\|AB\|_{p \times n} \leq \|A\|_{p \times m} \|B\|_{m \times n}. \quad (\text{A.99})$$

⁷Or consistency: when such property is satisfied the norm is called “matrix”.

Example A.11.3 Let $\mathbf{V} = \mathbb{R}^{p \times m}$ be the space of $p \times m$ real matrices. We introduce, for any values p and m , two matrix norms (that are therefore submultiplicative) for any $A = [a_{ij}] \in \mathbb{R}^{p \times m}$:

(i) [ℓ_1 NORM] $\|A\|_1 = \sum_{i,j} |a_{ij}|$. To verify the submultiplicativity, note that

$$\|AB\|_1 = \sum_{i,j} \left| \sum_k a_{ik} b_{kj} \right| \leq \sum_{i,k,j} |a_{ik} b_{kj}| \leq \sum_{ik} \sum_{mj} |a_{ik} b_{mj}| = \left(\sum_{ik} |a_{ik}| \right) \left(\sum_{mj} |b_{mj}| \right) = \|A\|_1 \|B\|_1$$

(ii) [ℓ_2 NORM OR EUCLIDEAN NORM⁸] $\|A\|_2 = \left(\sum_{i,j} a_{ij}^2 \right)^{1/2} = [\text{trace}(A^T A)]^{1/2}$. By the submultiplicativity, it holds

$$\|AB\|_2^2 = \sum_{i,j} \left| \sum_k a_{ik} b_{kj} \right|^2 \leq \sum_{i,j} \left(\sum_k |a_{ik}|^2 \right) \left(\sum_m |b_{mj}|^2 \right) = \left(\sum_{ik} |a_{ik}|^2 \right) \left(\sum_{mj} |b_{mj}|^2 \right) = \|A\|_2^2 \|B\|_2^2$$

By setting $\|A\|_\infty = \max_{i,j} \{|a_{ij}|, i = 1, \dots, p, j = 1, \dots, m\}$ we obtain a norm for matrices, similar to the ℓ_∞ norm for the vectors (thus satisfying Axioms (i), (ii), (iii) of the definition), but this is not submultiplicative. Indeed

$$A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}, \quad \|A\|_\infty = 1, \quad \|A^2\|_\infty = 2$$

Such a norm is called sometimes “vector norm” to highlight the loss of the subadditivity property, which characterizes the “matrix norms”.

When we consider matrices as linear operators between finite dimensional vector spaces, there exists a standard procedure to provide a submultiplicative norm, starting from the norms defined in the spaces where they act.

Generally speaking, if $\mathcal{A} : \mathbf{V} \rightarrow \mathbf{W}$ is a linear continuous transformation between normed spaces $\mathbf{V}$ and $\mathbf{W}$, the *norm of the operator* $\mathcal{A}$ can be defined by setting⁹

$$\|\mathcal{A}\| = \sup_{\mathbf{v} : \|\mathbf{v}\| \leq 1} \|\mathcal{A}\mathbf{v}\|. \quad (\text{A.100})$$

The space $\mathcal{L}(\mathbf{V}, \mathbf{W})$ of the linear continuous transformations of $\mathbf{V}$ into $\mathbf{W}$ inherits in this way a norm from $\mathbf{V}$ and from $\mathbf{W}$: indeed, conditions (i) and (ii) are trivially satisfied, namely $\|\mathcal{A}\| = 0 \Leftrightarrow \mathcal{A} = 0$ and $\|\alpha \mathcal{A}\| = |\alpha| \|\mathcal{A}\|$. Moreover, if $\mathcal{A}$ and $\mathcal{B}$ belong to $\mathcal{L}(\mathbf{V}, \mathbf{W})$,

$$\begin{aligned} \|\mathcal{A} + \mathcal{B}\| &= \sup_{\mathbf{v} : \|\mathbf{v}\| \leq 1} \|\mathcal{A}\mathbf{v} + \mathcal{B}\mathbf{v}\| \leq \sup_{\mathbf{v} : \|\mathbf{v}\| \leq 1} (\|\mathcal{A}\mathbf{v}\| + \|\mathcal{B}\mathbf{v}\|) \\ &\leq \sup_{\mathbf{v} : \|\mathbf{v}\| \leq 1} \|\mathcal{A}\mathbf{v}\| + \sup_{\mathbf{v} : \|\mathbf{v}\| \leq 1} \|\mathcal{B}\mathbf{v}\| = \|\mathcal{A}\| + \|\mathcal{B}\|, \end{aligned}$$

which proves (iii).

The *operator norm* (or *induced norm*) $\|\cdot\|$ is also submultiplicative: if $\mathcal{A} \in \mathcal{L}(\mathbf{V}, \mathbf{W})$ and $\mathcal{B} \in \mathcal{L}(\mathbf{W}, \mathbf{U})$, the operator norm of $\mathcal{B}\mathcal{A} \in \mathcal{L}(\mathbf{V}, \mathbf{U})$ satisfies the inequality

$$\|\mathcal{B}\mathcal{A}\| \leq \|\mathcal{B}\| \|\mathcal{A}\|. \quad (\text{A.101})$$

Indeed, if $\mathcal{A}$ is not identically null (in such a case (A.101) would trivially hold), it is

$$\|\mathcal{B}\mathcal{A}\| = \sup_{\mathbf{v} : \mathbf{v} \neq \mathbf{0}} \frac{\|(\mathcal{B}\mathcal{A})\mathbf{v}\|}{\|\mathbf{v}\|} = \sup_{\mathbf{v} : \mathcal{A}\mathbf{v} \neq \mathbf{0}} \frac{\|(\mathcal{B}\mathcal{A})\mathbf{v}\|}{\|\mathcal{A}\mathbf{v}\|} \frac{\|\mathcal{A}\mathbf{v}\|}{\|\mathbf{v}\|} \leq \sup_{\mathbf{v} : \mathcal{A}\mathbf{v} \neq \mathbf{0}} \frac{\|(\mathcal{B}\mathcal{A})\mathbf{v}\|}{\|\mathcal{A}\mathbf{v}\|} \sup_{\mathbf{v} : \mathcal{A}\mathbf{v} \neq \mathbf{0}} \frac{\|\mathcal{A}\mathbf{v}\|}{\|\mathbf{v}\|}$$

⁸Also: Frobenius norm, Schur norm, Hilbert-Schmidt norm, ...

⁹For the sake of clarity, for the operator norms in this paragraph the notation $\|\cdot\|$ is used. In the remainder of these Notes, for all norms the notation $\|\cdot\|$ has been adopted; when necessary, it has to be understood from the context which norm we are referring to.

$$\leq \sup_{\mathbf{w} \neq \mathbf{0}} \frac{\|\mathcal{B}\mathbf{w}\|}{\|\mathbf{w}\|} \sup_{\mathbf{v}: \mathcal{A}\mathbf{v} \neq \mathbf{0}} \frac{\|\mathcal{A}\mathbf{v}\|}{\|\mathbf{v}\|} = \|\mathcal{B}\| \|\mathcal{A}\|.$$

- EXERCISE A.11.6 (i) If the linear transformation $\mathcal{A}$ is not null, $\mathcal{A}/\|\mathcal{A}\|$ has unitary norm. (ii) If $\mathcal{A} : \mathbf{V} \rightarrow \mathbf{V}$ has norm (A.100) smaller than 1, for any $\epsilon > 0$ there exists an index n such that $\|\mathcal{A}^n \mathbf{v}\| \leq \epsilon \|\mathbf{v}\|$ for any $\mathbf{v} \in \mathbf{V}$.
- EXERCISE A.11.7 Verify the following equalities, which provide equivalent definitions of the norm of a linear operator (in the last case, if $\mathcal{A}$ is not identically null)

$$\|\mathcal{A}\| = \sup_{\mathbf{v}: \|\mathbf{v}\| \leq 1} \|\mathcal{A}\mathbf{v}\| = \sup_{\mathbf{v}: \|\mathbf{v}\| = 1} \|\mathcal{A}\mathbf{v}\| = \sup_{\mathbf{v} \neq \mathbf{0}} \frac{\|\mathcal{A}\mathbf{v}\|}{\|\mathbf{v}\|} = \sup_{\mathbf{v}: \mathcal{A}\mathbf{v} \neq \mathbf{0}} \frac{\|\mathcal{A}\mathbf{v}\|}{\|\mathbf{v}\|}$$

Example A.11.4 (i) [OPERATOR NORM OF A MATRIX IN ℓ_∞ SPACES] If $\mathbb{R}^p$ and $\mathbb{R}^m$ are both endowed with ℓ_∞ norm

$$\|\mathbf{v}\|_{\mathbb{R}^m} = \max_{i=1, \dots, m} |v_i|, \quad \|\mathbf{w}\|_{\mathbb{R}^p} = \max_{i=1, \dots, p} |w_i|,$$

the operator norm of a matrix $A \in \mathbb{R}^{p \times m}$ is given by

$$\|A\|_\infty = \sup_{\|\mathbf{v}\|_\infty \leq 1} \|\mathcal{A}\mathbf{v}\|_\infty = \sup_{|v_j| \leq 1} \left(\max_{i=1, \dots, p} \left| \sum_{j=1}^m a_{ij} v_j \right| \right) = \max_{i=1, \dots, p} \left(\sum_{j=1}^m |a_{ij}| \right)$$

Hence, the ℓ_∞ operator norm of A is *the largest among the ℓ_1 norms of the matrix rows*.

(ii) [OPERATOR NORM OF A MATRIX IN ℓ_1 SPACES] If $\mathbb{R}^p$ and $\mathbb{R}^m$ are both endowed with ℓ_1 norms

$$\|\mathbf{v}\|_{\mathbb{R}^m} = \sum_{i=1}^m |v_i|, \quad \|\mathbf{w}\|_{\mathbb{R}^p} = \sum_{i=1}^p |w_i|,$$

to compute the operator norm corresponding to a matrix $A \in \mathbb{R}^{p \times m}$ one can note that

$$\begin{aligned} \|A\|_1 &= \sup_{\|\mathbf{v}\|_1 \leq 1} \|\mathcal{A}\mathbf{v}\|_1 = \sup_{\sum_j |v_j| \leq 1} \left(\sum_{i=1}^p \left| \sum_{j=1}^m a_{ij} v_j \right| \right) \leq \sup_{\sum_j |v_j| \leq 1} \left(\sum_{j=1}^m \sum_{i=1}^p |a_{ij}| |v_j| \right) \\ &= \sup_{\sum_j |v_j| \leq 1} \left(\sum_{j=1}^m |v_j| \sum_{i=1}^p |a_{ij}| \right) = \max_{j=1, \dots, m} \left(\sum_{i=1}^p |a_{ij}| \right) \end{aligned}$$

It follows that the operator norm ℓ_1 of A does not exceed the largest of the ℓ_1 norms of the columns of A . In addition, if the column corresponding to the maximum ℓ_1 norm is the h -th, by choosing $\mathbf{v} = \mathbf{e}_h$ one finds $\|A\mathbf{e}_h\|_1 = \max_{j=1, \dots, m} (\sum_{i=1}^p |a_{ij}|)$. Hence $\|A\|_1$ is *the largest among the ℓ_1 norms of the columns of A* .

Example A.11.5 [OPERATOR NORM OF A MATRIX IN ℓ_2 SPACES, OR SPECTRAL NORM] If $\mathbb{R}^p$ and $\mathbb{R}^m$ are endowed with the ℓ_2 norms

$$\|\mathbf{v}\|_{\mathbb{R}^m} = \left(\sum_{i=1}^m |v_i|^2 \right)^{1/2} = (\tilde{\mathbf{v}}^T \mathbf{v})^{1/2}, \quad \|\mathbf{w}\|_{\mathbb{R}^p} = \left(\sum_{i=1}^p |w_i|^2 \right)^{1/2} = (\tilde{\mathbf{w}}^T \mathbf{w})^{1/2},$$

the operator norm of a matrix $A \in \mathbb{R}^{p \times m}$ is given by

$$\|A\|_2 = \sup_{\mathbf{v} \neq \mathbf{0}} \frac{\|\mathcal{A}\mathbf{v}\|}{\|\mathbf{v}\|} = \sup_{\mathbf{v} \neq \mathbf{0}} \left(\frac{\mathbf{v}^T A^T A \mathbf{v}}{\mathbf{v}^T \mathbf{v}} \right)^{1/2}. \quad (\text{A.102})$$

In the next paragraph we will see that the spectral norm of A coincides with the square root of the maximum eigenvalue of the symmetric matrix $A^T A$.

- EXERCISE A.11.8 The ℓ_2 norm (or Euclidean norm, defined in Example A.11.3) of the matrix $A = \begin{bmatrix} 1 & 2 \\ 0 & 2 \end{bmatrix}$ is equal to $\|A\|_2 = \sqrt{9}$, while its norm as an operator in ℓ_2 spaces (i.e. the spectral norm) is $\|A\|_2 = \sqrt{\frac{9+\sqrt{65}}{2}}$. It follows that, in general, Euclidean norm and spectral norm do not coincide.

Proposition A.11.5 [MATRICIAL NORM AND SPECTRAL RADIUS] *Let $\|\cdot\|$ be an arbitrary matrix norm (in particular an operator norm) in $\mathbb{R}^{n \times n}$. If $A \in \mathbb{R}^{n \times n}$ and $\rho(A) = \max\{|\lambda| : \lambda \in \Lambda(A)\}$ is the spectral radius of the matrix, then*

- (i) $\rho(A) \leq \|A\|$;
- (ii) $\rho(A) = \lim_{k \rightarrow +\infty} \|A^k\|^{\frac{1}{k}}$.

PROOF (i) If $\rho(A) = 0$ there is nothing left to prove. If $\lambda \neq 0$ is an eigenvalue with maximum absolute value, for any real number ϵ such that $0 < \epsilon < |\lambda|$, the matrix $B := (|\lambda| - \epsilon)^{-1}A$ has spectral radius larger than 1. According to what proved in Chapter 2 with respect to the matrix modes, once an arbitrary value $M > 0$ has been chosen, there is an integer h such that, for any $\nu > h$, each matrix B^ν contains at least one element whose absolute value is greater than M . Then, the matrix norm $\|B^k\|_\infty$ diverges when k increases (see Exercise A.11.5) and, by norm equivalence, also the matrix norm $\|B^k\|$ diverges. In particular, for sufficiently large k , it is also true that $1 < \|B^k\| = (|\lambda| - \epsilon)^{-k} \|A^k\|$. This implies $\|A\|^k \geq \|A^k\| > (|\lambda| - \epsilon)^k$ and then $\|A\| > (|\lambda| - \epsilon)$. Once recalled the arbitrariness of ϵ , we conclude by claiming

$$\|A\| \geq |\lambda| = \rho(A).$$

(ii) According to (i), it is $\|A^k\| \geq \rho(A^k) = \rho(A)^k$ and hence

$$\rho(A) = |\lambda| \leq \|A^k\|^{\frac{1}{k}} \quad (\text{A.103})$$

Let $B = (|\lambda| + \epsilon)^{-1}A$, with ϵ an arbitrary positive number. Therefore, since all eigenvalues in B have absolute values smaller than 1, the vector norm ℓ_∞ of B^k (that is the maximum absolute value among the entries of B^k) goes to zero when k grows, and therefore also the matrix norm $\|B^k\|$ tends to zero. Once $\epsilon > 0$ is fixed, there exists K such that for $k > K$ we have $\|B^k\| < 1$, namely $\|A^k\| < (|\lambda| + \epsilon)^k$ and then $\|A^k\|^{\frac{1}{k}} < (|\lambda| + \epsilon)$. Therefore, for any $\epsilon > 0$ it stands

$$\rho(A) \leq \|A^k\|^{\frac{1}{k}} < (\rho(A) + \epsilon),$$

provided that k is sufficiently high. As $\epsilon > 0$ is arbitrary, the quantity $\|A^k\|^{\frac{1}{k}}$ tends towards $\rho(A)$ as k diverges. ■

- EXERCISE A.11.9 Extend the results of the previous paragraph to vectors and matrices over the complex field.

A.12 Inner product spaces

Definition A.12.1 [INNER PRODUCT] *Let $\mathbf{V}$ be a vector space over the real field. An inner product on $\mathbf{V}$ is a map that associates to any pair of vectors a real number (their inner product)*

$$\langle \cdot, \cdot \rangle : \mathbf{V} \times \mathbf{V} \rightarrow \mathbb{R} : (\mathbf{v}_1, \mathbf{v}_2) \mapsto \langle \mathbf{v}_1, \mathbf{v}_2 \rangle$$

and that, for any choice of $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ in $\mathbf{V}$ and of α, β in $\mathbb{R}$, satisfies the conditions

- i) (symmetry) $\langle \mathbf{v}_1, \mathbf{v}_2 \rangle = \langle \mathbf{v}_2, \mathbf{v}_1 \rangle$;
- ii) (bilinearity) $\langle \alpha \mathbf{v}_1 + \beta \mathbf{v}_2, \mathbf{v}_3 \rangle = \alpha \langle \mathbf{v}_1, \mathbf{v}_3 \rangle + \beta \langle \mathbf{v}_2, \mathbf{v}_3 \rangle$;
- iii) (positive definite product) $\langle \mathbf{v}_1, \mathbf{v}_1 \rangle \geq 0$ and $\langle \mathbf{v}_1, \mathbf{v}_1 \rangle = 0$ only if $\mathbf{v}_1 = 0$.

A real vector space endowed with an inner product is also called “Euclidean space”.

By setting

$$\|\mathbf{v}\| := \langle \mathbf{v}, \mathbf{v} \rangle^{1/2}, \quad (\text{A.104})$$

we have that $\|\cdot\|$ defines a norm in $\mathbf{V}$, being satisfied the related axioms. In particular, the triangular inequality comes from the following

Proposition A.12.2 [CAUCHY-SCHWARZ INEQUALITY] *In a vector space endowed with inner product it is always true that*

$$|\langle \mathbf{v}, \mathbf{w} \rangle| \leq \|\mathbf{v}\| \cdot \|\mathbf{w}\|.$$

PROOF OF THE CAUCHY-SCHWARZ INEQUALITY Consider the second order polynomial function in the real variable x , non negative for all the values of x :

$$f(x) = \langle x\mathbf{v} + \mathbf{w}, x\mathbf{v} + \mathbf{w} \rangle = x^2 \langle \mathbf{v}, \mathbf{v} \rangle + 2x \langle \mathbf{v}, \mathbf{w} \rangle + \langle \mathbf{w}, \mathbf{w} \rangle = \|\mathbf{v}\|^2 x^2 + 2\langle \mathbf{v}, \mathbf{w} \rangle x + \|\mathbf{w}\|^2. \quad (\text{A.105})$$

The non-negativity condition for all $x \in \mathbb{R}$ implies that the discriminant of the trinomial is non positive, that is

$$4\langle \mathbf{v}, \mathbf{w} \rangle^2 - 4\|\mathbf{v}\|^2 \|\mathbf{w}\|^2 \leq 0. \quad \blacksquare$$

PROOF OF THE TRIANGULAR INEQUALITY

$$\|\mathbf{v} + \mathbf{w}\|^2 = \|\mathbf{v}\|^2 + 2\langle \mathbf{v}, \mathbf{w} \rangle + \|\mathbf{w}\|^2 \leq \|\mathbf{v}\|^2 + 2\|\mathbf{v}\| \cdot \|\mathbf{w}\| + \|\mathbf{w}\|^2 = (\|\mathbf{v}\| + \|\mathbf{w}\|)^2. \quad \blacksquare$$

Example A.12.1 In $\mathbb{R}^n$ the inner product between two n -tuples $\mathbf{v}$ and $\mathbf{w}$ is usually defined by posing

$$\langle \mathbf{v}, \mathbf{w} \rangle := \mathbf{v}^T \mathbf{w}. \quad (\text{A.106})$$

If $P \in \mathbb{R}^{n \times n}$ is a positive definite (and hence, symmetric) matrix, another inner product in $\mathbb{R}^n$ is defined by

$$\langle \mathbf{v}, \mathbf{w} \rangle_P := \mathbf{v}^T P \mathbf{w}. \quad (\text{A.107})$$

Since positive definite matrices are all and only those obtained through a factorization $P = M^T M$, with M an arbitrary invertible matrix in $\mathbb{R}^{n \times n}$ (see for example the following Example A.12.5), it is $\mathbf{v}^T P \mathbf{w} = \mathbf{v}^T M^T M \mathbf{w} = (M\mathbf{v})^T (M\mathbf{w})$, namely an inner product (A.107) can be seen as the product (A.106) provided that the space $\mathbb{R}^n$ is referred to a different basis.

Example A.12.2 In the space $\mathcal{C}_{[0,1]}$ of real valued scalar functions, continuous on the interval $[0, 1]$, an inner product is defined by setting $\langle f, g \rangle := \int_0^1 f(\sigma)g(\sigma)d\sigma$.

If the functions are defined in $\mathbb{R}^m$, an inner product is defined by $\langle \mathbf{f}, \mathbf{g} \rangle := \int_0^1 \mathbf{f}^T(\sigma)\mathbf{g}(\sigma)d\sigma$.

Proposition A.12.3 [EIGENVALUES AND INNER PRODUCTS] *Let $A \in \mathbb{R}^{n \times n}$.*

(i) *If α and β are two real constants such that $\alpha < \Re(\Lambda(A)) < \beta$, then there exists a positive definite matrix P such that, for all $\mathbf{v} \in \mathbb{R}^n$,*

$$\alpha \langle \mathbf{v}, \mathbf{v} \rangle_P \leq \langle \mathbf{v}, A\mathbf{v} \rangle_P \leq \beta \langle \mathbf{v}, \mathbf{v} \rangle_P \quad (\text{A.108})$$

(ii) *If A is invertible and γ and ζ are two real constants such that $0 < \gamma < |(\Lambda(A))| < \zeta$, there exists a positive definite matrix P such that, for all $\mathbf{v} \in \mathbb{R}^n$,*

$$\gamma^2 \langle \mathbf{v}, \mathbf{v} \rangle_P \leq \langle A\mathbf{v}, A\mathbf{v} \rangle_P \leq \zeta^2 \langle \mathbf{v}, \mathbf{v} \rangle_P. \quad (\text{A.109})$$

PROOF By means of a similarity transformation, induced by a suitable invertible real matrix R , A is transformed into the Jordan real form, $A_J = R^{-1}AR$, with diagonal blocks of the type (A.89).

(i) If ϵ is a positive number smaller than the smallest between the differences $\mathcal{Re}\{\lambda_i\} - \alpha$ and $\beta - \mathcal{Re}\{\lambda_i\}$, $i = 1, 2, \dots, n$, there exists a further similarity, induced by an invertible diagonal real matrix D (which one?), that multiplies by ϵ the off-diagonal unitary elements in the mini-blocks of A_J , bringing him to either one of the following two structures

$$\begin{bmatrix} \lambda & \epsilon & & \\ & \lambda & \epsilon & \\ & & \ddots & \ddots \\ & & & \lambda & \epsilon \\ & & & & \lambda \end{bmatrix}, \quad \begin{bmatrix} M & \epsilon I_2 & & \\ & M & \epsilon I_2 & \\ & & \ddots & \ddots \\ & & & M & \epsilon I_2 \\ & & & & M \end{bmatrix} \text{ con } M = \begin{bmatrix} \sigma & \omega \\ -\omega & \sigma \end{bmatrix}, \quad (\text{A.110})$$

depending on whether the eigenvalue(s) of the mini-block is a real number λ or the complex pair $\sigma \pm j\omega$. Denote by $A_{J,\epsilon}$ the modified Jordan matrix and set $T^{-1} = RD$ and $P = (T^{-1})^T T^{-1}$. Hence,

$$\begin{aligned} \langle \mathbf{v}, A\mathbf{v} \rangle_P &= \mathbf{v}^T (T^{-1})^T T^{-1} A \mathbf{v} = (T^{-1} \mathbf{v})^T T^{-1} A T (T^{-1} \mathbf{v}) = (T^{-1} \mathbf{v})^T A_{J,\epsilon} (T^{-1} \mathbf{v}) \\ \langle \mathbf{v}, \mathbf{v} \rangle_P &= \mathbf{v}^T (T^{-1})^T T^{-1} \mathbf{v} = (T^{-1} \mathbf{v})^T (T^{-1} \mathbf{v}). \end{aligned}$$

If $A_{J,\epsilon}$ consists of a single mini-block and $T^{-1} \mathbf{v} = \mathbf{w} = [w_1 \dots w_n]^T$, it follows that

$$\mathbf{w}^T A_{J,\epsilon} \mathbf{w} = \lambda(w_1^2 + \dots + w_n^2) + \epsilon(w_1 w_2 + w_2 w_3 + \dots + w_{n-1} w_n) \quad (\text{A.111})$$

$$\mathbf{w}^T A_{J,\epsilon} \mathbf{w} = \sigma(w_1^2 + \dots + w_n^2) + \epsilon(w_1 w_3 + w_2 w_4 + \dots + w_{n-2} w_n), \quad (\text{A.112})$$

depending the fact that the mini-block has the first or the second structure in (A.110). In the first case, being

$$|w_1 w_2 + w_2 w_3 + \dots + w_{n-1} w_n| = \left| \left\langle \begin{bmatrix} w_1 \\ \vdots \\ w_{n-1} \\ 0 \end{bmatrix}, \begin{bmatrix} w_2 \\ \vdots \\ w_n \\ 0 \end{bmatrix} \right\rangle \right| \leq \left\| \begin{bmatrix} w_1 \\ \vdots \\ w_{n-1} \\ 0 \end{bmatrix} \right\| \left\| \begin{bmatrix} w_2 \\ \vdots \\ w_n \\ 0 \end{bmatrix} \right\| \leq \mathbf{w}^T \mathbf{w},$$

it can be concluded that $(\lambda - \epsilon) \mathbf{w}^T \mathbf{w} \leq \mathbf{w}^T A_{J,\epsilon} \mathbf{w} \leq (\lambda + \epsilon) \mathbf{w}^T \mathbf{w}$. With similar reasoning, in the second case it can be concluded that $(\sigma - \epsilon) \mathbf{w}^T \mathbf{w} \leq \mathbf{w}^T A_{J,\epsilon} \mathbf{w} \leq (\sigma + \epsilon) \mathbf{w}^T \mathbf{w}$. In both cases, with the condition on ϵ , (A.108) is satisfied. This result can be extended to the case when more mini-blocks are present.

(ii) Choose a positive ϵ , small enough to satisfy conditions

$$\gamma < \min_i (|\lambda_i| - 3\epsilon) < \max_i (|\lambda_i| + 3\epsilon) < \zeta, \quad \epsilon < |\lambda_i|/5, \quad \forall_i. \quad (\text{A.113})$$

Compute, as in the previous case, the matrix T leading to the real Jordan form $A_{J,\epsilon}$, and set $P = (T^{-1})^T T^{-1}$. It follows that

$$\langle A\mathbf{v}, A\mathbf{v} \rangle_P = (T^{-1} \mathbf{v})^T A_{J,\epsilon}^T A_{J,\epsilon} (T^{-1} \mathbf{v}).$$

If $A_{J,\epsilon}$ is made of a single mini-block having the first form in (A.110) we obtain

$$\begin{aligned} \mathbf{w}^T A_{J,\epsilon}^T A_{J,\epsilon} \mathbf{w} &= \lambda^2(w_1^2 + \dots + w_n^2) + \epsilon^2(w_2^2 + \dots + w_n^2) + 2\lambda\epsilon(w_1 w_2 + \dots + w_{n-1} w_n) \\ &= \lambda^2 \mathbf{w}^T \mathbf{w} + \epsilon^2 \mathbf{w}^T \mathbf{w} - \epsilon^2 w_1^2 + 2\lambda\epsilon(w_1 w_2 + \dots + w_{n-1} w_n) \end{aligned} \quad (\text{A.114})$$

Being

$$\begin{aligned} 2\epsilon\lambda(w_1 w_2 + \dots + w_{n-1} w_n) - \epsilon^2 w_1^2 &\leq 2\epsilon|\lambda||w_1 w_2 + \dots + w_{n-1} w_n| \leq 2\epsilon|\lambda| \mathbf{w}^T \mathbf{w} \\ 2\epsilon\lambda(w_1 w_2 + \dots + w_{n-1} w_n) - \epsilon^2 w_1^2 &\geq -2\epsilon|\lambda||w_1 w_2 + \dots + w_{n-1} w_n| - \epsilon^2 w_1^2 \geq -2\epsilon|\lambda| \mathbf{w}^T \mathbf{w} - \epsilon^2 \mathbf{w}^T \mathbf{w}, \end{aligned}$$

from (A.114) we get to

$$\begin{aligned} \langle A\mathbf{v}, A\mathbf{v} \rangle_P &= \mathbf{w}^T A_{J,\epsilon}^T A_{J,\epsilon} \mathbf{w} \leq (|\lambda| + \epsilon)^2 \mathbf{w}^T \mathbf{w} \leq (|\lambda| + 3\epsilon)^2 \mathbf{w}^T \mathbf{w} \leq \zeta^2 \langle \mathbf{v}, \mathbf{v} \rangle_P \\ \langle A\mathbf{v}, A\mathbf{v} \rangle_P &\geq (|\lambda|^2 - 2\epsilon|\lambda|) \mathbf{w}^T \mathbf{w} = [(|\lambda| - 3\epsilon)^2 + 4\epsilon|\lambda| - 9\epsilon^2] \mathbf{w}^T \mathbf{w} \geq (|\lambda| - 3\epsilon)^2 \mathbf{w}^T \mathbf{w} \geq \gamma^2 \langle \mathbf{v}, \mathbf{v} \rangle_P \end{aligned}$$

and (A.109) is satisfied.

If $A_{J,\epsilon}$ is the second form of the mini-blocks in (A.110) and we write $|\lambda|^2 = \sigma^2 + \omega^2$, it can be obtained, instead,

$$\mathbf{w}^T A_{J,\epsilon}^T A_{J,\epsilon} \mathbf{w} = (\sigma^2 + \omega^2)(w_1^2 + \dots + w_n^2) + \epsilon^2(w_3^2 + \dots + w_n^2) + 2\epsilon\sigma(w_1 w_3 + w_2 w_4 + \dots + w_{n-2} w_n)$$

$$\begin{aligned}
& + 2\epsilon\omega((w_2w_3 - w_1w_4) + (w_4w_5 - w_3w_6) + \dots + (w_{n-2}w_{n-1} - w_{n-3}w_n)) \\
& = (\sigma^2 + \omega^2 + \epsilon^2)\mathbf{w}^T\mathbf{w} - \epsilon^2(w_1^2 + w_2^2) + 2\epsilon\sigma(w_1w_3 + w_2w_4 + \dots + w_{n-2}w_n) \\
& + 2\epsilon\omega((w_2w_3 - w_1w_4) + (w_4w_5 - w_3w_6) + \dots + (w_{n-2}w_{n-1} - w_{n-3}w_n)) \\
& := (|\lambda|^2 + \epsilon^2)\mathbf{w}^T\mathbf{w} + \tau
\end{aligned} \tag{A.115}$$

and τ satisfies the inequalities

$$\begin{aligned}
\tau & \leq 2\epsilon(|\sigma| + |\omega|)\mathbf{w}^T\mathbf{w} \leq 4\epsilon|\lambda|\mathbf{w}^T\mathbf{w} \\
\tau & \geq -\epsilon^2\mathbf{w}^T\mathbf{w} - 2\epsilon(|\sigma| + |\omega|)\mathbf{w}^T\mathbf{w} \geq (-\epsilon^2 - 4\epsilon|\lambda|)\mathbf{w}^T\mathbf{w},
\end{aligned}$$

and, taking into account (A.113), it follows

$$\begin{aligned}
\mathbf{w}^T A_{J,\epsilon}^T A_{J,\epsilon} \mathbf{w} & \leq (|\lambda|^2 + \epsilon^2)\mathbf{w}^T\mathbf{w} + 4\epsilon|\lambda|\mathbf{w}^T\mathbf{w} \leq (|\lambda| + 3\epsilon)^2\mathbf{w}^T\mathbf{w} \leq \zeta^2\langle\mathbf{v}, \mathbf{v}\rangle_P \\
\mathbf{w}^T A_{J,\epsilon}^T A_{J,\epsilon} \mathbf{w} & \geq (|\lambda|^2 + \epsilon^2)\mathbf{w}^T\mathbf{w} - (\epsilon^2 + 4\epsilon|\lambda|)\mathbf{w}^T\mathbf{w} = [(|\lambda| - 3\epsilon)^2 + 2\epsilon|\lambda| - 9\epsilon^2]\mathbf{w}^T\mathbf{w} \\
& \geq (|\lambda| - 3\epsilon)^2\mathbf{w}^T\mathbf{w} \geq \gamma^2\langle\mathbf{v}, \mathbf{v}\rangle_P.
\end{aligned}$$

It follows that (A.109) is satisfied. The discussion can be extended to the case when more mini-blocks are present. ■

Given a normed real space $(\mathbf{V}, \|\cdot\|)$, an inner product $\langle\cdot, \cdot\rangle$ in $\mathbf{V}$ that induces the norm $\|\cdot\|$ through (A.104) does not always exist. In other words, the spaces endowed with an inner product do not cover the whole set of normed spaces. In this context, it is important the following:

Proposition A.12.4 [PARALLELOGRAM IDENTITY] *In order for the norm of a normed real space $\mathbf{V}$ to be induced by an inner product, it is necessary and sufficient that for any two vectors $\mathbf{v}$ and $\mathbf{w}$ in $\mathbf{V}$ the following equality is verified*

$$\|\mathbf{v} + \mathbf{w}\|^2 + \|\mathbf{v} - \mathbf{w}\|^2 = 2(\|\mathbf{v}\|^2 + \|\mathbf{w}\|^2). \tag{A.116}$$

PROOF If $\mathbf{V}$ is a real space with inner product, it is straightforward to verify the identity

$$\langle\mathbf{v} + \mathbf{w}, \mathbf{v} + \mathbf{w}\rangle + \langle\mathbf{v} - \mathbf{w}, \mathbf{v} - \mathbf{w}\rangle = 2(\langle\mathbf{v}, \mathbf{v}\rangle + \langle\mathbf{w}, \mathbf{w}\rangle)$$

and then the parallelogram identity. Viceversa, if $\mathbf{V}$ is a normed space and (A.116) is verified, we pose

$$\langle\mathbf{v}, \mathbf{w}\rangle := \frac{1}{4}(\|\mathbf{v} + \mathbf{w}\|^2 - \|\mathbf{v} - \mathbf{w}\|^2). \tag{A.117}$$

If it can be shown that (A.117) satisfies all the axioms of the inner product, since for $\mathbf{v} = \mathbf{w}$ (A.117) becomes $\langle\mathbf{v}, \mathbf{v}\rangle = \|\mathbf{v}\|^2$, we can claim that the norm in $\mathbf{V}$ is induced by an inner product. It is immediate to verify that the product defined by (A.117) is symmetric and positive definite. Let us focus, then, on the proof of bilinearity of (A.117), proving preliminarily that the expression

$$\begin{aligned}
4(\langle\mathbf{u} + \mathbf{v}, \mathbf{w}\rangle - \langle\mathbf{u}, \mathbf{w}\rangle - \langle\mathbf{v}, \mathbf{w}\rangle) & = \|\mathbf{u} + \mathbf{v} + \mathbf{w}\|^2 - \|\mathbf{u} + \mathbf{v} - \mathbf{w}\|^2 \\
& \quad - \|\mathbf{u} + \mathbf{w}\|^2 - \|\mathbf{v} + \mathbf{w}\|^2 \\
& \quad + \|\mathbf{u} - \mathbf{w}\|^2 + \|\mathbf{v} - \mathbf{w}\|^2
\end{aligned} \tag{A.118}$$

is identically null for any $\mathbf{v}, \mathbf{u}, \mathbf{w} \in \mathbf{V}$. By resorting to the parallelogram identity

$$\|\mathbf{u} \pm \mathbf{w} + \mathbf{v}\|^2 = 2(\|\mathbf{u} \pm \mathbf{w}\|^2 + \|\mathbf{v}\|^2) - \|\mathbf{u} \pm \mathbf{w} - \mathbf{v}\|^2$$

and replacing it in the right hand side of (A.118), we get

$$\begin{aligned}
4(\langle\mathbf{u} + \mathbf{v}, \mathbf{w}\rangle - \langle\mathbf{u}, \mathbf{w}\rangle - \langle\mathbf{v}, \mathbf{w}\rangle) & = -\|\mathbf{u} - \mathbf{v} + \mathbf{w}\|^2 + \|\mathbf{u} - \mathbf{v} - \mathbf{w}\|^2 \\
& \quad - \|\mathbf{u} - \mathbf{w}\|^2 - \|\mathbf{v} + \mathbf{w}\|^2 \\
& \quad + \|\mathbf{u} + \mathbf{w}\|^2 + \|\mathbf{v} - \mathbf{w}\|^2
\end{aligned} \tag{A.119}$$

The sum of (A.118) and (A.119) divided by two provides

$$4(\langle\mathbf{u} + \mathbf{v}, \mathbf{w}\rangle - \langle\mathbf{u}, \mathbf{w}\rangle - \langle\mathbf{v}, \mathbf{w}\rangle) = \frac{1}{2}(\|\mathbf{u} + \mathbf{v} + \mathbf{w}\|^2 + \|\mathbf{u} - \mathbf{v} + \mathbf{w}\|^2)$$

$$\begin{aligned}
& - \frac{1}{2} \left(\|\mathbf{u} + \mathbf{v} - \mathbf{w}\|^2 + \|-\mathbf{u} + \mathbf{v} - \mathbf{w}\|^2 \right) \\
& - \|\mathbf{v} + \mathbf{w}\|^2 + \|\mathbf{v} - \mathbf{w}\|^2.
\end{aligned} \tag{A.120}$$

Let us apply again the parallelogram identity: at the right hand side of (A.120) the first summand can be replaced with $\|\mathbf{u}\|^2 + \|\mathbf{v} + \mathbf{w}\|^2$ and the second one with $-\|\mathbf{u}\|^2 - \|\mathbf{v} - \mathbf{w}\|^2$. Then $\langle \mathbf{u} + \mathbf{v}, \mathbf{w} \rangle - \langle \mathbf{u}, \mathbf{w} \rangle - \langle \mathbf{v}, \mathbf{w} \rangle$ is identically null.

We can now verify that, by arbitrarily choosing $\mathbf{v}, \mathbf{w} \in \mathbf{V}$, the function $f : \mathbb{R} \rightarrow \mathbb{R} : \alpha \mapsto f(\alpha)$ defined by

$$f(\alpha) = \langle \alpha \mathbf{v}, \mathbf{w} \rangle - \alpha \langle \mathbf{v}, \mathbf{w} \rangle \tag{A.121}$$

is identically null. Clearly, from (A.117) it follows that $f(0) = f(-1) = 0$. For any integer n , it is then

$$\langle n\mathbf{v}, \mathbf{w} \rangle = \langle \text{sgn}(n) \underbrace{(\mathbf{v} + \dots + \mathbf{v})}_{|n| \text{ summands}}, \mathbf{w} \rangle = \text{sgn}(n) \left(\langle \mathbf{v}, \mathbf{w} \rangle + \dots + \langle \mathbf{v}, \mathbf{w} \rangle \right) = n \langle \mathbf{v}, \mathbf{w} \rangle$$

and f is null on the integers. If p and q are arbitrary integers, with $q \neq 0$, it then follows that

$$\left\langle \frac{p}{q} \mathbf{v}, \mathbf{w} \right\rangle = p \left\langle \frac{1}{q} \mathbf{v}, \mathbf{w} \right\rangle = \frac{p}{q} q \left\langle \frac{1}{q} \mathbf{v}, \mathbf{w} \right\rangle = \frac{p}{q} \left\langle \frac{q}{q} \mathbf{v}, \mathbf{w} \right\rangle = \frac{p}{q} \langle \mathbf{v}, \mathbf{w} \rangle$$

and f is null on rationals. Being $f(\cdot)$ a continuous function (why is this the case?), it is identically null. ■

Example A.12.3 In $\mathbb{R}^2$ the norm $\|\mathbf{v}\|_\infty = \max_i |v_i|$ is not induced by an inner product. Indeed, by choosing

$$\mathbf{v} = \begin{bmatrix} 1 \\ -2 \end{bmatrix}, \quad \mathbf{w} = \begin{bmatrix} -2 \\ 1 \end{bmatrix}$$

it follows that $\|\mathbf{v}\|_\infty = 2$, $\|\mathbf{w}\|_\infty = 2$, $\|\mathbf{v} + \mathbf{w}\|_\infty = 1$, $\|\mathbf{v} - \mathbf{w}\|_\infty = 3$ and the parallelogram identity is not satisfied.

This result can be extended to spaces $\mathbb{R}^n$, for all $n \geq 2$.

Definition A.12.5 [ORTOGONALITY] Two vectors $\mathbf{v}_1$ and $\mathbf{v}_2$ of the real space $\mathbf{V}$ endowed with an inner product are called orthogonal if $\langle \mathbf{v}_1, \mathbf{v}_2 \rangle = 0$.

The following proposition summarizes some fundamental facts on the structure of the set of vectors orthogonal to a given subspace.

Proposition A.12.6 [ORTHOGONAL SUBSPACE] If $\mathbf{U}$ is an arbitrary subspace of $\mathbf{V}$, then

- i) $\mathbf{U}^\perp := \{\mathbf{v} \in \mathbf{V} : \langle \mathbf{u}, \mathbf{v} \rangle = 0, \forall \mathbf{u} \in \mathbf{U}\}$ is a subspace of $\mathbf{V}$, called the orthogonal of $\mathbf{U}$;
- ii) $\mathbf{U} \cap \mathbf{U}^\perp = \{\mathbf{0}\}$;
- iii) $\mathbf{U} \subseteq (\mathbf{U}^\perp)^\perp$.

If $\mathbf{U}$ has a finite or at least a countable basis, then

- iv) $\mathbf{U}$ has a (finite or countable) “orthonormal” basis $(\mathbf{c}_1, \mathbf{c}_2, \dots, \mathbf{c}_n, \dots)$ (namely $\langle \mathbf{c}_i, \mathbf{c}_j \rangle = \delta_{ij}$). This basis can be obtained by applying Gram-Schmidt orthogonalization (Gram-Schmidt process) to a generic basis in $\mathbf{U}$.

If $\dim(\mathbf{U}) = n < \infty$ (or, more in general, if $\mathbf{U}$ is a closed subspace of $\mathbf{V}$), then

- v) $\mathbf{V} = \mathbf{U} \oplus \mathbf{U}^\perp$ and therefore $\dim(\mathbf{V}) = \dim(\mathbf{U}) + \dim(\mathbf{U}^\perp)$;
- vi) $\mathbf{U} = (\mathbf{U}^\perp)^\perp$.

PROOF Statements (i) - (iii) are straightforward.

The orthonormal basis of statement (iv) is obtained from a generic basis $(\mathbf{u}_1, \mathbf{u}_2, \dots)$ choosing

$$\begin{aligned} \mathbf{c}_1 &= \frac{\mathbf{u}_1}{\|\mathbf{u}_1\|} \\ \mathbf{c}'_2 &= \mathbf{u}_2 - \langle \mathbf{u}_2, \mathbf{c}_1 \rangle \mathbf{c}_1; \quad \mathbf{c}_2 = \frac{\mathbf{c}'_2}{\|\mathbf{c}'_2\|} \\ \dots &\dots\dots \\ \mathbf{c}'_k &= \mathbf{u}_k - \langle \mathbf{u}_k, \mathbf{c}_1 \rangle \mathbf{c}_1 - \dots - \langle \mathbf{u}_k, \mathbf{c}_{k-1} \rangle \mathbf{c}_{k-1}; \quad \mathbf{c}_k = \frac{\mathbf{c}'_k}{\|\mathbf{c}'_k\|} \\ \dots &\dots\dots \end{aligned}$$

Statement (v) can be proved by considering a generic vector $\mathbf{v} \in \mathbf{V}$ and verifying that the vector $\mathbf{v} - \sum_{i=1}^n \langle \mathbf{v}, \mathbf{c}_i \rangle \mathbf{c}_i$ belongs to $\mathbf{U}^\perp$.

Finally, by applying the previous statement, if $\mathbf{w} \in (\mathbf{U}^\perp)^\perp$, it can be written as $\mathbf{w} = \mathbf{u} + \mathbf{t}$, with $\mathbf{u} \in \mathbf{U}$ and $\mathbf{t} \in \mathbf{U}^\perp$. From $\mathbf{t} \in \mathbf{U}^\perp$ and $\mathbf{w} \in (\mathbf{U}^\perp)^\perp$ it follows $0 = \langle \mathbf{w}, \mathbf{t} \rangle$, while from $\mathbf{u} \in \mathbf{U}$ and $\mathbf{t} \in \mathbf{U}^\perp$ it follows $0 = \langle \mathbf{u}, \mathbf{t} \rangle$. Then $\langle \mathbf{t}, \mathbf{t} \rangle = \langle \mathbf{w} - \mathbf{u}, \mathbf{t} \rangle = 0$, hence $\mathbf{t}$ is null and $\mathbf{w} \in \mathbf{U}$. This proves the inclusion $(\mathbf{U}^\perp)^\perp \subseteq \mathbf{U}$, which, together with the reverse inclusion reported at statement (iii), proves statement (vi). ■

- EXERCISE A.12.1 [PYTHAGOREAN THEOREM] Prove that if $\langle \mathbf{v}, \mathbf{w} \rangle = 0$, then

$$\|\mathbf{v}\|^2 + \|\mathbf{w}\|^2 = \|\mathbf{v} + \mathbf{w}\|^2.$$

Given a vector $\mathbf{v}$ and a closed subspace $\mathbf{U}$ of the space $\mathbf{V}$ endowed with inner product, as $\mathbf{u}$ varies in $\mathbf{U}$ the “distance” $\|\mathbf{v} - \mathbf{u}\|$ reaches the minimum value in correspondence to a single vector $\mathbf{v}_U \in \mathbf{U}$. To find this latter, it is sufficient to decompose $\mathbf{v}$ into the form

$$\mathbf{v} = \mathbf{v}_U + \mathbf{v}_{U^\perp}, \quad \mathbf{v}_U \in \mathbf{U}, \quad \mathbf{v}_{U^\perp} \in \mathbf{U}^\perp$$

and observe that

$$\langle \mathbf{v} - \mathbf{u}, \mathbf{v} - \mathbf{u} \rangle = \langle \mathbf{v}_U + \mathbf{v}_{U^\perp} - \mathbf{u}, \mathbf{v}_U + \mathbf{v}_{U^\perp} - \mathbf{u} \rangle = \langle \mathbf{v}_U - \mathbf{u}, \mathbf{v}_U - \mathbf{u} \rangle + \langle \mathbf{v}_{U^\perp}, \mathbf{v}_{U^\perp} \rangle$$

assumes the minimum value if and only if $\mathbf{u} = \mathbf{v}_U$. The difference $\mathbf{v} - \mathbf{v}_U$ is a vector orthogonal to the subspace $\mathbf{U}$. The linear map $P_U : \mathbf{V} \rightarrow \mathbf{U} : \mathbf{v} \mapsto \mathbf{v}_U$ and the vector $\mathbf{v}_U \in \mathbf{U}$ are called the “orthogonal projection” of the space $\mathbf{V}$ and the vector $\mathbf{v}$ onto the subspace $\mathbf{U}$.

Definition A.12.7 [ORTHOGONAL MATRIX] A matrix $M \in \mathbb{R}^{n \times n}$ is called *orthogonal* if its columns constitute an orthonormal basis of $\mathbb{R}^n$, endowed with the “usual” inner product $\langle \mathbf{v}, \mathbf{w} \rangle = \mathbf{v}^T \mathbf{w}$.

It can be immediately obtained from the definition that M is orthogonal if and only if

$$MM^T = M^T M = I_n,$$

namely if the transpose and the inverse of M coincide.

Example A.12.4 [SYMMETRIC MATRICES] Let $A \in \mathbb{R}^{n \times n}$ be a symmetric matrix.

(i) If $\mathbf{v}_1$ and $\mathbf{v}_2$ are eigenvectors of A corresponding to distinct eigenvalues λ_1 and λ_2 (real, according to Exercise A.5.1), then $\mathbf{v}_1$ and $\mathbf{v}_2$ are orthogonal with respect to the usual inner product of $\mathbb{R}^n$. Indeed, from $\mathbf{v}_2^T \lambda_1 \mathbf{v}_1 = \mathbf{v}_2^T A \mathbf{v}_1 = \mathbf{v}_2^T A^T \mathbf{v}_1 = \lambda_2 \mathbf{v}_2^T \mathbf{v}_1$ it follows that $\mathbf{v}_2^T \mathbf{v}_1 = 0$.

(ii) If in each eigenspace $\mathbf{U}_{\lambda_i}$ of A an orthonormal basis $(\mathbf{v}_{i,1}, \mathbf{v}_{i,2}, \dots, \mathbf{v}_{i,n_i})$ is chosen, the set of n eigenvectors of A thus obtained makes an orthonormal basis for $\mathbb{R}^n$ and the matrix M listing these n eigenvectors as its columns is orthogonal. Then, every symmetric real matrix can be diagonalized by similarity by an orthogonal matrix.

(iii) A is null if and only if it is nilpotent.

Example A.12.5 [POSITIVE SEMIDEFINITE MATRICES] (i) **Non negative eigenvalues.** The symmetric matrix $A \in \mathbb{R}^{n \times n}$ is positive definite (semidefinite) if and only if its eigenvalues are all positive (non negative). Indeed symmetry implies the possibility to represent A in the form

$$A = M \text{diag}\{\lambda_1, \lambda_2, \dots, \lambda_n\} M^T, \quad (\text{A.122})$$

with $M \in \mathbb{R}^{n \times n}$ an orthogonal matrix. By choosing $\mathbf{y} = M^T \mathbf{x}$, the condition

$$\mathbf{x}^T A \mathbf{x} > 0 \ (\geq 0), \quad \forall \mathbf{x} \neq \mathbf{0}$$

is equivalent to the condition

$$\mathbf{y}^T \text{diag}\{\lambda_1, \lambda_2, \dots, \lambda_n\} \mathbf{y} > 0 \ (\geq 0), \quad \forall \mathbf{y} \neq \mathbf{0},$$

which is satisfied if and only if all real numbers λ_i are positive (non negative).

(ii) **Factorization.** If A is positive semidefinite, (A.122) can be rewritten as

$$A = [M \text{diag}\{\sqrt{\lambda_1}, \sqrt{\lambda_2}, \dots, \sqrt{\lambda_n}\}] [M \text{diag}\{\sqrt{\lambda_1}, \sqrt{\lambda_2}, \dots, \sqrt{\lambda_n}\}]^T.$$

If A has rank equal to r , by deleting in $[M \text{diag}\{\sqrt{\lambda_1}, \sqrt{\lambda_2}, \dots, \sqrt{\lambda_n}\}]$ the $n - r$ null columns the factorization $A = C^T C$ is obtained, with $C \in \mathbb{R}^{r \times n}$ a matrix of rank r .

(iii) **Existence and uniqueness of the square root.** If A is positive semidefinite, there exists a positive semidefinite matrix $B \in \mathbb{R}^{n \times n}$ such that $A = B^2$. It is sufficient to consider the factorization (A.122) and choose

$$B = M \text{diag}\{\lambda_1^{\frac{1}{2}}, \lambda_2^{\frac{1}{2}}, \dots, \lambda_n^{\frac{1}{2}}\} M^T.$$

B is called the square root of A , it is denoted by $A^{\frac{1}{2}}$ and has the same rank as A . It is unique. Indeed¹⁰, if a positive semidefinite matrix $C \neq B$ existed satisfying the equation $C^2 = A$, the difference $B - C \neq 0$, symmetric and not null, would have an eigenvalue $\lambda \neq 0$. If $\mathbf{v}$ is the corresponding eigenvector, from

$$\begin{aligned} 0 &= \langle \mathbf{v}, (B^2 - C^2) \mathbf{v} \rangle = \langle \mathbf{v}, B(B - C) \mathbf{v} + (B - C) C \mathbf{v} \rangle = \langle \mathbf{v}, B(B - C) \mathbf{v} \rangle + \langle (B - C) \mathbf{v}, C \mathbf{v} \rangle \\ &= \langle \mathbf{v}, B \lambda \mathbf{v} \rangle + \langle \lambda \mathbf{v}, C \mathbf{v} \rangle = \lambda [\langle \mathbf{v}, B \mathbf{v} \rangle + \langle \mathbf{v}, C \mathbf{v} \rangle] \end{aligned}$$

it follows that, being both non negative, the products $\langle \mathbf{v}, B \mathbf{v} \rangle$ and $\langle \mathbf{v}, C \mathbf{v} \rangle$ are both null. But then it would be $0 = \langle \mathbf{v}, (B - C) \mathbf{v} \rangle = \lambda \langle \mathbf{v}, \mathbf{v} \rangle$, which is clearly unreasonable.

(iv) **k -th roots** If A is positive semidefinite, from the factorization (A.122) it follows that the positive semidefinite matrix

$$B = M \text{diag}\{\lambda_1^{\frac{1}{k}}, \lambda_2^{\frac{1}{k}}, \dots, \lambda_n^{\frac{1}{k}}\} M^T$$

satisfies the identity $B^k = A$ and therefore it is a k -th root of A . Similarly to the square root, also the k -th root of A is unique¹¹ and is indicated with $A^{\frac{1}{k}}$.

(v) **Ordering and inverse.** If A and B are positive semidefinite, the notation $A \succeq B$ ($A \succ B$) means that $A - B$ is positive semidefinite (definite). Moreover, if the positive definite matrix A is factorized into the form $A = M D M^T$, with M orthogonal and D diagonal, then A^{-1} is also positive definite (why is it the case?) and is factorized into the form $A^{-1} = M D^{-1} M^T$.

If A and B in $\mathbb{R}^{n \times n}$ are both positive definite, from $A \succeq B$ it follows $B^{-1} \succeq A^{-1}$. Indeed,

- if $B = I_n$ and $A \succeq I_n$, the factorization $A = M D M^T$ holds, and from $M D M^T \succeq M I_n M^T$ it follows $D \succeq I_n$, hence $D^{-1} \preceq I_n$, and finally $A^{-1} = M D^{-1} M^T \preceq I_n$.
- otherwise, choose $B = N \tilde{D} N^T$, with N orthogonal and $\tilde{D}$ diagonal, and from the hypotheses it follows $N^T A N \succeq \tilde{D}$, and hence $\tilde{D}^{-\frac{1}{2}} N^T A N \tilde{D}^{-\frac{1}{2}} \succeq I_n$.

Choose now $\tilde{D}^{-\frac{1}{2}} N^T A N \tilde{D}^{-\frac{1}{2}} = M D M^T$, so that

$$\begin{aligned} D &= M^T \tilde{D}^{-\frac{1}{2}} N^T A N \tilde{D}^{-\frac{1}{2}} M \succeq I_n \\ D^{-1} &= M^T \tilde{D}^{\frac{1}{2}} N^T A^{-1} N \tilde{D}^{\frac{1}{2}} M \preceq I_n \\ M D^{-1} M^T &= \tilde{D}^{\frac{1}{2}} N^T A^{-1} N \tilde{D}^{\frac{1}{2}} \preceq I_n \\ &\quad A^{-1} \preceq N \tilde{D}^{-1} N^T = B^{-1} \end{aligned}$$

¹⁰M.Koebe, U.Schäfer *The unique square root of a positive semidefinite matrix*, IJMest, pp.990-92, 2004

¹¹For a proof, see R.Horn, C.Johnson, *Matrix Analysis*, vol. 1, pp.405-06, Cambridge U.P. (1985).

• EXERCISE A.12.2 SIMULTANEOUS DIAGONALIZATION

‡ *Solution:* let $D = \text{diag}\{\lambda_1, \lambda_2, \dots, \lambda_n\}$. Then (A.122) can be used to obtain $TD^{-\frac{1}{2}} = I_n$. Moreover, $D^{-\frac{1}{2}}MBM^TD^{-\frac{1}{2}}$ is positive semidefinite, implying that there exists an orthogonal matrix N for which it holds that $(ND^{-\frac{1}{2}}M)B(M^TD^{-\frac{1}{2}}N^T) = \text{diag}\{\mu_1, \mu_2, \dots, \mu_n\}$. This means that letting $T = M^TD^{-\frac{1}{2}}N^T$ we can obtain $T^TBT = \text{diag}\{\mu_1, \mu_2, \dots, \mu_n\}$, and $T^TAT = (ND^{-\frac{1}{2}}M)A(M^TD^{-\frac{1}{2}}N^T) = NI_nN^T = I_n$.

Note that if A is a symmetric matrix and T is an invertible matrix, the matrices A and T^TAT are said to be congruent. This means that given a positive definite matrix A and a positive semidefinite matrix B , the matrix $T = M^TD^{-\frac{1}{2}}N^T$ induces a congruency that diagonalizes both matrices.

As a last part of the exercise we let the reader check whether the matrices A and T^TAT are similar or not.

Example A.12.6 [GRAMIAN MATRIX OF n LINEARLY INDEPENDENT VECTORS] Consider $\mathbb{R}^n$ together with the scalar product $\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x}^T \mathbf{y}$ and the corresponding induced norm. Consider moreover the basis $(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n)$ and define the *Gramian matrix* $\mathcal{G} = [\langle \mathbf{v}_i, \mathbf{v}_j \rangle]_{i,j=1,\dots,n}$. If $\mathbf{x} = \sum_{i=1}^n \alpha_i \mathbf{v}_i$ is an arbitrary vector of $\mathbb{R}^n$ with coordinates that refer to the basis introduced above, then its norm can be expressed through the Gramian matrix

$$\|\mathbf{x}\|^2 = \left(\sum_{i=1}^n \alpha_i \mathbf{v}_i^T \right) \left(\sum_{i=1}^n \alpha_i \mathbf{v}_i \right) = [\alpha_1 \quad \alpha_2 \quad \dots \quad \alpha_n] \begin{bmatrix} \mathbf{v}_1^T \mathbf{v}_1 & \mathbf{v}_1^T \mathbf{v}_2 & \dots & \mathbf{v}_1^T \mathbf{v}_n \\ \mathbf{v}_2^T \mathbf{v}_1 & \mathbf{v}_2^T \mathbf{v}_2 & \dots & \mathbf{v}_2^T \mathbf{v}_n \\ \dots & \dots & \dots & \dots \\ \mathbf{v}_n^T \mathbf{v}_1 & \mathbf{v}_n^T \mathbf{v}_2 & \dots & \mathbf{v}_n^T \mathbf{v}_n \end{bmatrix} \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_n \end{bmatrix}, \quad (\text{A.123})$$

and since $\|\mathbf{x}\|$ is positive for every $\mathbf{x} \neq \mathbf{0}$, i.e., for any choice of the coordinates α_i (as soon as they all are not null) the Gramian matrix is positive definite, and all its eigenvalues are strictly positive.

If M is instead an orthogonal matrix such that $\mathcal{G} = M^T \text{diag}\{\lambda_1, \lambda_2, \dots, \lambda_n\} M$, letting

$$[\alpha_1 \quad \alpha_2 \quad \dots \quad \alpha_n] M^T = [\beta_1 \quad \beta_2 \quad \dots \quad \beta_n] \quad (\text{A.124})$$

we can rewrite (A.123) as

$$\|\mathbf{x}\|^2 = [\alpha_1 \quad \alpha_2 \quad \dots \quad \alpha_n] M^T \text{diag}\{\lambda_1, \lambda_2, \dots, \lambda_n\} M \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_n \end{bmatrix} = \sum_{i=1}^n \lambda_i \beta_i^2 \quad (\text{A.125})$$

Considering that $M^T M = I_n$, following from (A.124) one has that $\sum_{i=1}^n \alpha_i^2 = \sum_{i=1}^n \beta_i^2$, and from (A.125) it follows that

$$\min_i(\lambda_i) \sum_{i=1}^n \alpha_i^2 \leq \|\mathbf{x}\|_2^2 \leq \max_i(\lambda_i) \sum_{i=1}^n \alpha_i^2. \quad (\text{A.126})$$

Since $\mathbf{x}$ belongs to the sphere $S(\mathbf{0}, r)$, it holds that

$$\sum_{i=1}^n \alpha_i^2 < \frac{r^2}{\max_i(\lambda_i)} \Rightarrow \mathbf{x} \in S(\mathbf{0}, r), \quad \mathbf{x} \in S(\mathbf{0}, r) \Rightarrow \sum_{i=1}^n \alpha_i^2 < \frac{r^2}{\min_i(\lambda_i)}. \quad (\text{A.127})$$

Note that the first condition is a “sufficient” condition, while the second is a “necessary” one.

Example A.12.7 [IDEMPOTENT MATRICES, PROJECTORS, AND ORTHOGONAL PROJECTORS] Let $A \in \mathbb{R}^{n \times n}$. Then:

(i) A is an *idempotent matrix* if $A = A^2$. Note that A is idempotent if and only if $I_n - A$ is idempotent, too. It can also be proven that the only possible eigenvalues for an idempotent matrix are 0 and 1. The only idempotent matrix that is also invertible is the identity: indeed from $A^2 = A$ and $A^{-1}A = I_n$ it follows that $I = A^{-1}A = A^{-1}A^2 = A$.

(ii) A is a *projector*, a.k.a. a *projection matrix* on the subspace $\mathbf{U}$ of $\mathbb{R}^n$ if $A\mathbf{v} \in \mathbf{U}, \forall \mathbf{v} \in \mathbb{R}^n$, and $A\mathbf{u} = \mathbf{u}, \forall \mathbf{u} \in \mathbf{U}$. If $\mathbf{U}$ has a basis $(\mathbf{u}_1, \dots, \mathbf{u}_k)$ and if $(\mathbf{u}_1, \dots, \mathbf{u}_k, \mathbf{v}_{k+1}, \dots, \mathbf{v}_n)$ is a basis of $\mathbb{R}^n$, then for each choice of vectors $\mathbf{u}_{k+1}, \dots, \mathbf{u}_n$ in $\mathbf{U}$ it holds that

$$A = [\mathbf{u}_1 \ \dots \ \mathbf{u}_k \ \mathbf{v}_{k+1} \ \dots \ \mathbf{v}_n]^{-1} [\mathbf{u}_1 \ \dots \ \mathbf{u}_k \ \mathbf{u}_{k+1} \ \dots \ \mathbf{u}_n]$$

is a projection matrix on $\mathbf{U}$.

If the columns of $M_1 := [\mathbf{u}_1 \ \dots \ \mathbf{u}_k]$ and of $[M_1 \ M_2] := [\mathbf{u}_1 \ \dots \ \mathbf{u}_k \ \mathbf{v}_{k+1} \ \dots \ \mathbf{v}_n]$ are an orthonormal basis for $\mathbf{U}$ and $\mathbb{R}^n$, respectively, then

$$A = [M_1 \ M_2]^{-1} [M_1 \ M_1 M_1^T M_2] = [M_1 \ M_2]^T [M_1 \ M_1 M_1^T M_2] = \begin{bmatrix} I_k & M_1^T M_2 \\ M_2^T M_1 & M_2^T M_1 M_1^T M_2 \end{bmatrix}$$

is a symmetric projection matrix on $\mathbf{U}$.

Conversely, if A is a projection matrix on $\mathbf{U}$, then for every $\mathbf{v} \in \mathbb{R}^n$ it holds that $A^2\mathbf{v} = A(A\mathbf{v}) = A\mathbf{v}$, and thus $A^2 = A$, implying that A is idempotent. On the other hand, if A is idempotent and we let $\mathbf{U} = \text{Im}(A)$, then for every $A\mathbf{v} \in \text{Im}A$ it holds that $A(A\mathbf{v}) = A^2\mathbf{v} = A\mathbf{v}$ and A is a projection matrix on $\text{Im}(A)$. This implies that a matrix is idempotent if and only if it is a projection matrix.

(iii) **Decomposing $\mathbb{R}^n$.** If A is a projection matrix on $\text{Im}(A)$, then $I_n - A$ is the corresponding projection matrix on $\text{Im}(I_n - A)$, and on $\ker(A)$: indeed $\mathbf{v} \in \ker(A)$ implies $(I_n - A)\mathbf{v} = \mathbf{v}$, while $\mathbf{v} \in \mathbb{R}^n$ implies $A(I_n - A)\mathbf{v} = \mathbf{0}$ (i.e., $(I_n - A)\mathbf{v} \in \ker(A)$). Thus $\text{Im}(I_n - A) = \ker(A)$. If $\mathbf{v} \in \ker(A) \cap \text{Im}(A)$ then $(I_n - A)\mathbf{v} = \mathbf{v} = \mathbf{0}$, thus $\ker(A) \cap \text{Im}(A) = \{\mathbf{0}\}$ and

$$\mathbb{R}^n = \ker(A) \oplus \text{Im}(A) = \text{Im}(I_n - A) \oplus \text{Im}(A).$$

(iv) **Uniqueness of the symmetric projection.** A part from the trivial cases $\mathbf{U} = \mathbb{R}^n$ and $\mathbf{U} = \{\mathbf{0}\}$, the subspace $\mathbf{U}$ in general does not uniquely define the projection matrix A on $\mathbf{U}$. However, it does, if one constrains A to be a symmetric projection matrix. Indeed, if $A = A^T \in \mathbb{R}^n$ and $B = B^T \in \mathbb{R}^n$ are both projection matrices on $\mathbf{U}$, then for every $\mathbf{v} \in \mathbb{R}^n$ it holds that $AB\mathbf{v} = B\mathbf{v}$ and $BA\mathbf{v} = A\mathbf{v}$ and thus, simultaneously, $AB = B$ and $BA = A$. This however implies also that

$$(A - B)^T(A - B) = A^T A - A^T B - B^T A + B^T B = AA - AB - BA + BB = A - B - A + B = 0$$

and thus that $A - B = 0$. If A is idempotent and symmetric, then for each $\mathbf{v}, \mathbf{w} \in \mathbb{R}^n$ it holds that $\langle A\mathbf{v}, (I_n - A)\mathbf{w} \rangle = \mathbf{v}^T A(I_n - A)\mathbf{w} = \mathbf{v}^T (A - A^2)\mathbf{w} = \mathbf{0}$, and hence $\text{Im}(A) \perp \text{Im}(I_n - A)$. The decomposition of every vector $\mathbf{v}$ as

$$\mathbf{v} = A\mathbf{v} + (I_n - A)\mathbf{v} = \mathbf{v}_{\text{Im}(A)} + \mathbf{v}_{\text{Im}(A)^\perp}$$

then ensures that $A\mathbf{v}$ is the orthogonal projection of $\mathbf{v}$ on the subspace $\text{Im}(A)$. On the contrary, if A induces an orthogonal projection on a subspace $\mathbf{U}$ of $\mathbb{R}^n$, then A is idempotent and symmetric. Indeed the orthogonality of the projection implies that, for each $\mathbf{v} \in \mathbb{R}^n$ it holds that $(I_n - A)\mathbf{v} = \mathbf{v} - A\mathbf{v} \in \mathbf{U}^\perp$, which implies $\text{Im}(A) = \mathbf{U}$ and $\text{Im}(I_n - A) = \mathbf{U}^\perp$ and, for every i and j ,

$$0 = \langle A\mathbf{e}_i, (I_n - A)\mathbf{e}_j \rangle = \mathbf{e}_i^T A^T (I_n - A)\mathbf{e}_j = \mathbf{e}_i^T A^T \mathbf{e}_j - \mathbf{e}_i^T A^T A\mathbf{e}_j.$$

Exchanging $\mathbf{e}_i$ with $\mathbf{e}_j$ it is thus possible to obtain $0 = \mathbf{e}_j^T A^T \mathbf{e}_i - \mathbf{e}_j^T A^T A\mathbf{e}_i$ and, considering also the previous identity, $\mathbf{e}_j^T A^T \mathbf{e}_i = \mathbf{e}_i^T A^T \mathbf{e}_j$. This means that A is symmetric.

If $A \in \mathbb{R}^{p \times m}$ is a rectangular matrix, (A.102) describes how to derive its operator norm (a.k.a. spectral norm) in ℓ_2 :

$$\|A\|_2 = \sup_{\mathbf{v} \neq \mathbf{0}} \left(\frac{\mathbf{v}^T A^T A \mathbf{v}}{\mathbf{v}^T \mathbf{v}} \right)^{1/2}.$$

The matrix $A^T A$, whose dimension is $m \times m$, is symmetric and positive semidefinite, thus diagonalizable by similarity with an orthogonal matrix $M = (M^T)^{-1}$. This implies that

$A^T A$ has real non-negative eigenvalues. Letting then σ_i be the arithmetic roots of such eigenvalues, it is possible to obtain

$$M^T A^T A M = \text{diag}\{\sigma_1^2, \sigma_2^2, \dots, \sigma_m^2\} \quad (\text{A.128})$$

and, letting $\mathbf{v} = M\mathbf{w}$,

$$\begin{aligned} \|A\|_2 &= \sup_{\mathbf{v} \neq 0} \left(\frac{\mathbf{v}^T A^T A \mathbf{v}}{\mathbf{v}^T \mathbf{v}} \right)^{1/2} = \sup_{\mathbf{w} \neq 0} \left(\frac{\mathbf{w}^T M^T (A^T A) M \mathbf{w}}{\mathbf{w}^T M^T M \mathbf{w}} \right)^{1/2} \\ &= \sup_{\mathbf{w} \neq 0} \left(\frac{\mathbf{w}^T \text{diag}\{\sigma_1^2, \sigma_2^2, \dots, \sigma_m^2\} \mathbf{w}}{\mathbf{w}^T \mathbf{w}} \right)^{1/2} = \sup_{\sum_{i=1}^m w_i^2 = 1} \left(\sum_{i=1}^m \sigma_i^2 w_i^2 \right)^{1/2} = \max_{i=1,2,\dots,m} \{\sigma_i\}. \end{aligned} \quad (\text{A.129})$$

The eigenvalues σ_i^2 may always be thought sorted so that their values are decreasing, i.e., $\sigma_1^2 \geq \sigma_2^2 \geq \dots \geq \sigma_m^2$, provided that one opportunely re-sort the columns of M . In this case it holds that:

- $\|A\|_2 = \sigma_1$;
- if $r = \text{rank}(A) = \text{rank}(A^T A)$, then $\sigma_{r+1} = \dots = \sigma_m = 0$;
- if $M = [M_1 \ M_2]$, with $M_1 \in \mathbb{R}^{m \times r}$, then (A.128) implies both $AM_2 = 0$ and $M_1^T A^T A M_1 = \text{diag}\{\sigma_1^2, \dots, \sigma_r^2\}$.

From the last point it follows that

$$\underbrace{\text{diag}\{\sigma_1^{-1}, \dots, \sigma_r^{-1}\} M_1^T A^T}_{U_1^T} \underbrace{A M_1 \text{diag}\{\sigma_1^{-1}, \dots, \sigma_r^{-1}\}}_{U_1} = I_r = U_1^T U_1.$$

Postmultiplying, we then obtain $AM = A[M_1 \ M_2] = [U_1 \text{diag}\{\sigma_1, \dots, \sigma_r\} \ 0]$ for M^T , and

$$A = [U_1 \text{diag}\{\sigma_1, \dots, \sigma_r\} \ 0] M^T = U_1 \text{diag}\{\sigma_1, \dots, \sigma_r\} M_1^T. \quad (\text{A.130})$$

Since the r columns of U_1 are a set of orthonormal vectors in $\mathbb{R}^p$, we can complete the rectangular matrix U_1 to form an orthogonal matrix $U = [U_1 \ U_2]$ by adding suitable $p - r$ columns. In this case (A.130) becomes

$$A = U \text{diag}\{\sigma_1, \dots, \sigma_r, 0, \dots, 0\} M^T. \quad (\text{A.131})$$

In other words, in this way we proved the following:

Proposition A.12.8 [SINGULAR VALUES DECOMPOSITION] *Let $A \in \mathbb{R}^{p \times m}$ be of rank r . Then there exist two orthogonal matrices $U \in \mathbb{R}^{p \times p}$ and $M \in \mathbb{R}^{m \times m}$ for which*

- *A factorizes as $A = U \text{diag}\{\sigma_1, \dots, \sigma_r, 0, \dots, 0\} M^T$;*
- *the positive numbers $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_r$, also called the singular values of A , are the arithmetic roots of the non-null eigenvalues of $A^T A$.*

As we saw, the spectral norm of A is the largest of its singular values, i.e., the largest among the square roots of the eigenvalues of $A^T A$.

If $\mathbf{V}$ is a vector space over the complex field $\mathbb{C}$, then the inner product can be defined substituting in definition A.12.1 condition (i) about the symmetry of the operation with the corresponding condition

$$\text{i')} \quad \langle \mathbf{v}_1, \mathbf{v}_2 \rangle = \langle \mathbf{v}_2, \mathbf{v}_1 \rangle^*.$$

Note that it is here necessary to use the conjugacy in order to keep the overall product positive definite: if $\langle \mathbf{v}, \mathbf{v} \rangle$ is a positive real number, then with condition (i) one would obtain $\langle j\mathbf{v}, j\mathbf{v} \rangle = -\langle \mathbf{v}, \mathbf{v} \rangle < 0$.

Notation-wise, a complex vector space endowed with an inner product is called a *unitary space*.

Definition A.12.9 A matrix $M \in \mathbb{C}^{n \times n}$ is called

normal if $\check{M}^T M = M \check{M}^T$ (in the real case, a matrix is normal if and only if it commutes with its transposed);

Hermitian if $M = \check{M}^T$ (in the real case, a matrix is Hermitian if and only if it is symmetric);

unitary if its columns define an orthonormal basis of $\mathbb{C}^n$. Thus M is unitary if $\check{M}^T M = M \check{M}^T = I_n$ (in the real case, a matrix is unitary if and only if it is orthogonal).

It is immediate to verify that Hermitian and unitary matrices are also normal.

The definitions and the results that we obtained in this paragraph for the Euclidean spaces and for the matrices over the real field can be extended to generic unitary spaces and to matrices over the complex field. In the various proofs and definitions it is indeed enough to substitute symmetric matrices with Hermitian ones, and orthogonal matrices with unitary ones. We let the reader develop further the details as an exercise.

Proposition A.12.10 [TOEPLITZ THEOREM] Let $M \in \mathbb{C}^{n \times n}$. Then

- i) M is normal if and only if there exists a unitary matrix U that reduces it by similarity to a diagonal matrix;
- ii) M is Hermitian if and only if the corresponding form is real;
- iii) M is unitary if and only if the diagonal elements of such a form have unitary modulus.

A.13 Adjoint transformation

To introduce the notion of adjoint transformation, it is convenient to consider some elementary facts about a linear transformation among real finite dimensional spaces.

Let $\mathbb{R}^m$ and $\mathbb{R}^p$ be both endowed with the usual inner product $\langle \mathbf{u}, \mathbf{v} \rangle := \mathbf{u}^T \mathbf{v}$ and let A be a matrix in $\mathbb{R}^{p \times m}$. This latter induces a linear transformation from $\mathbb{R}^m$ to $\mathbb{R}^p$, whose image space is generated by the columns of A .

The null space of the transformation is constituted by the columns of $\mathbb{R}^m$ whose pre-multiplication by the rows of A gives a null result, that is by the $\mathbb{R}^m$ vectors orthogonal

to the columns of A^T , namely by the vectors of the image space of A^T . We have thus verified that

$$\ker A = (\operatorname{Im} A^T)^\perp. \quad (\text{A.132})$$

The finite dimension hypothesis implies then

$$(\ker A)^\perp = \operatorname{Im} A^T \quad (\text{A.133})$$

and, exchanging the roles of A and A^T ,

$$(\operatorname{Im} A)^\perp = \ker A^T. \quad (\text{A.134})$$

From the previous relations, one gets that the image and the null space of the matrix A^T , acting from $\mathbb{R}^p$ to $\mathbb{R}^m$, can be obtained by orthogonality relations from the null space and the image of A , acting from $\mathbb{R}^m$ to $\mathbb{R}^p$.

Note that, with respect to the inner products of $\mathbb{R}^m$ and $\mathbb{R}^p$, the matrices A and A^T satisfy the equations

$$\langle A\mathbf{v}, \mathbf{w} \rangle_{\mathbb{R}^p} = (A\mathbf{v})^T \mathbf{w} = \mathbf{v}^T (A^T \mathbf{w}) = \langle \mathbf{v}, A^T \mathbf{w} \rangle_{\mathbb{R}^m} \quad (\text{A.135})$$

for all $\mathbf{v} \in \mathbb{R}^m$ and $\mathbf{w} \in \mathbb{R}^p$.

- EXERCISE A.13.1 Let $A \in \mathbb{R}^{p \times m}$. The map induced by the matrix, $A : \mathbb{R}^m \rightarrow \mathbb{R}^p$, when it is restricted to the subspace $\operatorname{Im} A^T \subseteq \mathbb{R}^m$, i.e., $A : \operatorname{Im} A^T \rightarrow \operatorname{Im} A : \mathbf{v} \mapsto A\mathbf{v}$, is a bijection.

The aforementioned facts can be generalized also to transformations between infinite dimensional spaces, by introducing the notion of adjoint transformation.

Definition A.13.1 [ADJOINT TRANSFORMATION] *Let $\mathbf{V}$ and $\mathbf{W}$ be two vector spaces over the real field, with inner products $\langle \cdot, \cdot \rangle_{\mathbf{W}}$ and $\langle \cdot, \cdot \rangle_{\mathbf{V}}$, and consider the linear transformation*

$$\mathcal{A} : \mathbf{V} \rightarrow \mathbf{W}.$$

A linear transformation

$$\mathcal{A}^* : \mathbf{W} \rightarrow \mathbf{V}$$

is called adjoint of $\mathcal{A}$ if, for any choice of $\mathbf{v} \in \mathbf{V}$ and of $\mathbf{w} \in \mathbf{W}$, it results

$$\langle \mathcal{A}\mathbf{v}, \mathbf{w} \rangle_{\mathbf{W}} = \langle \mathbf{v}, \mathcal{A}^*\mathbf{w} \rangle_{\mathbf{V}}. \quad (\text{A.136})$$

If there exists an adjoint transformation of $\mathcal{A}$, this is unique. Indeed, if $\mathcal{A}^*$ and $\mathcal{A}^\bullet$ are two adjoint transformations of $\mathcal{A}$, then, for any $\mathbf{v} \in \mathbf{V}$ and any $\mathbf{w} \in \mathbf{W}$

$$\begin{aligned} \langle \mathbf{v}, \mathcal{A}^*\mathbf{w} \rangle &= \langle \mathbf{v}, \mathcal{A}^\bullet\mathbf{w} \rangle = \langle \mathcal{A}\mathbf{v}, \mathbf{w} \rangle \\ \langle \mathbf{v}, \mathcal{A}^*\mathbf{w} - \mathcal{A}^\bullet\mathbf{w} \rangle &= 0 \end{aligned}$$

then, for all $\mathbf{w} \in \mathbf{W}$,

$$\mathcal{A}^*\mathbf{w} - \mathcal{A}^\bullet\mathbf{w} = 0$$

which proved the equality $\mathcal{A}^* = \mathcal{A}^\bullet$.

Example A.13.1 Set $A : \mathbb{R}^m \rightarrow \mathbb{R}^p : \mathbf{v} \mapsto A\mathbf{v}$. According to (A.135) it follows that $\langle A\mathbf{v}, \mathbf{w} \rangle_{\mathbb{R}^p} = \langle \mathbf{v}, A^T \mathbf{w} \rangle_{\mathbb{R}^m}$ and then, by the definition of adjoint, $A^* = A^T$.

Example A.13.2 Let $\mathcal{U}$ be the space of piecewise continuous functions on the interval $[\tau, t]$, taking values in $\mathbb{R}^m$, and consider the linear transformation

$$\mathcal{A} : \mathcal{U} \rightarrow \mathbb{R}^p : \mathbf{u}(\cdot) \mapsto \int_{\tau}^t M(\sigma) \mathbf{u}(\sigma) d\sigma$$

where $M(\cdot)$ is a $p \times m$ matrix whose elements are continuous functions on $[\tau, t]$. Assume that the inner product in $\mathcal{U}$ is defined by the relation

$$\langle \mathbf{u}(\cdot), \mathbf{v}(\cdot) \rangle_{\mathcal{U}} := \int_{\tau}^t \mathbf{u}^T(\sigma) \mathbf{v}(\sigma) d\sigma$$

and in $\mathbb{R}^p$ it is the usual inner product. For any $\mathbf{w} \in \mathbb{R}^p$ and any $\mathbf{u}(\cdot) \in \mathcal{U}$ it is then

$$\langle \mathcal{A}\mathbf{u}(\cdot), \mathbf{w} \rangle_{\mathbb{R}^p} = \left(\int_{\tau}^t M(\sigma) \mathbf{u}(\sigma) d\sigma \right)^T \mathbf{w} = \int_{\tau}^t \mathbf{u}^T(\sigma) M^T(\sigma) \mathbf{w} d\sigma = \langle \mathbf{u}(\cdot), M^T(\cdot) \mathbf{w} \rangle_{\mathcal{U}}.$$

It follows that the linear transformation

$$\mathcal{A}^* : \mathbb{R}^p \rightarrow \mathcal{U} : \mathbf{w} \mapsto M^T(\cdot) \mathbf{w}$$

is the adjoint of $\mathcal{A}$.

The linear transformations for which the adjoint exists have properties that generalize the corresponding properties of the matrix case:

- i) $(\mathcal{A}^*)^* = \mathcal{A}$;
- ii) if $\mathcal{A} : \mathbf{V} \rightarrow \mathbf{W}$ and $\mathcal{B} : \mathbf{W} \rightarrow \mathbf{U}$ then

$$(\mathcal{B}\mathcal{A})^* = \mathcal{A}^* \mathcal{B}^*;$$

- iii) if $\mathcal{A}$ has the inverse, $(\mathcal{A}^{-1})^* = (\mathcal{A}^*)^{-1}$;
- iv) $(\mathcal{A}^* \mathcal{A})^* = \mathcal{A}^* \mathcal{A}$ and $(\mathcal{A} \mathcal{A}^*)^* = \mathcal{A} \mathcal{A}^*$.

A transformation that coincides with its own adjoint is called *self-adjoint*: hence, $\mathcal{A}^* \mathcal{A}$ and $\mathcal{A} \mathcal{A}^*$ are self-adjoint.

- v) $\ker \mathcal{A} = (\operatorname{Im} \mathcal{A}^*)^{\perp}$ and $\ker \mathcal{A}^* = (\operatorname{Im} \mathcal{A})^{\perp}$;
- vi) $\operatorname{Im} \mathcal{A} \subseteq (\ker \mathcal{A}^*)^{\perp}$ and $\operatorname{Im} \mathcal{A}^* \subseteq (\ker \mathcal{A})^{\perp}$;
- vi') if $\operatorname{Im} \mathcal{A}$ has finite dimension, then $\operatorname{Im} \mathcal{A} = (\ker \mathcal{A}^*)^{\perp}$;
if $\operatorname{Im} \mathcal{A}^*$ has finite dimension, then $\operatorname{Im} \mathcal{A}^* = (\ker \mathcal{A})^{\perp}$;
- vii) $\ker \mathcal{A} = \ker(\mathcal{A}^* \mathcal{A})$;
- viii) $(\operatorname{Im} \mathcal{A})^{\perp} = (\operatorname{Im}(\mathcal{A} \mathcal{A}^*))^{\perp}$;
- viii') if $\operatorname{Im} \mathcal{A}$ has finite dimension, then $\operatorname{Im} \mathcal{A} = \operatorname{Im}(\mathcal{A} \mathcal{A}^*)$.

PROOF of (v) $\mathbf{v} \in \ker \mathcal{A} \Leftrightarrow \mathcal{A}\mathbf{v} = \mathbf{0} \Leftrightarrow \langle \mathcal{A}\mathbf{v}, \mathbf{w} \rangle = 0, \forall \mathbf{w} \in \mathbf{W} \Leftrightarrow \langle \mathbf{v}, \mathcal{A}^* \mathbf{w} \rangle = 0, \forall \mathbf{w} \in \mathbf{W} \Leftrightarrow \mathbf{v} \in (\text{Im} \mathcal{A}^*)^\perp$.

PROOF of (vi) and (vi') $\text{Im} \mathcal{A} \subseteq ((\text{Im} \mathcal{A})^\perp)^\perp = (\ker \mathcal{A}^*)^\perp$.

If $\text{Im} \mathcal{A}$ has finite dimension, the inclusion $\text{Im} \mathcal{A} \subseteq ((\text{Im} \mathcal{A})^\perp)^\perp$ stands as an equality relation.

PROOF of (vii) $\mathbf{v} \in \ker \mathcal{A} \Leftrightarrow \mathcal{A}\mathbf{v} = \mathbf{0} \Rightarrow \mathcal{A}^* \mathcal{A}\mathbf{v} = \mathbf{0} \Leftrightarrow \mathbf{v} \in \ker(\mathcal{A}^* \mathcal{A})$.

Viceversa, $\mathbf{v} \in \ker(\mathcal{A}^* \mathcal{A}) \Leftrightarrow \mathcal{A}^* \mathcal{A}\mathbf{v} = \mathbf{0} \Rightarrow \langle \mathbf{v}, \mathcal{A}^* \mathcal{A}\mathbf{v} \rangle = 0 \Leftrightarrow \langle \mathcal{A}\mathbf{v}, \mathcal{A}\mathbf{v} \rangle = 0 \Leftrightarrow \mathcal{A}\mathbf{v} = \mathbf{0}$.

PROOF of (viii) and (viii') $(\text{Im} \mathcal{A})^\perp = \ker \mathcal{A}^* = \ker((\mathcal{A}^*)^* \mathcal{A}^*) = \ker(\mathcal{A} \mathcal{A}^*) = (\text{Im}(\mathcal{A} \mathcal{A}^*))^\perp$.

If $\text{Im} \mathcal{A}$ has finite dimension, then $\text{Im} \mathcal{A} = ((\text{Im} \mathcal{A})^\perp)^\perp$ and, by considering the orthogonal spaces of the first and the last elements of the previous equality, $\text{Im} \mathcal{A} = ((\text{Im} \mathcal{A})^\perp)^\perp = ((\text{Im}(\mathcal{A} \mathcal{A}^*))^\perp)^\perp = \text{Im}(\mathcal{A} \mathcal{A}^*)$. ■

Resorting also to some properties of the transpose matrix (that are the same, as we know, than those of the adjoint transformation when we operate with finite dimensional spaces) we will discuss about some characteristics of the pseudoinverse matrices and their application to the study of linear equation systems.

Proposition A.13.2 [PSEUDOINVERSES AND MOORE-PENROSE PSEUDOINVERSE] *Let $A \in \mathbb{R}^{p \times m}$ be a matrix with rank $r > 0$.*

- (i) *There exist $X \in \mathbb{R}^{p \times r}$ and $Y \in \mathbb{R}^{r \times m}$ of rank r , satisfying the factorization $A = XY$.*
- (ii) *Matrix $X^T A Y^T = (X^T X)(Y Y^T) \in \mathbb{R}^{r \times r}$ is invertible.*
- (iii) *Matrix $Z \in \mathbb{R}^{p \times m}$ defined as*

$$Z := Y^T (X^T A Y^T)^{-1} X^T = Y^T (Y Y^T)^{-1} (X^T X)^{-1} X^T \quad (\text{A.137})$$

satisfies the following conditions (Penrose's equations¹²)

- (a) $AZA = A$
- (b) $ZAZ = Z$
- (c) $(AZ)^T = AZ$
- (d) $(ZA)^T = ZA$

- (iv) *If $Z \in \mathbb{R}^{m \times p}$ satisfies conditions (a),(b),(c),(d), it coincides with (A.137).*

The matrix defined by the right hand side of (A.137) - the only one that simultaneously satisfies the set of four Penrose's equations - is called "Moore-Penrose pseudoinverse" of A and is denoted by the symbol $A^\dagger$.

PROOF (i) Let us build X by choosing r independent columns of A . The i -th column of A can be expressed in a unique way as a combination of columns of X : the coefficients constitute the i -th column of Y and Y is necessarily of rank r .

(ii) By property (viii) of the adjoint matrix, $Y Y^T$ has the same rank as Y , therefore it is invertible; by property (vii), $X^T X$ has empty kernel, like X , hence it is invertible.

(iii) Straightforward verification.

(iv) If Z and $\tilde{Z}$ satisfy, respectively,

$$\begin{array}{ll} (a) & AZA = A \\ (b) & ZAZ = Z \\ (c) & (AZ)^T = AZ \\ (d) & (ZA)^T = ZA \end{array} \quad \begin{array}{ll} (\tilde{a}) & A\tilde{Z}A = A \\ (\tilde{b}) & \tilde{Z}A\tilde{Z} = \tilde{Z} \\ (\tilde{c}) & (A\tilde{Z})^T = A\tilde{Z} \\ (\tilde{d}) & (\tilde{Z}A)^T = \tilde{Z}A \end{array}$$

¹²These are always simultaneously satisfied when A is an invertible matrix and Z is its inverse.

then

$$\begin{aligned}
 Z &=_{(b)} ZAZ =_{(d)} Z(AZ)^T = ZZ^T Z^T =_{(\tilde{a})} Z(Z^T A^T)(\tilde{Z}^T AT) = Z(AZ)^T (A\tilde{Z})^T \\
 &=_{(d),(\tilde{d})} Z(AZ)(A\tilde{Z}) =_{(b)} ZA\tilde{Z} =_{(\tilde{b})} (ZA)\tilde{Z}A\tilde{Z} =_{(c),(\tilde{c})} (ZA)^T (\tilde{Z}A)^T \tilde{Z} = (A^T Z^T A^T) \tilde{Z}^T \tilde{Z} \\
 &=_{(a)} A^T \tilde{Z}^T \tilde{Z} = (\tilde{Z}A)^T \tilde{Z} =_{(\tilde{c})} \tilde{Z}A\tilde{Z} =_{(\tilde{b})} \tilde{Z}
 \end{aligned}
 \quad \blacksquare$$

In general, if Z does not satisfy all four Penrose's equations, but only some of them, the uniqueness of their solution is no longer guaranteed.

- EXERCISE A.13.2 If $A \in \mathbb{R}^{p \times m}$ is the null matrix, the only matrix $Z \in \mathbb{R}^{m \times p}$ that satisfies all four Penrose's equations is the null matrix.
- EXERCISE A.13.3 Let $A \in \mathbb{R}^{p \times m}$. If $Z \in \mathbb{R}^{m \times p}$ satisfies the Penrose's equation (a), then AZ is a projection matrix onto $\text{Im}(A)$, and an orthogonal projection matrix if Z satisfies also equation (c).
 $\sharp$ *Solution.* $AZA = A$ implies that AZ is idempotent, and hence it is a projection matrix. Moreover, for any $\mathbf{v} \in \mathbb{R}^m$, it is $AZ\mathbf{v} \in \text{Im}(A)$ and $AZ(A\mathbf{v}) = A\mathbf{v}$. Consequently, AZ is a projector onto $\text{Im}(A)$. If also equation (c) stands, $AZ = (AZ)^T$ is symmetric, therefore point (iv) of Example A.12.7 is true and $AZ\mathbf{v}$ is the orthogonal projection of $\mathbf{v}$ onto $\text{Im}(A)$.
- EXERCISE A.13.4 Let $A \in \mathbb{R}^p \times m$ be a matrix of rank $r > 0$, $A = XY$ a factorization of A with $X \in \mathbb{R}^{p \times r}$ and $Y \in \mathbb{R}^{r \times m}$, $A^\dagger \in \mathbb{R}^{m \times p}$ the Moore-Penrose inverse of A . Then,
 i) $YA^\dagger X = I_r$ and $A^\dagger$ has rank r ;
 ii) $\text{Im}(A^\dagger) = \text{Im}(Y^T) = \text{Im}(A^T) = (\ker A)^\perp$ and $\ker(A^\dagger) = \ker(X^T) = \ker(A^T) = (\text{Im } A)^\perp$.
 $\sharp$ *Solution.* (i) Apply (A.137).
 ii) $\text{Im}(A^\dagger) = \text{Im}(Y^T(YY^T)^{-1}(X^T X)^{-1}X^T)$ and $\text{Im}(A^T) = \text{Im}(Y^T X^T)$ are both subspaces of $\text{Im}(Y^T)$, having the same dimension r . According to the second chain of equalities, $\ker(A^\dagger) = \ker(Y^T(YY^T)^{-1}(X^T X)^{-1}X^T)$ and $\ker A^T = \ker(Y^T X^T)$ both include $\ker(X^T)$, and have the same dimension $n - r$.

Proposition A.13.3 [SOLUTION TO A LINEAR SYSTEM AND PSEUDOINVERSE MATRICES]
 Consider the system of equations

$$A\mathbf{x} = \mathbf{b}, \quad A \in \mathbb{R}^{p \times m} \quad (\text{A.138})$$

in the unknown $\mathbf{x} \in \mathbb{R}^m$.

- i) Let $Z \in \mathbb{R}^{m \times p}$ be a solution of Penrose's equation (a), $AZA = A$. Then,
 - $A\mathbf{x} = \mathbf{b}$ has solution if and only if vector $\mathbf{b}$ satisfies condition $AZ\mathbf{b} = \mathbf{b}$,
 - if $AZ\mathbf{b} = \mathbf{b}$, then $Z\mathbf{b}$ is a particular solution of (A.138); if $\bar{\mathbf{x}}$ is an arbitrary particular solution of (A.138), its general solution is

$$\mathbf{x} = \bar{\mathbf{x}} + (I_m - ZA)\mathbf{y}, \quad \forall \mathbf{y} \in \mathbb{R}^m. \quad (\text{A.139})$$

- ii) Let $Z \in \mathbb{R}^{m \times p}$ be a solution of Penrose's equations (a) and (d), $AZA = A$ and $ZA = (ZA)^T$. Then, if $AZ\mathbf{b} = \mathbf{b}$, vector $\mathbf{x} = Z\mathbf{b}$ is the minimum norm solution of the system $A\mathbf{x} = \mathbf{b}$.
- iii) Let $Z \in \mathbb{R}^{m \times p}$ be a solution of the Penrose's equations (a) and (c), $AZA = A$ and $AZ = (AZ)^T$. Then, vector $\mathbf{x} = Z\mathbf{b}$ minimizes $\|A\mathbf{x} - \mathbf{b}\|$.
- iv) If $A^\dagger$ is the Moore-Penrose pseudoinverse of A , for any $\mathbf{b} \in \mathbb{R}^m$ the vector $\mathbf{x} = A^\dagger \mathbf{b}$ is, among the vectors that minimize $\|A\mathbf{x} - \mathbf{b}\|$, that of minimum norm.

PROOF (i) If system (A.138) admits a solution, $\mathbf{b}$ belongs to $\text{Im}(A)$, that is $\mathbf{b} = A\mathbf{h}$, $\exists \mathbf{h} \in \mathbb{R}^m$. Then, $AZ\mathbf{b} = AZA\mathbf{h} = A\mathbf{h} = \mathbf{b}$. Viceversa, from $AZ\mathbf{b} = \mathbf{b}$ it follows that $\mathbf{x} = Z\mathbf{b}$ is a solution of (A.138).

By left-multiplying by the matrix A both sides of (A.139), it follows that all elements $\mathbf{x}$ obtained by varying $\mathbf{y}$ are solutions of (A.138). Conversely, if $\mathbf{x}$ is a solution of (A.138), it satisfies $A\mathbf{x} = \mathbf{b}$, hence $\mathbf{x} = ZAx + (I_m - ZA)\mathbf{x} = Z\mathbf{b} + (I_m - ZA)\mathbf{x}$ is given by (A.139).

ii) If $AZ\mathbf{b} = \mathbf{b}$, vector $Z\mathbf{b}$ is a particular solution and $\mathbf{x} = Z\mathbf{b} + (I_m - ZA)\mathbf{y}$ is the general solution of (A.138). Since ZA is idempotent and symmetric, according to Example A.12.7.(iv) the subspaces $\text{Im}(ZA)$ and $\text{Im}(I_m - ZA)$ are orthogonal, hence $Z\mathbf{b} = Z(AZ\mathbf{b}) = ZA(Z\mathbf{b})$ is orthogonal to $(I_m - ZA)\mathbf{y}$ and

$$\|Z\mathbf{b} + (I_m - ZA)\mathbf{y}\|^2 = \|Z\mathbf{b}\|^2 + \|(I_m - ZA)\mathbf{y}\|^2 \geq \|Z\mathbf{b}\|^2, \quad \forall \mathbf{y} \in \mathbb{R}^m.$$

iii) Since AZ is idempotent and symmetric, the subspaces $\text{Im}(AZ)$ and $\text{Im}(I_p - AZ)$ are orthogonal. For any choice of $\mathbf{x} \in \mathbb{R}^m$, in the decomposition $A\mathbf{x} - \mathbf{b} = AZA\mathbf{x} - AZ\mathbf{b} - (I_p - AZ)\mathbf{b}$ the vectors $(I_p - AZ)\mathbf{b}$ and $AZ(A\mathbf{x}) - AZ\mathbf{b}$ are orthogonal. The choice $\mathbf{x} = Z\mathbf{b}$, makes the second vector null and, by Pythagorean theorem, minimizes the quantity $\|A\mathbf{x} - \mathbf{b}\|$.

iv) $A^\dagger$ is a solution of Penrose's equations (a) and (c), and hence $\mathbf{b} = AA^\dagger\mathbf{b} + (I_p - AA^\dagger)\mathbf{b}$ provides a decomposition of $\mathbf{b}$ into the subspaces $\text{Im}(A)$ and $(\text{Im}(A))^\perp$. To minimize

$$\|A\mathbf{x} - \mathbf{b}\|^2 = \|A\mathbf{x} - AA^\dagger\mathbf{b} - (I_p - AA^\dagger)\mathbf{b}\|^2 = \|A\mathbf{x} - AA^\dagger\mathbf{b}\|^2 + \|(I_p - AA^\dagger)\mathbf{b}\|^2,$$

it is necessary and sufficient to choose a vector $\mathbf{x}$ that satisfies the equation (which obviously admits solution) $A\mathbf{x} = AA^\dagger\mathbf{b} := \bar{\mathbf{b}}$. Since $AA^\dagger\bar{\mathbf{b}} = AA^\dagger(AA^\dagger\mathbf{b}) = AA^\dagger\mathbf{b} = \bar{\mathbf{b}}$ and since $A^\dagger$ is a solution of the Penrose's equations (a), (b) and (d), by (ii) the minimum norm solution of the equation $A\mathbf{x} = \bar{\mathbf{b}}$ is given by $\bar{\mathbf{x}} = A^\dagger\bar{\mathbf{b}} = A^\dagger AA^\dagger\mathbf{b} = A^\dagger\mathbf{b}$. ■

A.14 Special matrices

In this section we mention some properties of some special square matrices that we have introduced in the previous sections.

A.14.1 Permutation matrices

A *permutation* σ of the set $\{1, 2, \dots, n\}$ is an injective (and thus surjective) map from $\{1, 2, \dots, n\}$ onto itself. To specify a permutation there is the need to define the values $\sigma(1) = i_1, \sigma(2) = i_2, \dots, \sigma(n) = i_n$; the standard approach is to indicate such values through the notation

$$\sigma = \begin{pmatrix} 1 & 2 & \dots & n \\ i_1 & i_2 & \dots & i_n \end{pmatrix}, \quad (\text{A.140})$$

where the second row defines the transformed elements corresponding to the elements of the first row. The inverse permutation associated to σ , i.e., the permutation that brings the elements back to their original positions, is given by

$$\sigma^{-1} = \begin{pmatrix} i_1 & i_2 & \dots & i_n \\ 1 & 2 & \dots & n \end{pmatrix}.$$

The composition of two permutations σ and τ of $\{1, 2, \dots, n\}$ is the resulting permutation

$$\sigma\tau : \{1, 2, \dots, n\} \mapsto \{1, 2, \dots, n\} : k \mapsto \tau(k) \mapsto \sigma(\tau(k))$$

represented in an expanded form by

$$\sigma\tau = \begin{pmatrix} 1 & 2 & \dots & n \\ \sigma(\tau(1)) & \sigma(\tau(2)) & \dots & \sigma(\tau(n)) \end{pmatrix}.$$

A cyclic permutation γ , also called a *cycle*, is a permutation that cyclically permutes a subset $\{i_1, i_2, \dots, i_r\}$ of $\{1, 2, \dots, n\}$, in the sense that

$$\gamma(i_1) = i_2, \gamma(i_2) = i_3, \dots, \gamma(i_{r-1}) = i_r, \gamma(i_r) = i_1,$$

and that it leaves unchanged the other $n - r$ numbers. Such cyclic permutations are typically denoted by the symbol $\gamma = (i_1 i_2 \dots i_r)$. Two cycles are also said to be disjoint if their symbols do not contain common letters. It is remarkable that every permutation σ can be written as the composition of a set of disjoint cycles, and these cycles are uniquely identified by σ .

Let $\mathbf{e}_j$, $j = 1, 2, \dots, n$, be the column vectors defining the canonical basis of $\mathbb{R}^n$. Then one can associate to the generic permutation σ in (A.140) the so-called *permutation matrix*

$$\Pi_\sigma = [\mathbf{e}_{i_1} \quad \mathbf{e}_{i_2} \quad \dots \quad \mathbf{e}_{i_n}] \quad (\text{A.141})$$

in which the element $[\Pi_\sigma]_{h,k}$ is 1 if $h = \sigma(k)$, and is 0 if $h \neq \sigma(k)$. In other words this means that in such a permutation matrix the elements that are equal to 1 are all and only the elements whose position is $(\sigma(\nu), \nu)$ (or, equivalently, whose position is $(\nu, \sigma^{-1}(\nu))$) for each $\nu = 1, 2, \dots, n$. All the other elements are, instead, equal to 0. Summarizing, Π_σ is such that

- the elements of the k -th column are null, except for those whose index is $\sigma(k) = i_k$; this element has indeed value 1;
- the elements of the h -th row are null, except for those whose index is $\sigma^{-1}(h)$; this element has indeed value 1.

If $A \in \mathbb{R}^{q \times n}$, then the k -th column of the matrix $A\Pi_\sigma$ is equal to the $\sigma(k)$ -th column of A . Indeed,

$$[A\Pi_\sigma]_{h,k} = \sum_j [A]_{h,j} [\Pi_\sigma]_{j,k} = [A]_{h,\sigma(k)} [\Pi_\sigma]_{\sigma(k),k} = [A]_{h,\sigma(k)}, \quad h = 1, 2, \dots, q.$$

Example A.14.1 Consider the permutation of $\{1, 2, 3, 4\}$ defined by $\sigma = (1 \ 2 \ 3 \ 43 \ 4 \ 2 \ 1)$. To this permutation we can associate the permutation matrix

$$\Pi_\sigma = [\mathbf{e}_3 \quad \mathbf{e}_4 \quad \mathbf{e}_2 \quad \mathbf{e}_1] = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}.$$

If we then consider the 2×4 matrix

$$A = \begin{bmatrix} e & 6 & 2 & \pi \\ 5 & \sqrt{2} & 4 & 9 \end{bmatrix},$$

then the k -th column in

$$A\Pi_\sigma = \begin{bmatrix} e & 6 & 2 & \pi \\ 5 & \sqrt{2} & 4 & 9 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 2 & \pi & 6 & e \\ 4 & 9 & \sqrt{2} & 5 \end{bmatrix}$$

is equal to the $\sigma(k)$ -th column in A .

We now define some properties of permutation matrices.

- i) [PRODUCT OF PERMUTATION MATRICES] If σ and τ are two permutations of $\{1, 2, \dots, n\}$, then $\Pi_{\tau\sigma} = \Pi_{\sigma}\Pi_{\tau}$. Indeed it follows immediately that

$$[\Pi_{\sigma}\Pi_{\tau}]_{h,k} = \sum_j [\Pi_{\sigma}]_{h,j} [\Pi_{\tau}]_{j,k} = [\Pi_{\sigma}]_{h,\sigma^{-1}(h)} [\Pi_{\tau}]_{\sigma^{-1}(h),k} = 1$$

if and only if $\sigma^{-1}(h) = \tau(k)$ (i.e., if and only if $h = \sigma(\tau(k))$). For all the other possible choices of h and k the corresponding element $[\Pi_{\sigma}\Pi_{\tau}]_{h,k}$ is instead equal to zero. Thus $\Pi_{\sigma}\Pi_{\tau}$ coincides with $\Pi_{\tau\sigma}$.

- ii) [INVERSE OF A PERMUTATION MATRIX] If Π_{σ} is a permutation matrix, then $\Pi_{\sigma}^T = \Pi_{\sigma}^{-1} = \Pi_{\sigma^{-1}}$. Indeed $[\Pi_{\sigma}^T]_{h,\sigma(h)} = [\Pi_{\sigma}]_{\sigma(h),h} = 1, \forall h$, while all the remaining elements of the matrix are null. Letting thus $h = \sigma^{-1}(k)$, for every k it follows that $1 = [\Pi_{\sigma}^T]_{\sigma^{-1}(k),k}$, while all the remaining elements of the matrix are null. Thus $\Pi_{\sigma}^T = \Pi_{\sigma^{-1}}$, and this equals Π_{σ}^{-1} thanks to the properties of the product of permutation matrices above.

- iii) [SIMILARITY RELATIONS INDUCED BY PERMUTATION MATRICES] Let $A \in \mathbb{R}^{n \times n}$, and let Π_{σ} be a permutation matrix. From the previous relations it follows that $\Pi_{\sigma}^{-1}A\Pi_{\sigma} = \Pi_{\sigma^{-1}}A\Pi_{\sigma}$, and thus that

$$\begin{aligned} [\Pi_{\sigma^{-1}}A\Pi_{\sigma}]_{h,k} &= \sum_{i,j} [\Pi_{\sigma^{-1}}]_{h,i} [A]_{i,j} [\Pi_{\sigma}]_{j,k} = \sum_i [\Pi_{\sigma^{-1}}]_{h,i} A_{i,\sigma(k)} [\Pi_{\sigma}]_{\sigma(k),k} \\ &= [\Pi_{\sigma^{-1}}]_{h,\sigma(h)} A_{\sigma(h),\sigma(k)} [\Pi_{\sigma}]_{\sigma(k),k} = [A]_{\sigma(h),\sigma(k)}. \end{aligned}$$

The similarity relations induced by permutations matrices have the special name of *cogredience transformations*.

Among the permutation matrices of size $n \times n$ there exist some that play special roles. For example, consider

$$S = \begin{bmatrix} 0 & 1 & 0 & 0 & \dots & 0 \\ 0 & 0 & 1 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots & & \vdots \\ 0 & 0 & 0 & 0 & \dots & 1 \\ 1 & 0 & 0 & 0 & \dots & 0 \end{bmatrix}.$$

This matrix, called the *shift matrix* S associated to the cycle $(1, 2, \dots, n)$, and the associated *counteridentity*

$$R = \begin{bmatrix} 0 & 0 & 0 & \dots & 0 & 1 \\ 0 & 0 & 0 & \dots & 1 & 0 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & 1 & 0 & \dots & 0 & 0 \\ 1 & 0 & 0 & \dots & 0 & 0 \end{bmatrix}.$$

- EXERCISE A.14.1 Verify that the shift matrix $S \in \mathbb{R}^{n \times n}$ is such that:
 - i) $I_n, S, S^2, \dots, S^{n-1}$ are linearly independent, and moreover $S^n = I_n$, so that S has s^{n-1} as its minimal polynomial and the n roots of 1 as eigenvalues;
 - ii) $S^T = S^{-1} = S^{n-1}$;
 - iii) $[S^{-1}MS]_{h,k} = [M]_{h-1,k-1}$, where the indexes should be intended modulo n (so that the element 0 should be intended as n).

- EXERCISE A.14.2 Verify that the counteridentity $R \in \mathbb{R}^{n \times n}$ is such that:
 - i) $R = R^T$, $R^2 = I_n$, so that $R = R^{-1}$;
 - ii) since $s^2 - 1$ is the polynomial annihilator, R has two distinct eigenvalues, $+1$ and -1 , except for the case when $n = 1$.
 - iii) Find which permutation is associated to R , and what is its decomposition in disjoint cycles.
- EXERCISE A.14.3 Let σ be a permutation of $\{1, 2, \dots, n\}$, and let $\sigma = \gamma_1, \gamma_2, \dots, \gamma_t$ be its decomposition in disjoint cycles. Verify that:
 - i) there exists a similarity relation induced by a permutation matrix that brings π_σ to a block-diagonal form that has as diagonal blocks all the shift matrices associated to the individual cycles γ_i ;
 - ii) all the eigenvalues of Π_σ are roots of one;
 - iii) how are the characteristic polynomial, the minimal polynomial, and the Jordan canonical form of Π_σ connected to the decomposition of σ into the disjunct cycles?

A.15 Bibliography

The monographies on Linear Algebra are very numerous, and we cite here only books that cover in a systematic way all the elements that have been presented in this appendix.

The following books take an axiomatic approach that is based on considering abstract vector spaces (in the last two references also considering infinite dimensions):

1. P.Halmos “*Finite Dimensional Vector Spaces*”, Springer, 1974
2. S.Lang “*Algebra linear*”, Boringhieri, 1970
3. W.Greub “*Linear Algebra*” (quarta edizione), Springer, 1975
4. N.Jacobson “Lectures in Abstract Algebra: vol 2, Linear algebra”, Van Nostrand, 1953

The following books are instead mainly focused on matrices and operations on matrices:

5. F.R.Gantmacher “*Matrix Theory*”, 2 volumi, Chelsea, 1964
6. R.Bellman “*Introduction to Matrix Analysis*”, McGraw Hill, 1960
7. R.Horn, C.Johnson “*Matrix Analysis*”, vol.1 and 2, Cambridge U.P., 1985
8. G.Golub, C.Van Loan “*Matrix Computations*”, J.Hopkins U.P., 1989
9. A.Ben-Israel, T. Greville “*Generalized Inverses*”, Springer, 2002

We also mention the following encyclopedic book, containing synthetic but accurate descriptions of numerous aspects of Linear Algebra (both theoretical and applications-driven), together with an extensive bibliography for each of the treated arguments:

10. “*Handbook of Linear Algebra*”, curato da L.Hogben, Chapman & Hall/CRC, 2007,